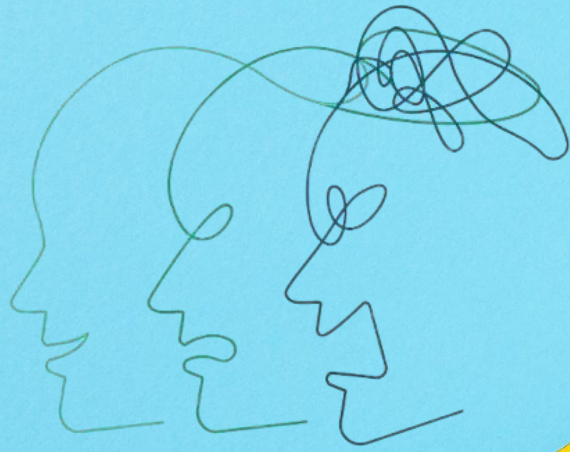


# Hard launching : Introduction to Psychology



Course instructor:  
Shweta Chawak  
IIT Bombay  
Humanities and Social Sciences

# Hello there!

## **Shweta Chawak (she/her/hers)**

- PhD in Health & Medical Psychology (IIT Hyderabad)
- Research interests: psycho-oncology, preventive health, and women's health
- “How do people with cancer diagnosis can openly talk to their doctor and family?”

## **Things that do not make sense to me!**

- Bitter coffee
- Sigma boy song, Skibidi, 67!

Reach me at: [shwetachawak@iitb.ac.in](mailto:shwetachawak@iitb.ac.in)

Student hours: By appointment

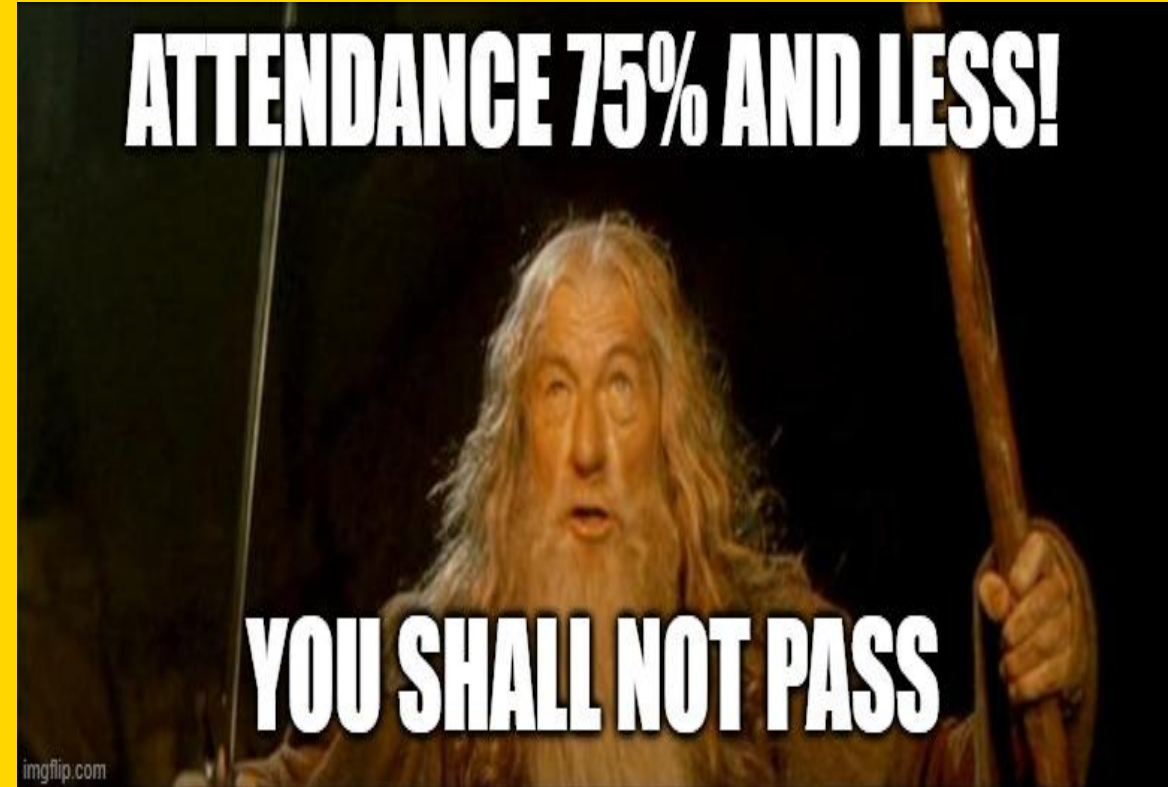
✨ Feel free to drop in memes and pop culture references to Psychology

# Important reminder:

- We will be covering issues that can potentially be sensitive to some individuals
- Everyone has their own experiences, perspectives, stories, and frames on which they view things. **I only ask that you be respectful**
- Safe Space and Respect for Diversity

# Attendance policy!

- 75% attendance is appreciated
- DX grade will be granted
- Life happens: No questions asked can avail two leaves (but need to inform the faculty)
- Attending the lecture is not equivalent to participation



# Some class rules:

- **Use of Phone is discouraged** (unless needed for a class activity- Prof Decides!)
- Any special support needed- Feel free to reach out



Using  
phones to  
click pictures  
of slides



Using  
phones for  
class activity

# Course Outline and Assessment

| Weeks      | Unit                                          | Quiz     |
|------------|-----------------------------------------------|----------|
| Week 1-2   | What is Psychology and Research in Psychology | Quiz I   |
| Week 3-4   | Learning                                      |          |
| Week 5-7   | Memory                                        | Quiz II  |
| Week 8-9   | Intelligence                                  | Quiz III |
| Week 10-11 | Motivation and Emotions                       | Quiz IV  |
| Week 12-13 | Personality                                   |          |

- Assessment I: Quiz (Best of 2 from 4)- 20 Marks
- Assessment II: Mid-sem – 30 Marks
- Assessment III: End-sem- 50 Marks

# Course material

## Core Textbook:

- **Ciccarelli, S. K., & Meyer, G. E. (2006). *Psychology*. Pearson Education.**

## Additional Texts:

- Coon, D., & Mitterer, J. O. (2010). *Introduction to psychology* (12th ed.). Wadsworth Cengage Learning.
- OpenStax, & CNX. (2015). *Introduction to psychology*. OpenStax.
- Feldman, R. S. (2009). *Essentials of understanding psychology* (8th ed.). McGraw-Hill.

Students are free to refer any textbook or journal article for further review

# What is Psychology according to you?

# What is it to be a Psychology student?

*Can you read my mind?*

*Need me some psych advice*

*I always find people watching interesting*

*Oh well, I better not say anything else, or you'll psychoanalyse me!*

*We'll be great friends because I can be a case study for you!  
I have lots of problems!*

*I don't want you to fix me*

Source: Trust me bro  
Reddit

# ACTUALLY...



# Professor debunks Psychology!



# What is psychology?

- Greek word: psyche = mind
- Logos = knowledge or study
- Scientific study of **behaviour** and **mental processes** → Overt behaviours: directly observable actions and responses  
Covert behaviours: internal

Attempts to study **affect**, **behaviour**, and **cognition** (ABC's)

- Affect: emotions; feelings; urges
- Behaviour: (un)observable actions and responses taken by the individual
- Cognitions: our inner thoughts; our rationales; our beliefs



*Why do you do what you do?*

(Ciccarelli & Meyer, 2006; Coon & Mitterer, 2010; Feldman, 2009)



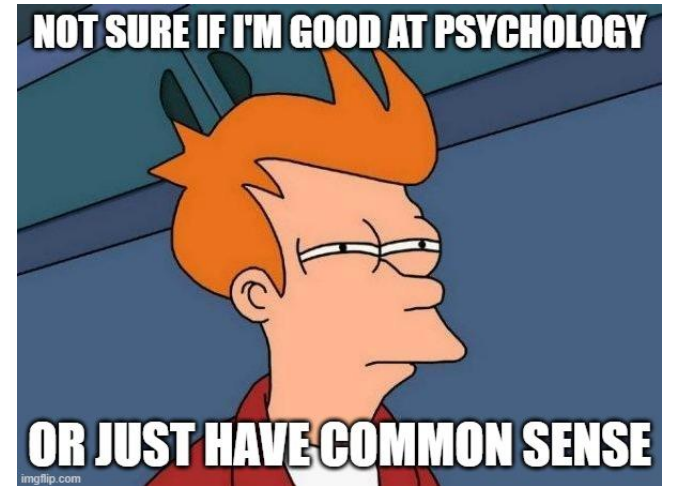
# Is Psychology similar to common-sense?

- Punishing children Strongly is the most effective way to stop bad behaviour
- We are generally good at judging other people's personality after meeting them briefly
- What works psychologically for one person will usually work for most others
- Human memory works like a video recording that can be replayed accurately

# Why is psychology called as science?

- Is psychology similar to common-sense?
- Emphasis on empirical evidence (direct observation or experiment)
- Systematic and intersubjective

(Coon & Mitterer, 2010; Feldman, 2009)



# Conclusion

- Psychology
  - Greek word: psyche = mind
  - Logos = knowledge or study
- Scientific study of behaviour and mental processes
  - Overt and Covert behaviour
- Study of ABC: Affect, Behaviour, and Cognition
- Not common sense:
  - Focuses on evidence
  - Systematic and intersubjective

# The origin of psychology

Course instructor:  
Shweta Chawak  
IIT Bombay  
Humanities and Social Sciences

# RECAP!

- Psychology
  - Greek word: psyche = mind
  - Logos = knowledge or study
- Scientific study of behaviour and mental processes
  - Overt and Covert behaviour
- Study of ABC: Affect, Behaviour, and Cognition
- Not common sense:
  - Focuses on evidence
  - Systematic and intersubjective

***Long Time Ago... In Galaxy not so  
far away!  
Long before Psychology became  
Science, humans were trying to  
understand  
How are mind body connected?***



# In the Beginning!

- Aristotle: soul is related to the body
- Plato and Rene Descartes: soul is separate from the body known as the dualism
- Philosophers: human mind and its connection to the physical body
- Physicians and physiologists: physical connection between the body and the brain

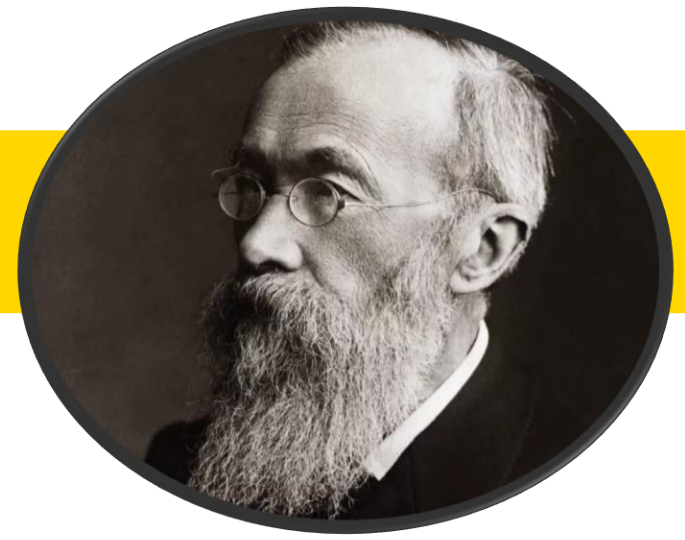


First question people ask when you say you study  
Psychology....

*Are you analysing me?*

*Do you know what I am thinking right now?*

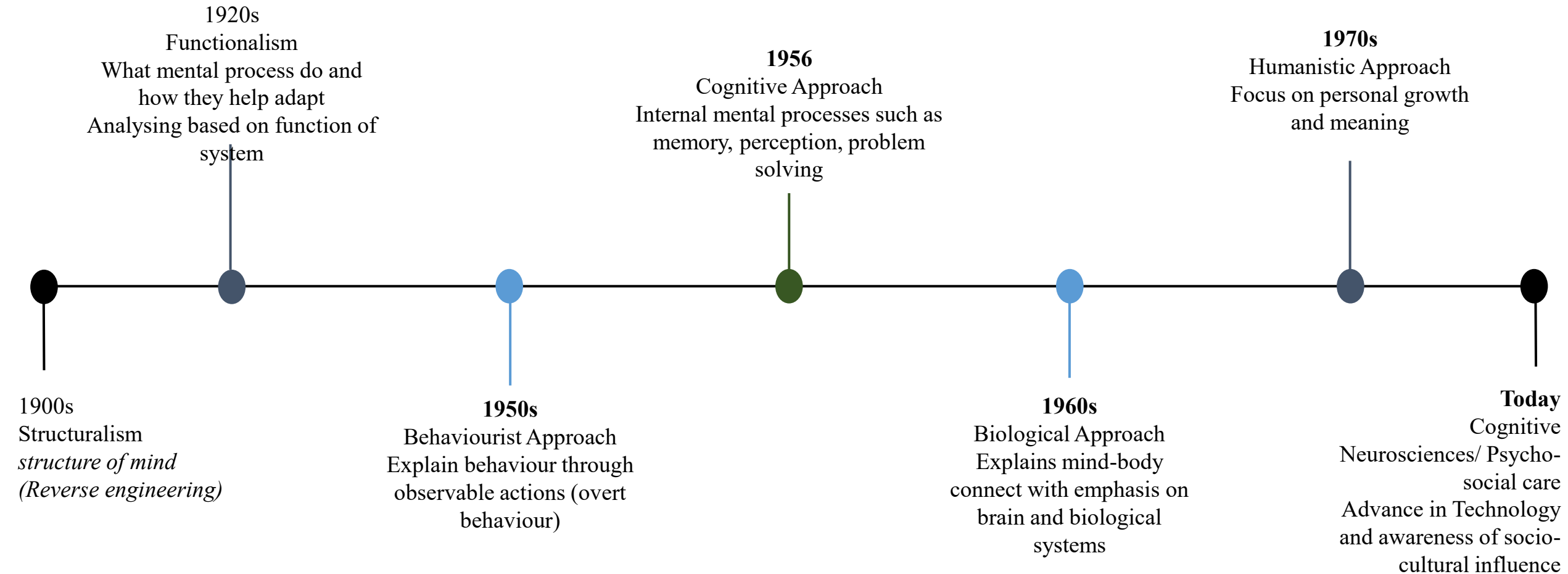
# Birth of Scientific Psychology



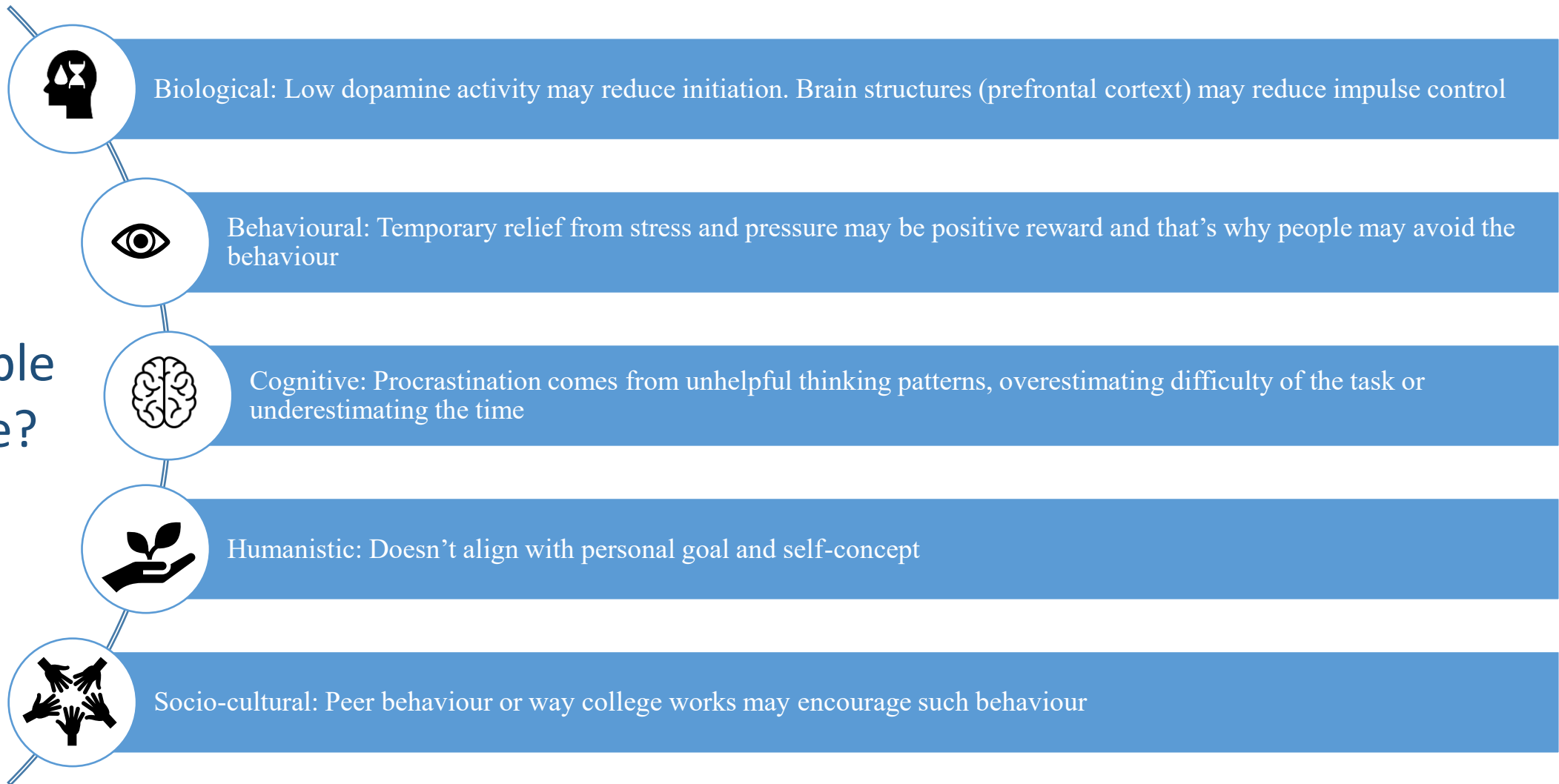
- Structuralism: Wilhelm Wundt aka **Father of Psychology**
- Application of scientific principles to the study of human mind: “What is the structure of the mind?”
- *Introspection:*
  - *Observe their experiences*
  - *Describe sensations precisely (intensity/ quality/ duration)*
  - *Avoid interpretation or meaning making*
- *Laboratory based study!*

- F: Father of Psychology
- A: Analytical introspection
- T: Trained first Psychology students
- H: Helped establish Psychology as science
- E: Established lab at Leipzig, Germany (1879)
- R: Research on Consciousness (Sensations, Images, Feelings)

# Timeline of psychology approaches



# Why do people procrastinate?



## Activity

Why do people get addicted to their phones?

# Conclusion

- Father of Psychology: Wilhem Wundt (Structuralism)
- **Approaches:**
  - Biological: dopamine, reward pathway, prefrontal cortex
  - Behavioral: reinforcement, habit learning, cues
  - Humanistic: needs, meaning, self-growth
  - Cognitive: beliefs, attention, interpretation
  - Sociocultural: norms, peer influence, family context

# Fields of Psychology

Course instructor:  
Shweta Chawak  
IIT Bombay  
Humanities and Social Sciences



## Approaches Recap!

### Biological Approach

- Focuses on influences of hormones, brain structures, and chemicals
- Human and animal behaviour is seen as a direct result of events in the body

### Behavioural Approach

- Focuses on observable behaviour learned through environment interaction

### Cognitive Approach

- Focuses on internal mental processes
- Memory, Intelligence, perception, thought processes, language, and problem solving

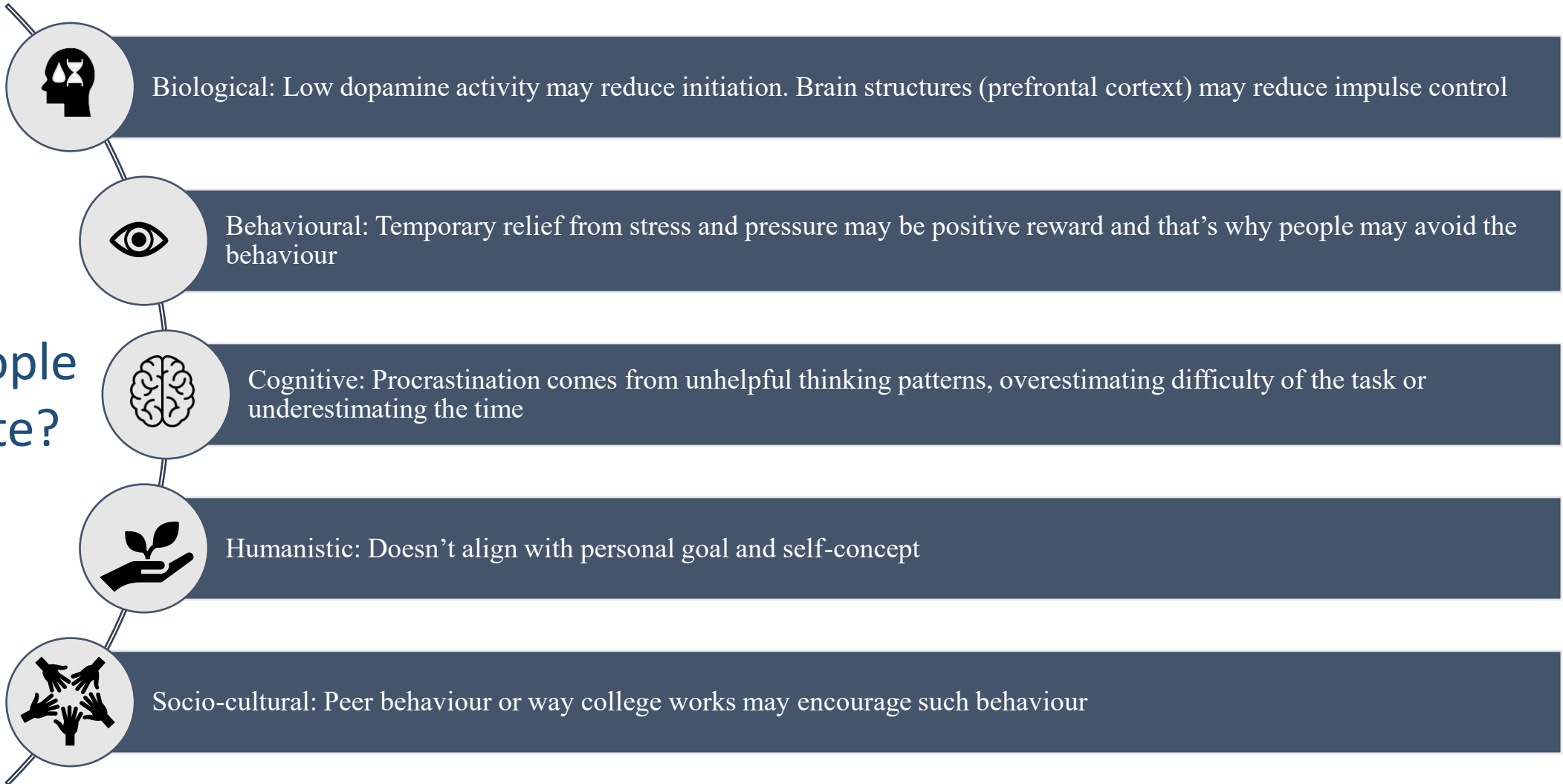
### Humanistic Approach

- Focuses on human potential, free will, and self-actualisation
- Personal goals and growth

### Sociocultural Approach

- Focuses on behaviour of individual as a result of presence of other people, being part of large group, or large culture

## Why do people procrastinate?



## Activity

Why do people get addicted to their phones?

<https://www.instagram.com/uroojashfaq/>



# Types of psychological professionals

🌟 Who is a psychiatrist? Who is a psychologist? 🌟

| Type of professional   | Psychiatrist                                                                                                                                          | Psychologist                                                                                                                                                                                  | Psychiatric social worker                                                                                                  |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Educational degree     | Doctor of Medicine (MD)                                                                                                                               | Doctorate degree                                                                                                                                                                              | Master of Social Work (MSW)                                                                                                |
| Training               | Medical doctor (MBBS)                                                                                                                                 | Academic                                                                                                                                                                                      | Academic                                                                                                                   |
| Specialisation         | <ul style="list-style-type: none"> <li>▪ Diagnosis and treatment of psychological disorders</li> <li>▪ May provide therapy and counselling</li> </ul> | <ul style="list-style-type: none"> <li>▪ Master's degree: counsellor, therapist, researcher</li> <li>▪ Clinical psychologist: license by the Rehabilitation Council of India (RCI)</li> </ul> | <ul style="list-style-type: none"> <li>▪ Environmental conditions</li> <li>▪ E.g., poverty, stress, drug misuse</li> </ul> |
| Prescribe medications? | Yes                                                                                                                                                   | No                                                                                                                                                                                            | No                                                                                                                         |
| Workplace setting      | Private practice or hospital settings                                                                                                                 | Multiple                                                                                                                                                                                      | Clinics, hospitals, social service organisations                                                                           |

# Fields of psychology



Human behaviour  
is multilayered

|                                       |                                    |                                                                                   |
|---------------------------------------|------------------------------------|-----------------------------------------------------------------------------------|
| Foundational Fields                   | Biological Psychology/Neuroscience | Examines the biological bases of behavior and mental processes                    |
|                                       | Cognitive Psychology               | Studies mental processes like memory, perception, and problem-solving             |
|                                       | Development Psychology             | Examines psychological growth and changes throughout the lifespan                 |
|                                       | Social Psychology                  | Explores how people influence and relate to each other                            |
|                                       | Behavioural Psychology             | Analyses observable behavior and its modification                                 |
|                                       | Personality Psychology             | Investigates individual differences in personality traits                         |
| Applied Fields                        | Clinical                           | Focuses on diagnosing and treating mental health disorder                         |
|                                       | Counselling                        | Helps individuals cope with personal and interpersonal challenges                 |
|                                       | Health                             | Studies psychological factors affecting health and illness                        |
|                                       | Industrial/Organisational          | Focuses on human behavior in the workplace to improve productivity and well-being |
|                                       | Educational                        | Focuses on how people learn and the effectiveness of educational interventions    |
|                                       | Forensic                           | Applies psychology to legal and criminal justice issues                           |
|                                       | Sports                             | Optimises athletic performance and psychological well-being in sports             |
| Interdisciplinary and emerging fields | Cultural Psychology                | Cultural context shapes cognition, emotion, and behaviour                         |
|                                       | Environmental                      | Studies the interaction between people and their physical environment             |
|                                       | Positive                           | Focus on well-being, strengths, resilience, flourishing                           |
|                                       | Psycholinguistics                  | Focus on how we acquire and use language                                          |
|                                       | Neuropsychology                    | Brain and behaviour in a clinical setting                                         |

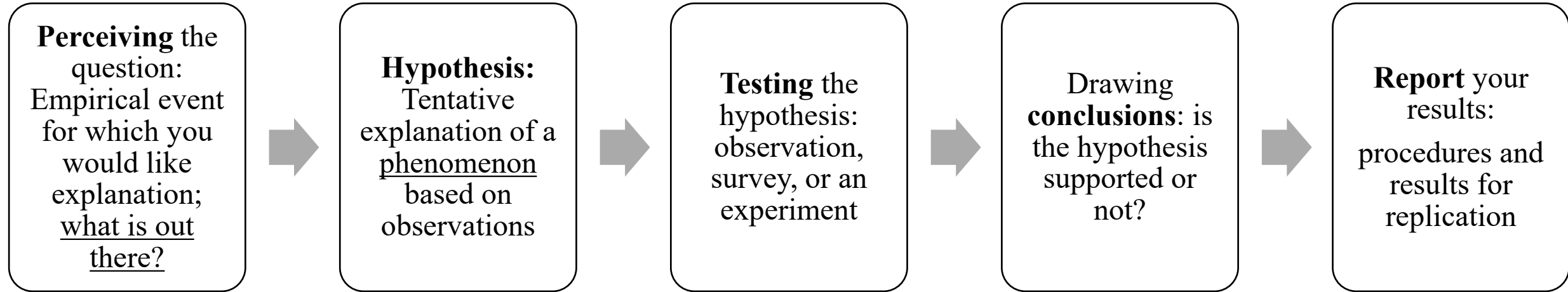
# RECAP

- Approaches: Biological, Behavioural, Cognitive, Humanistic, Sociocultural
- Types of psychological professional: Psychologists vs Psychiatrist
- Fields of Psychology
  - Foundational
  - Applied
  - Interdisciplinary



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IIT Bombay  
Humanities and Social Sciences

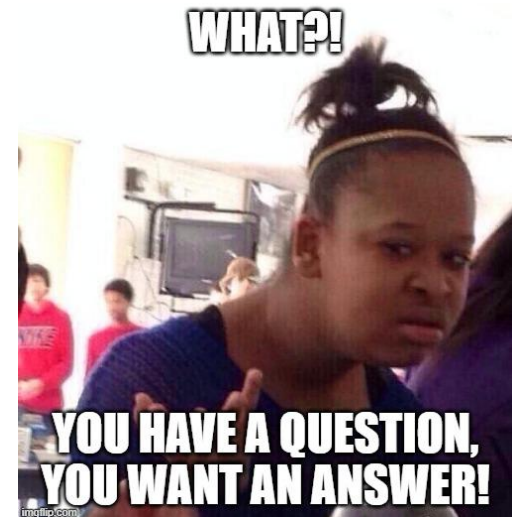
# What is this Scientific Method?



Question: Does frequent use of Instagram impact the attention span of Gen Z college students?

Hypothesis: Increased daily use of Instagram is associated with shorter attention spans in Gen Z college students

(Ciccarelli & Meyer, 2006; Feldman, 2009)



# Methods in Psychology



(Ciccarelli & Meyer, 2006)

# Descriptive Method

- Systematic approach used to **observe**, **describe**, and **document** aspects of human behaviour, phenomena, or populations as they naturally occur
- Key Feature: Focus on observations
- Types of descriptive research
  - Naturalistic observation
  - Laboratory observation
  - Clinical case study
  - Survey
  - Cross-sectional and Longitudinal research

# Descriptive Method: Naturalistic Observation

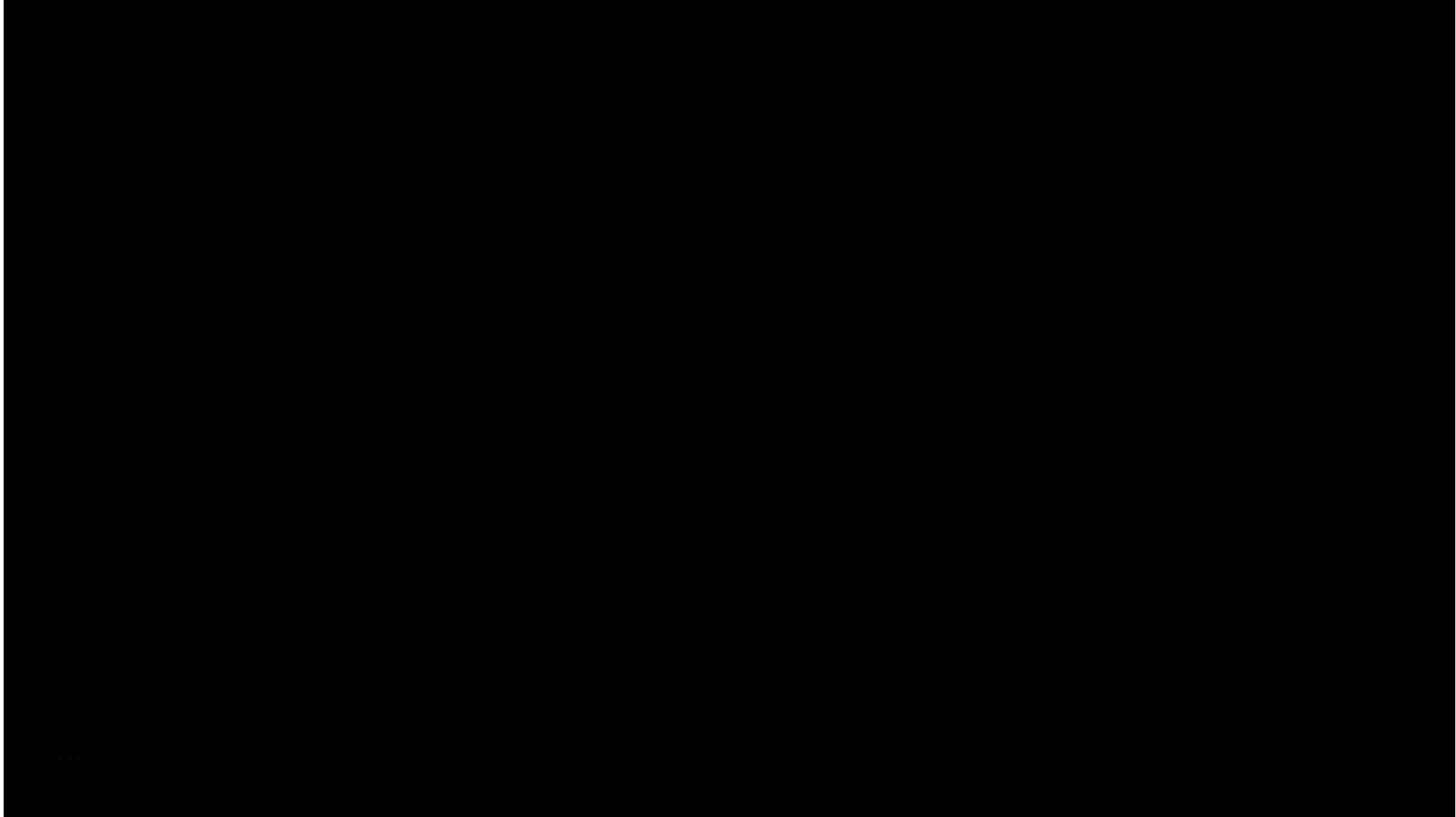
- Behaviour(s) of people and animal in their **natural environment**
- **Observer effect**: tendency to behave differently when ‘they’ know that they are being observed
- **Participant observation**: researcher becomes a participant in the group being observed
- Strength:
  - First-hand look
- Drawbacks:
  - Observer bias: tendency of researchers to see what they expect to see
  - Hawthorne effect: Increase in effective behaviour when being observed
  - Cannot be replicated

(Ciccarelli & Meyer, 2006)



“What is happening out there?”

## Hawthorne effect



<https://www.youtube.com/watch?v=o4XX90lqT6E&t=22s>

# Descriptive Method: Laboratory Observation

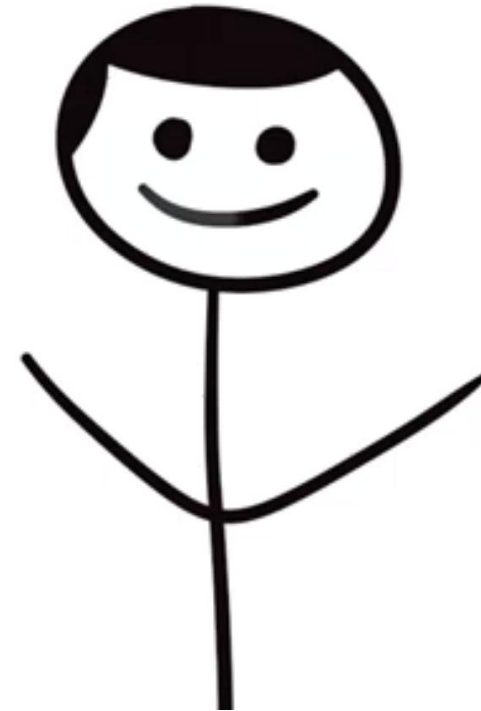
- Practicality issue is observing in natural setting
- Difficult in setting up screening equipment
- Trying to control various environmental factors



<https://www.youtube.com/watch?v=QTsewNrHUHU>

# Descriptive Method: Clinical Case Study

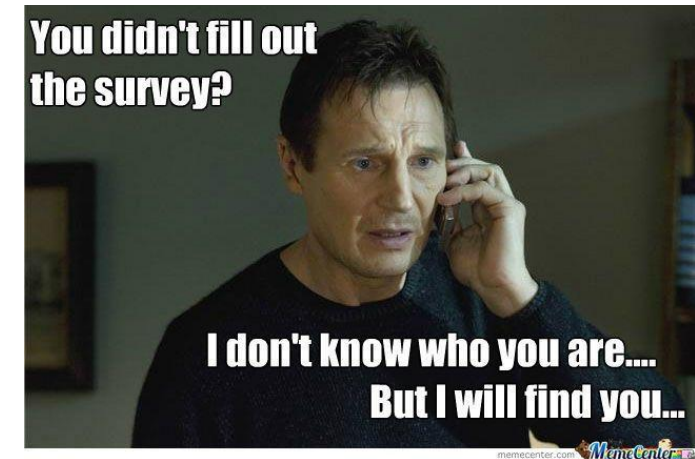
- Individual studies in great details, try learning everything about one individual
- E.g., Phineas Gage
- Strength:
  - In-depth
- Drawback:
  - Cannot be replicated



(Ciccarelli & Meyer, 2006)

# Descriptive Method: Survey

- Series of questions (mostly anonymous)
- Mode: in person, interview, or a questionnaire (online or offline)
- Covert Behaviours are assessed or screened
- **Representative sample**: randomly selected sample of participants from a larger population of participants
  - Example: Students of IIT Bombay
- Strength:
  - Can gather large data within a short period



**Random: equal chance of being chosen**



# Descriptive Method: Survey (cont'd...)

<https://www.instagram.com/khapiter.guys>

15/01/2026

HS 110 | Introduction to Psychology



10

# Descriptive Method: Survey (cont'd...)

## *Biases in survey*

### **Courtesy bias:**

Accuracy of response  
(responding in socially  
accurate manner)

What is socially desirable  
Tests involves Lie scale or  
repeated items

### **Wording bias:**

The way question is  
framed distorts the answers  
than the belief of  
individual

Examples: Leading  
questions, Emotional  
questions, Complex or  
confusing words

### **Question order bias:**

It occurs when sequence of  
questions in a survey  
influences how  
respondents answer them.

For example: Priming  
effect, fatigue effect, order  
effect (leading to specific  
way of answering)

# Quick Recap!

- Scientific Method
- Perceiving the question – Hypothesis – Hypothesis Testing – Conclusion – Report
- Types of research methods
  - Descriptive
  - Correlational
  - Experimental
- Descriptive types
  - Naturalistic observation
  - Laboratory observation
  - Clinical case study
  - Survey



Course instructor:  
Shweta Chawak  
IIT Bombay  
Humanities and Social Sciences

# Quick Recap!

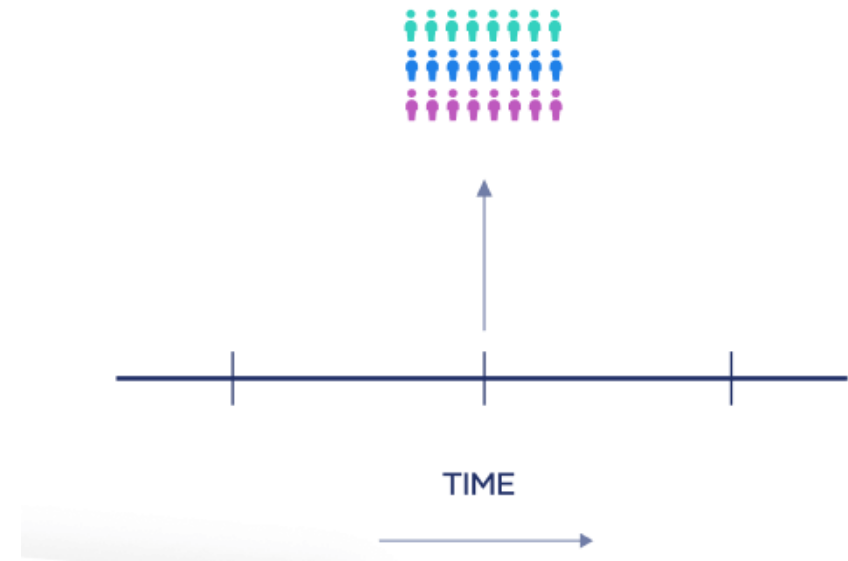
- Scientific Method
- Perceiving the question – Hypothesis – Hypothesis Testing – Conclusion – Report
- Types of research methods
  - Descriptive
  - Correlational
  - Experimental
- Descriptive types
  - Naturalistic observation
  - Laboratory observation
  - Clinical case study
  - Survey

# Descriptive Method: Cross-Sectional & Longitudinal Research

- Cross-sectional research: analysing information about a population at a **specific point in time**
- Strength:
  - Cost-effective
  - Can incorporate multiple variables and groups
- Drawbacks:
  - Reporting bias: sharing selective information from the participant and/or researcher
  - Single time point: difficult to understand long term trends or analyse change over time

## Cross-sectional study

Data collected at one point in time



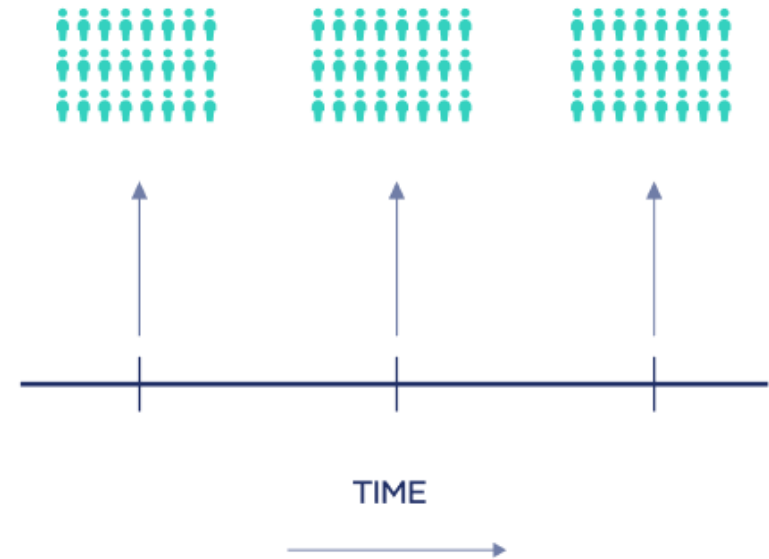
(Simply Psychology, 2023)

# Descriptive Method: Cross-Sectional & Longitudinal Research

- Longitudinal research: analysing information about a specific population over an **extended period of time**
- **Retrospective study:** you collect data on events that have already happened
- **Prospective study:** you choose a group of subjects and follow them over time, collecting data in real time
- Strength:
  - Detect patterns and changes in the population
- Drawbacks:
  - Expensive and time-consuming

## Longitudinal study

Data collected repeatedly over time



(Simply Psychology, 2023)



# Descriptive Method: Cross-Sectional & Longitudinal Research (cont'd...)

| Research Type            | What It Means                                          | Example                                                                                                                |
|--------------------------|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Cross-sectional research | Looks at different groups at one moment in time        | Surveying high schoolers, college students, and new grads <b><u>all in 2025</u></b> to see who uses Instagram the most |
| Longitudinal research    | Studies the same group of people over a period of time | Tracking the same group of Gen Z teens <b><u>from 2020 to 2025</u></b> to see how their favorite social apps change    |



*Cross-sectional research is like taking a group selfie: everyone at once, different ages or groups*



*Longitudinal research is like a series of throwback selfies: same people, but over a few years*

# Correlational Research: Finding Relationships

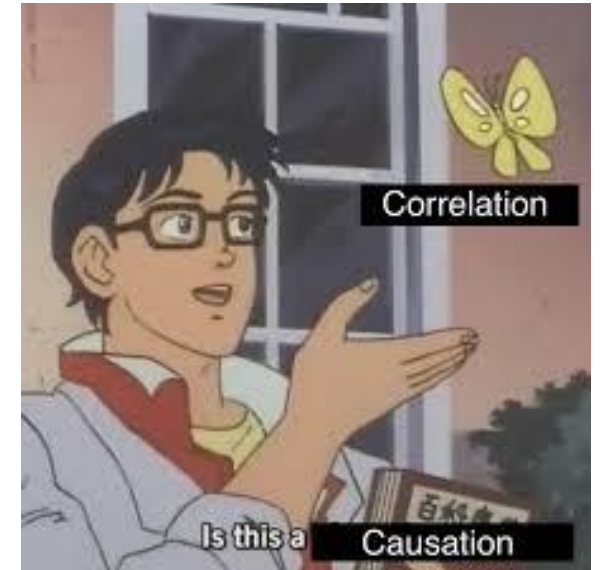
- Correlational research is a statistical technique
- Looks for **patterns** in the data
- A measure of the **relationship** between two or more variables
- **Variable**: anything that can change or vary



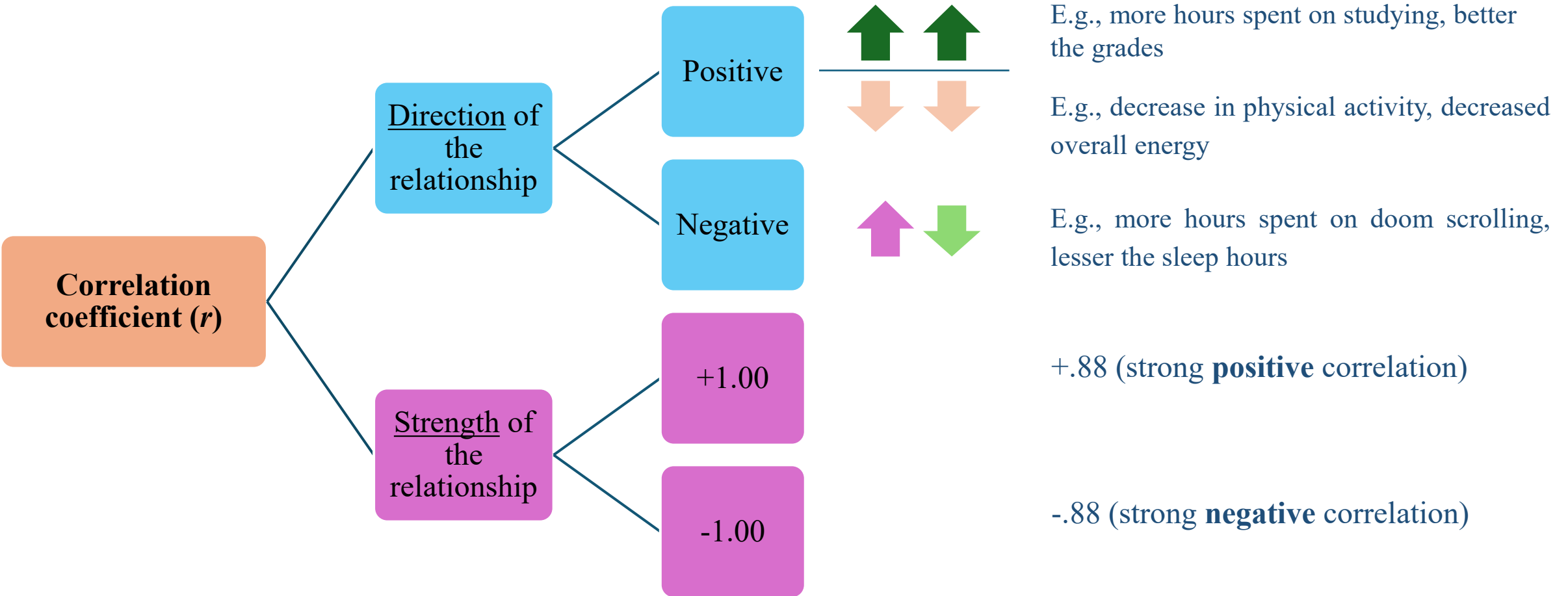
E.g.,

- Ice cream sales and drowning both increase in the summer
- Students who spend more time on social media often feel more anxious

**Correlation does not imply causation!**



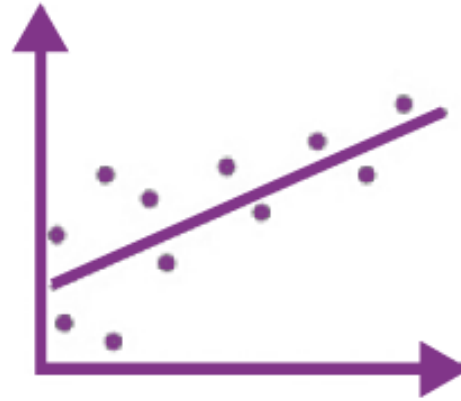
# Finding Relationships: Correlational Research (cont'd...)



✨ Closer to zero: weaker relationship ✨



**Strong positive correlation**



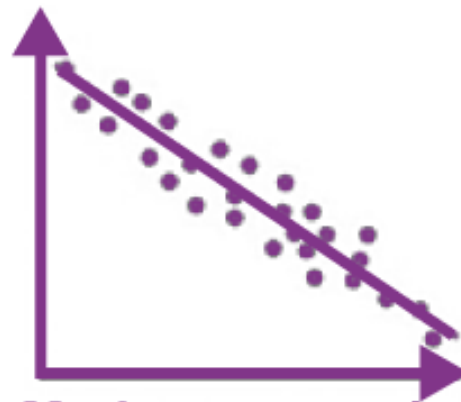
**Weak positive correlation**



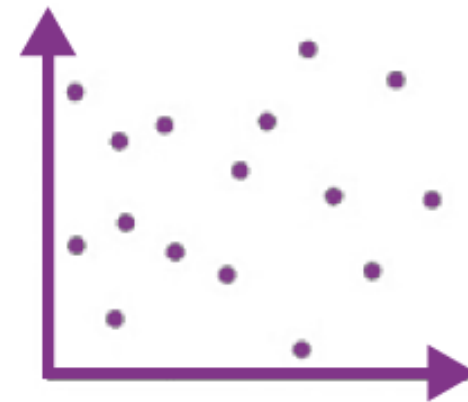
**Strong negative correlation**



**Weak negative correlation**



**Moderate negative correlation**



**No correlation**

# Quick Recap!

## Cross sectional research

- Observational study that collects data from diverse population at a single point in time
- Helps understand prevalence and describe characteristics

## Longitudinal research

- A study method that involves collecting data from same individual or group over extended period of time from weeks to decades

## Correlational research

- Non experimental method that measures the statistical relationship between two or more variables
- It seen if change in one corresponds change in another without manipulation



Course instructor:  
Shweta Chawak  
IIT Bombay  
Humanities and Social Sciences

# Experimental Research: Cause and Effect Relationships

## Experiment

Direct manipulation of a variable to see if corresponding changes in behaviour result, allowing the determination of cause and effect relationships

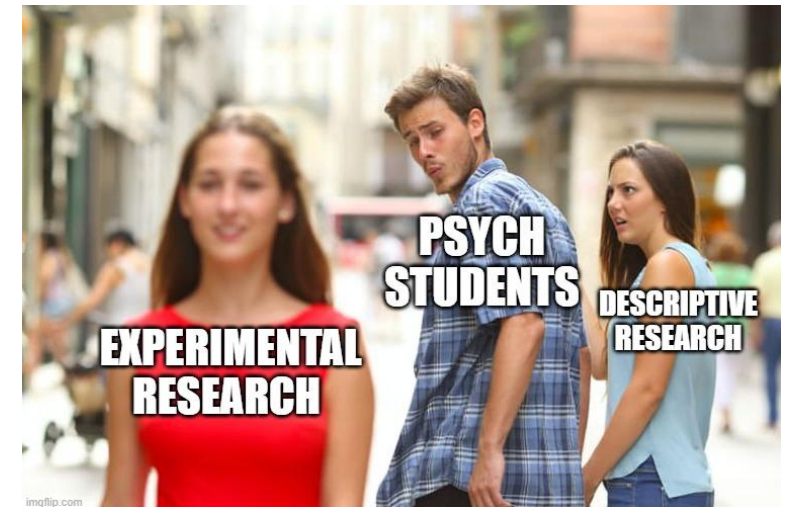


## Operational definition

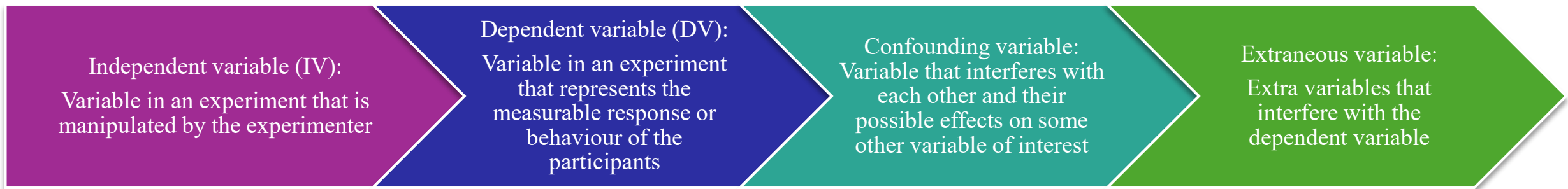
Definition of a variable of interest that allows it to be directly measured; specific

## Variable

Anything that can be measured, manipulated, or controlled

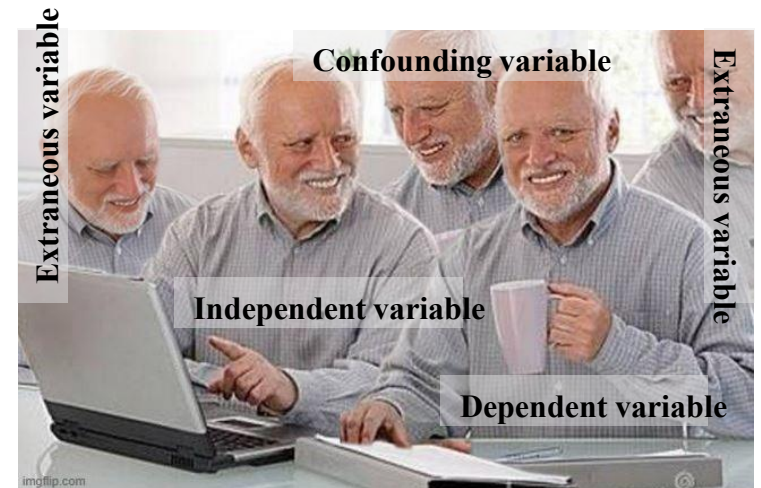


# Experimental Research: Cause and Effect Relationships (cont'd...)



**Experimental group:** independent variable (receives the experimental manipulation)

**Control group:** participants in an experiment who are not subjected to the independent variable and who may receive a placebo treatment



# Cause and Effect Relationships: Experimental Research (cont'd...)

| Section                                                                                                   | Details                                                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  Title                   | <ul style="list-style-type: none"> <li>The Impact of Social Media Use on Self-Esteem in Gen Z College Students</li> </ul>                                                                               |
|  Research Question       | <ul style="list-style-type: none"> <li>Does limiting social media usage improve self-esteem in Gen Z college students?</li> </ul>                                                                       |
|  Hypothesis              | <ul style="list-style-type: none"> <li>Limiting social media to 30 min/day for 2 weeks will increase self-esteem compared to usual usage</li> </ul>                                                     |
|  Independent Variable    | <ul style="list-style-type: none"> <li>Daily social media use: <ul style="list-style-type: none"> <li>- Experimental: <math>\leq 30</math> min/day</li> <li>- Control: usual use</li> </ul> </li> </ul> |
|  Dependent Variable      | <ul style="list-style-type: none"> <li>Self-esteem measured by Rosenberg Self-Esteem Scale before and after two weeks</li> </ul>                                                                        |
|  Groups                  | <ul style="list-style-type: none"> <li>Experimental: limit use (monitored)</li> <li>Control: no change</li> </ul>                                                                                       |
|  Confounding Variables | <ul style="list-style-type: none"> <li>Mental health, life events, personality, other digital activities</li> <li>Controlled via screening and design</li> </ul>                                        |
|  Extraneous Variables  | <ul style="list-style-type: none"> <li>Testing environment, time of day, mood, platform type</li> <li>Minimised by standardising conditions</li> </ul>                                                  |
|  Design Summary        | <ul style="list-style-type: none"> <li>Random assignment, pre/post-tests, compliance monitoring, 2-week duration</li> </ul>                                                                             |



**Independent Variable (IV) :**

Two potion ingredients: 1. Jobberknoll feather, 2. Bicorn horn

**Dependent Variable (DV):**

Transparent potion

**Control Variable (CV):**

Other ingredients

Hermione is trying to prepare a veritas serum. However, she is unable to recollect all the ingredients. She is mixed between Jobberknoll feather and Bicorn horn.

One way to figure out this is by trying to add each of the ingredients and check which one works.

Adding the correct ingredient will make the potion make it transparent.

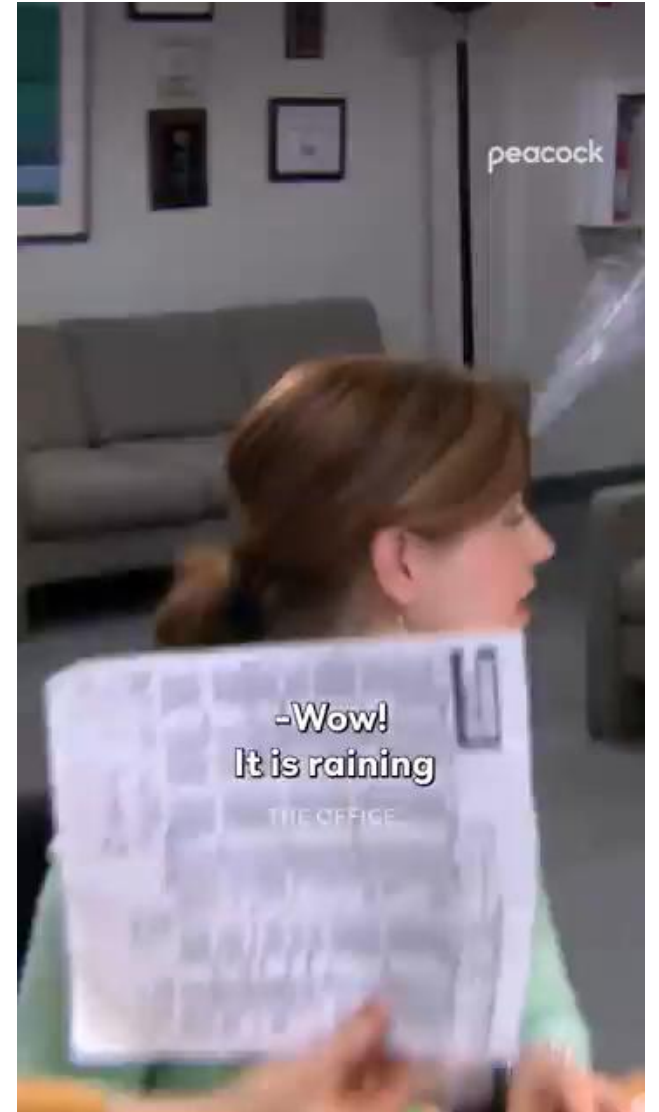


1. Go to [wooclap.com](https://www.wooclap.com)
2. Enter the event code in the top banner  
Event code  
[FYPMEX](#)

# Experimental Research: Drawbacks (cont'd...)

## **Experimenter effect:**

tendency of the experimenter's expectations for a study to unintentionally influence the results of the study



The Office

# Cause and Effect Relationships: Experimental Research (cont'd...)

## Single-anonymous study:

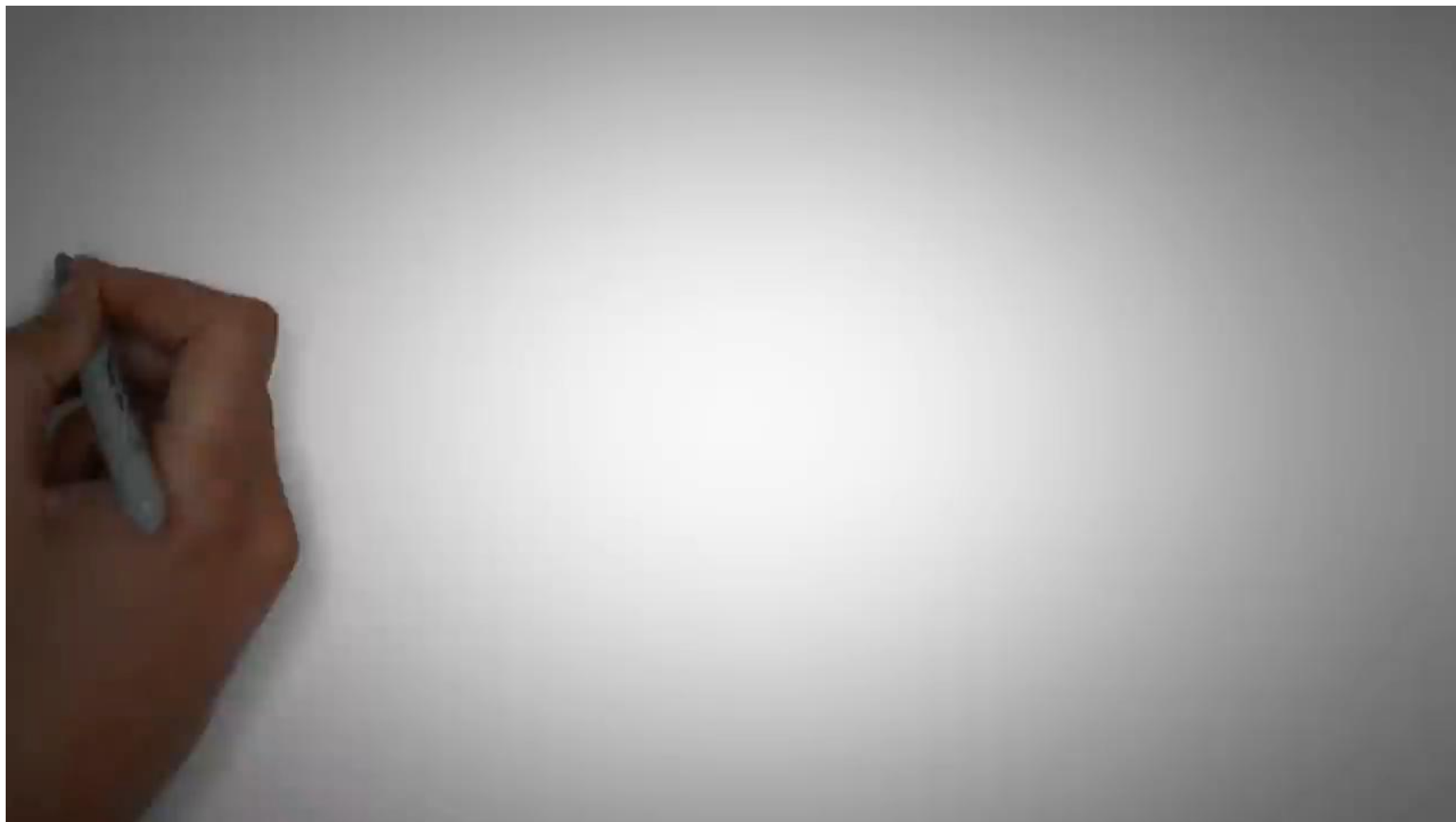
a study in which the participants do not know if they are in the experimental or the control group

## Double-anonymous study:

a study in which neither the experimenter nor the participants knows if the participants are in the experimental or control group



# **Ethical Issues in Psychological Research**



# Ethical Issues in Psychological Research

- **Rights and well-being:** weighed against the study's value to science
- **Informed consent** for participation
- **Debriefing:** Deception must be justified; reveal true nature of the study and expectations of results
- **Right to withdraw** from the study at any time
- Safeguard from **potential risks/emotional distress**
- **Confidentiality**



# Key Stuff to Lock In



- Hypothesis
- Clinical case studies
- Naturalistic observation
- Observer effect
- Participant observation
- Observer bias
- Surveys
- Representative sample
- Courtesy bias
- Wording Bias

- Question-order bias
- Cross-sectional research
- Longitudinal research
- Correlational research
- Experimental research
- Operational definition
- Variables
- Independent variables
- Dependent variables
- Confounding bias

- Confounding variables
- Extraneous variables
- Experimental group
- Control group
- Experimenter effect
- Single-anonymous study
- Double-anonymous study
- Informed consent
- Debriefing
- Confidentiality

# References

- Ciccarelli, S. K., & Meyer, G. E. (2006). *Psychology*. Pearson Education.
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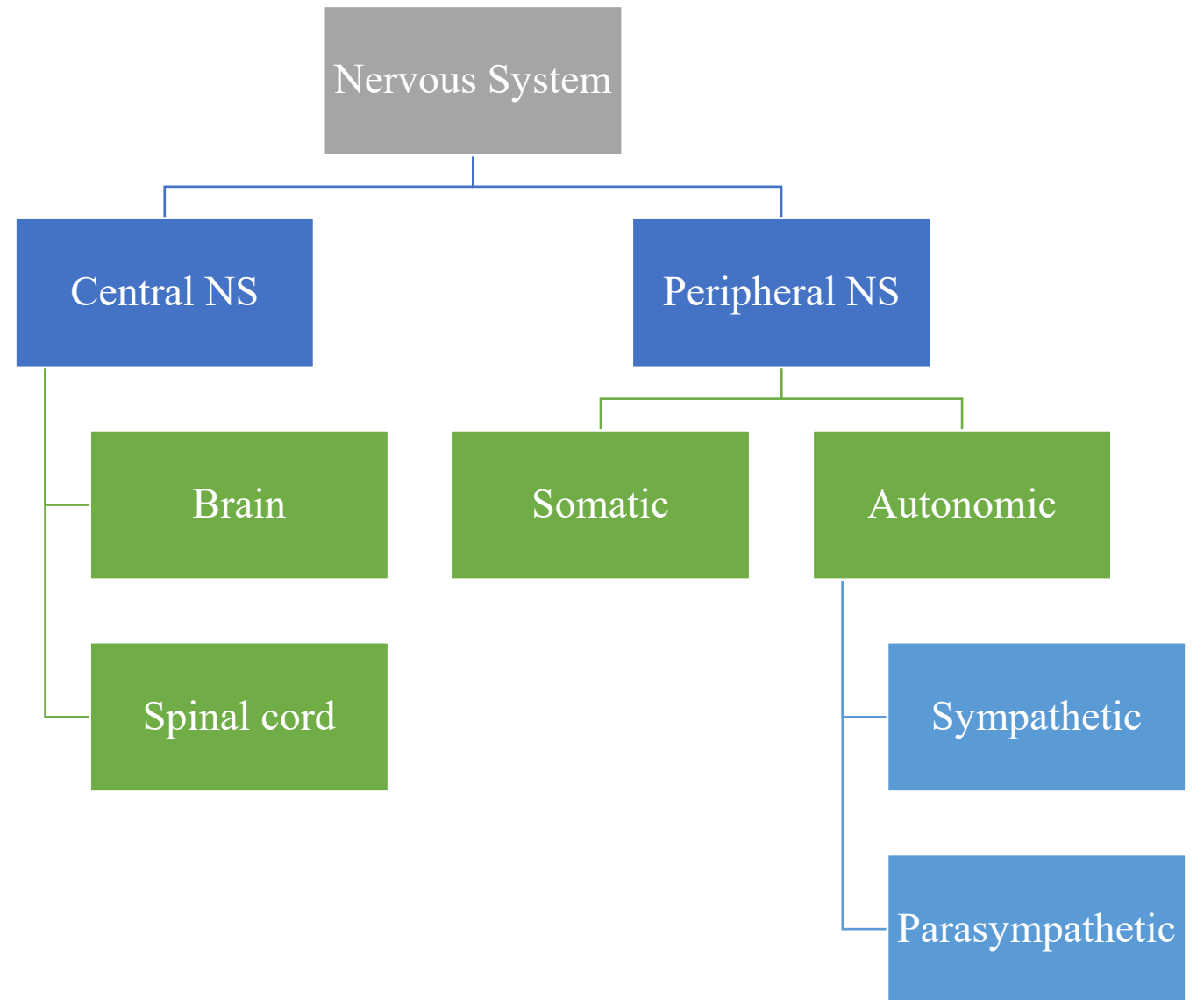
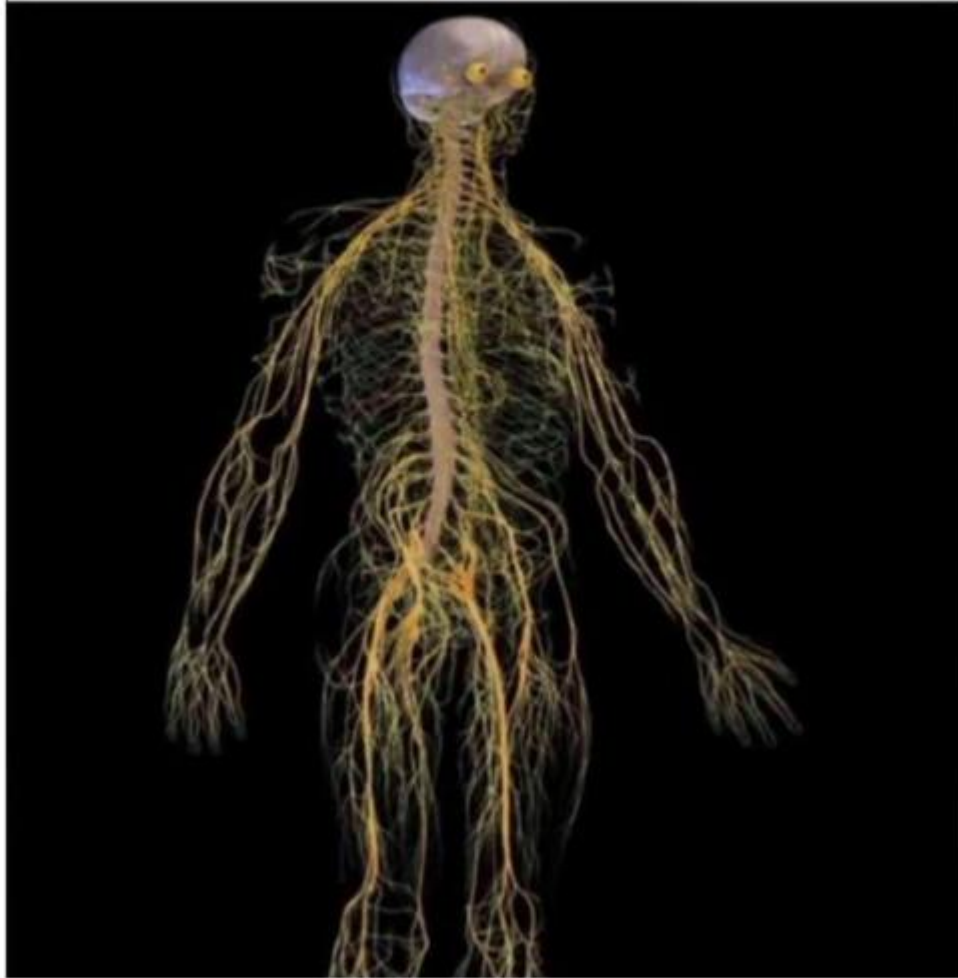


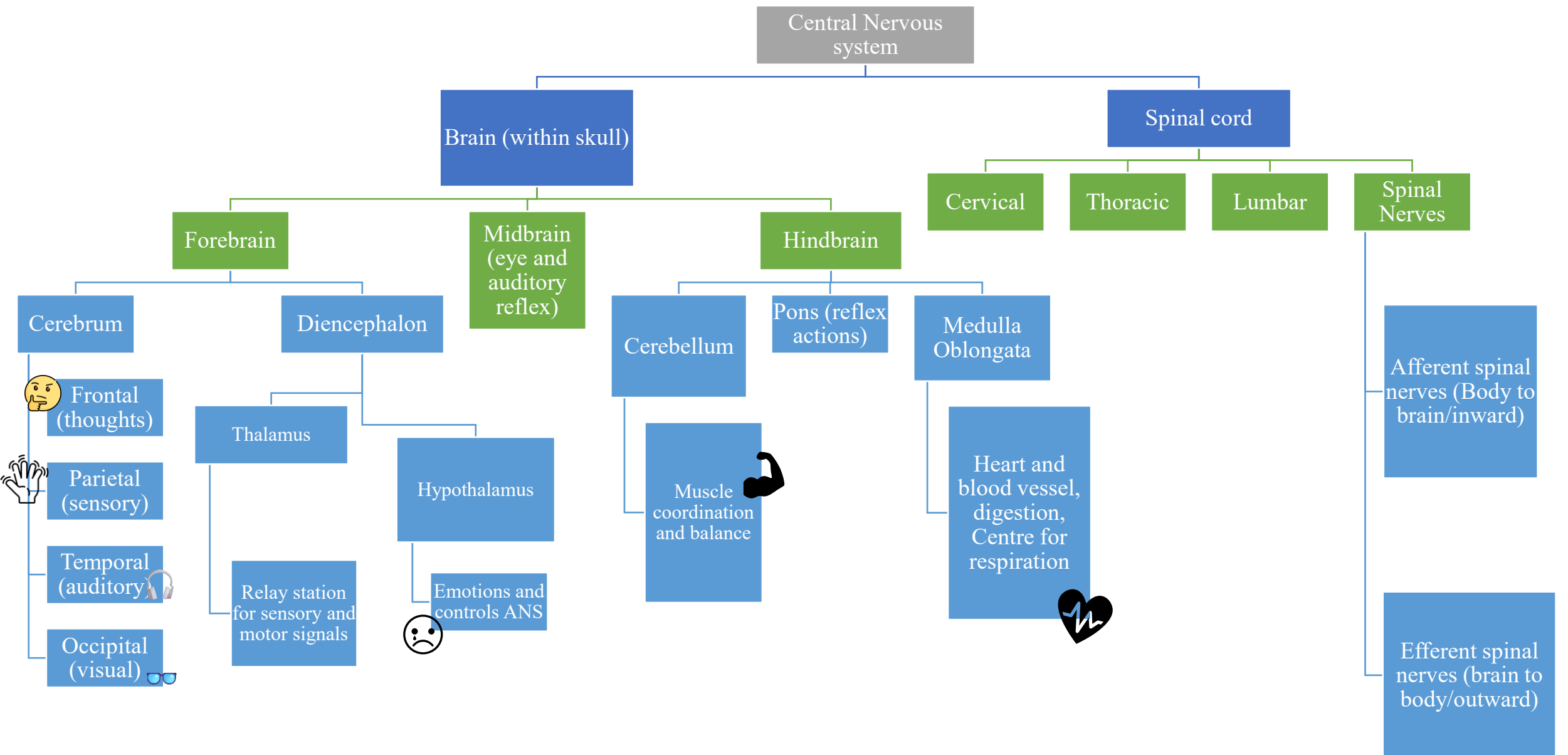
# Biological Bases of Behaviour

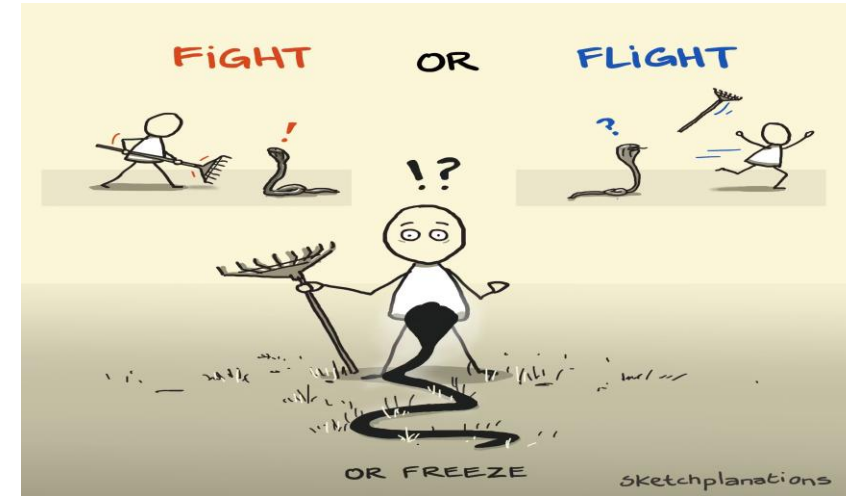
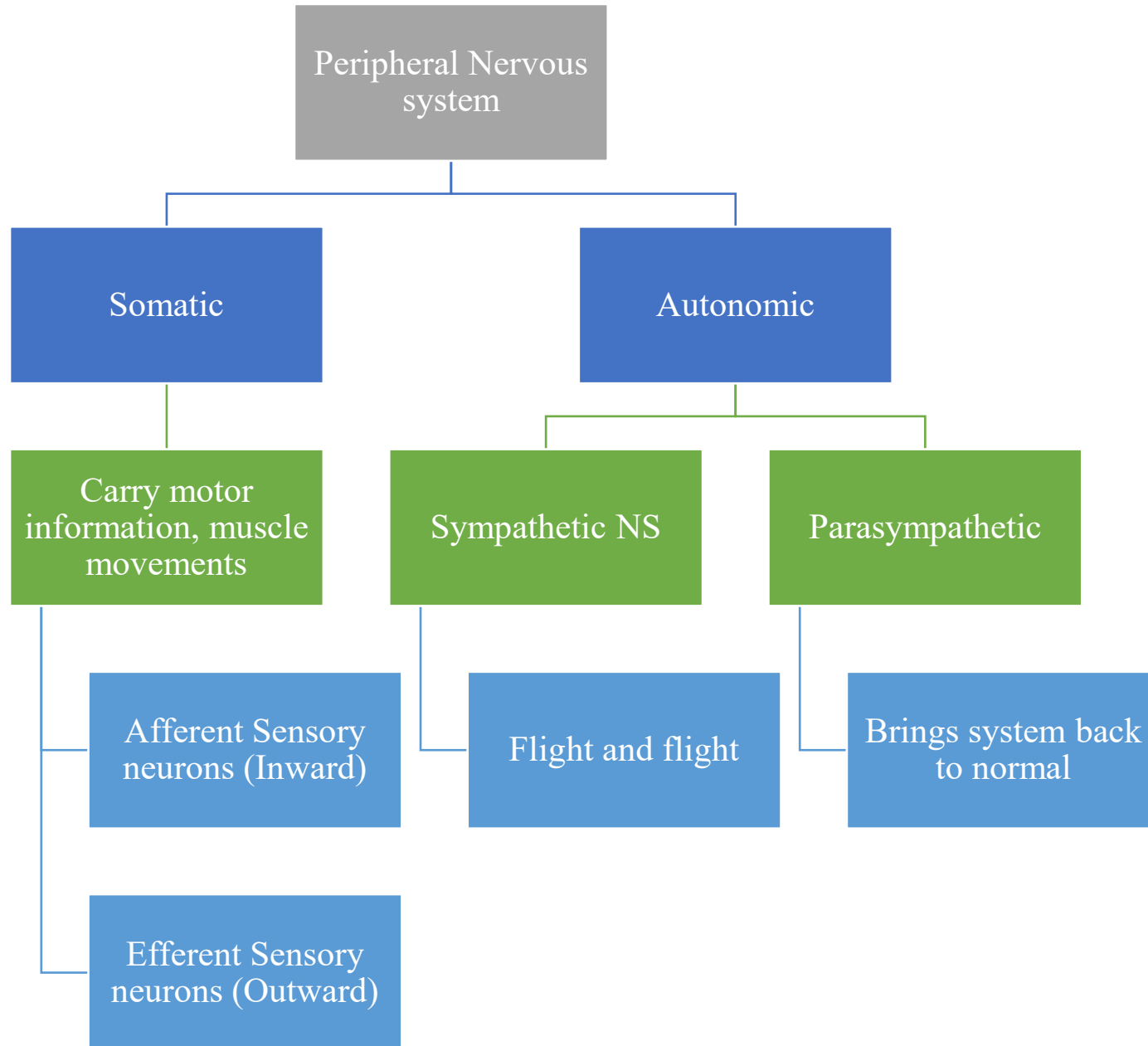
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Dr Shweta Chawak  
Assistant Professor  
Humanities and Social Sciences  
IIT Bombay



**"you look nervous"**  
**No way really**



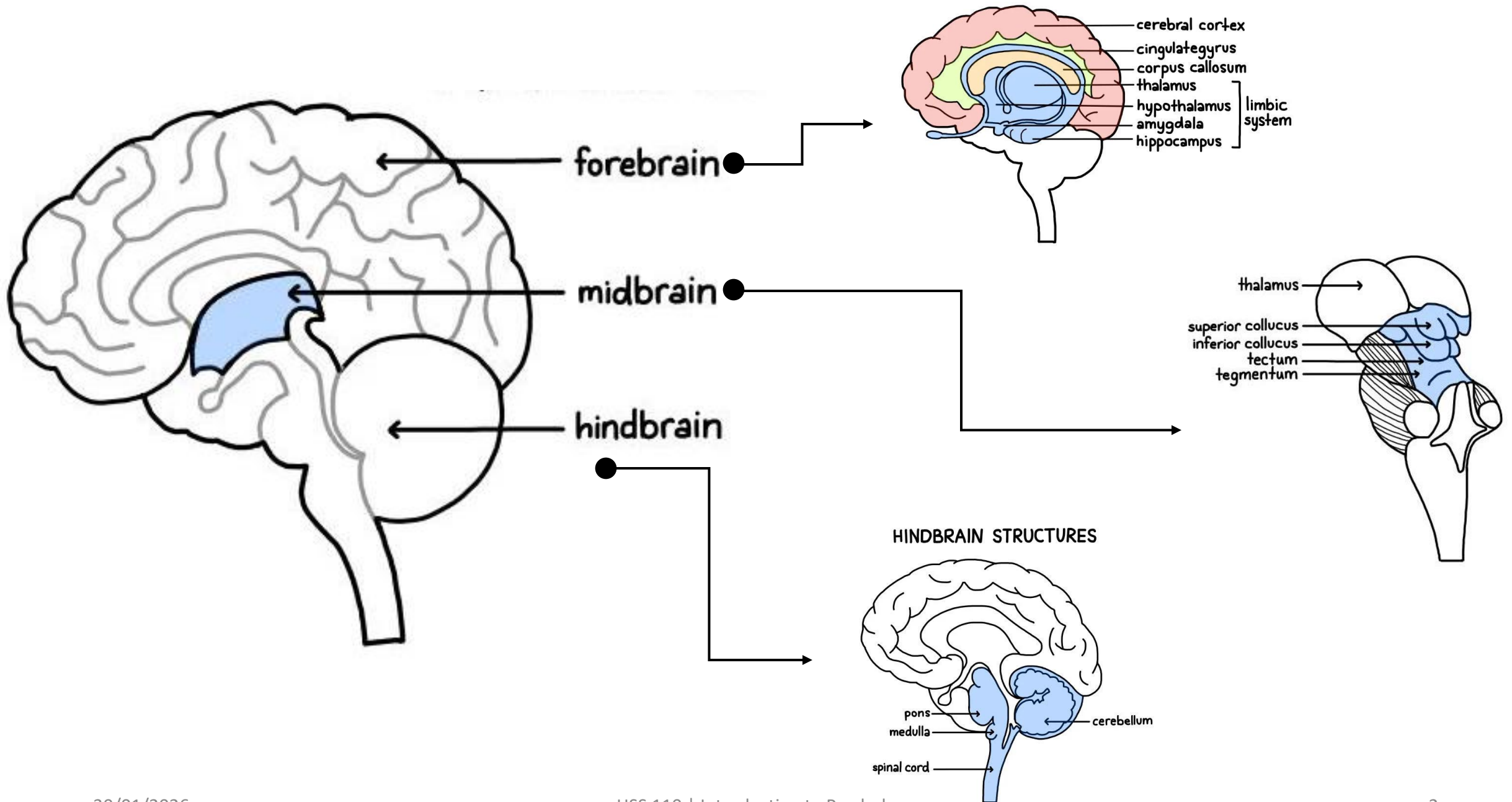






# Biological Bases of Behaviour

Course instructor:  
Dr Shweta Chawak  
Assistant Professor  
Humanities and Social Sciences  
IIT Bombay



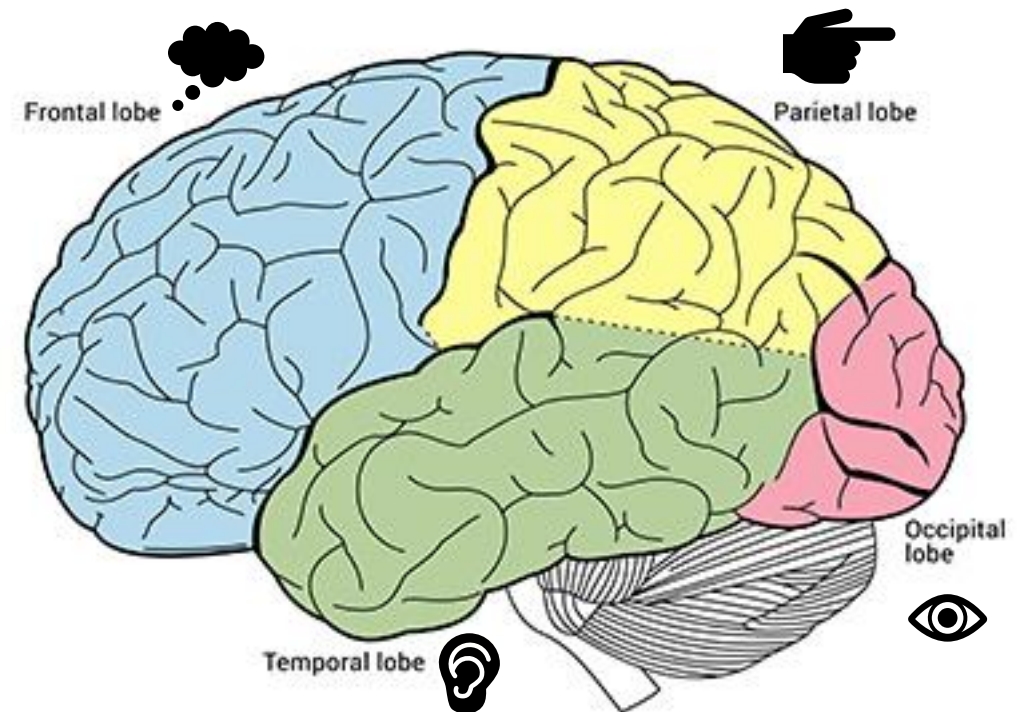
# Structure of the Brain: The Forebrain

## Cortex, the basal ganglia, and the limbic system

- Cortex: Outermost part of the brain
- Divided into two sections called the cerebral hemispheres: right & left
- Connected by corpus callosum
- Each hemisphere is divided into four lobes or sections

### Specialisation of the two hemispheres

| Left Hemisphere           | Right Hemisphere                  |
|---------------------------|-----------------------------------|
| Controls the right hand   | Controls the left hand            |
| Spoken language           | Nonverbal                         |
| Written language          | Visual-spatial perception         |
| Mathematical calculations | Music and artistic processing     |
| Logical thought processes | Emotional thought and recognition |
| Analysis of detail        | Processes the whole               |
| Reading                   | Pattern recognition               |
|                           | Facial recognition                |



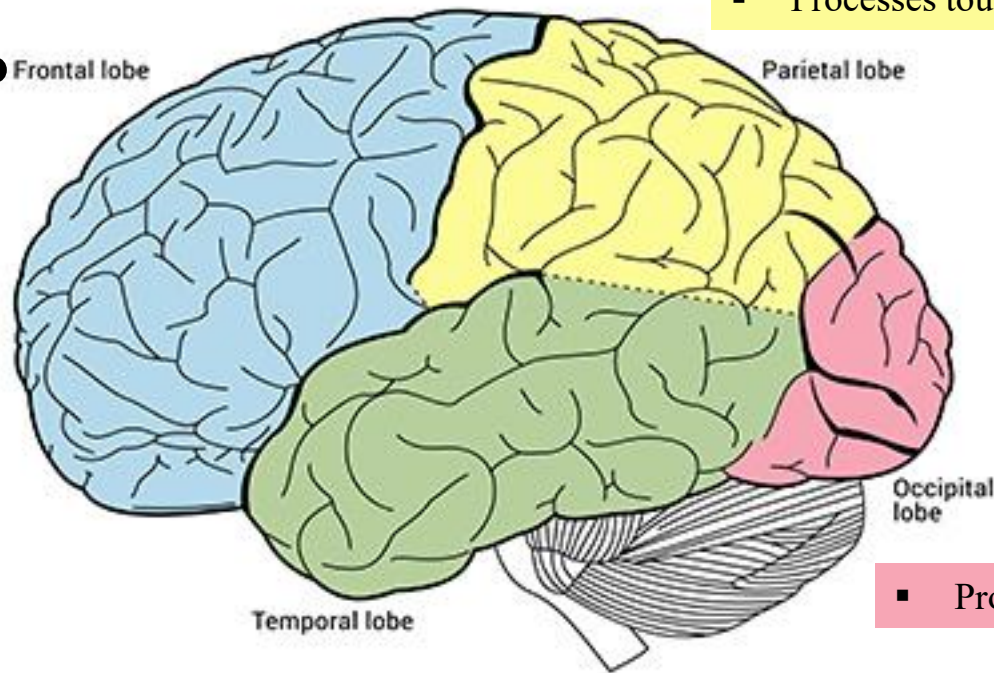
# Structure of the Brain: The Forebrain

Cortex, the basal ganglia, and the limbic system

- Processes higher mental functions
- E.g., planning, personality, memory storage, complex decision making, language
- Helps control emotions through the limbic system

Damage: motor tasks

E.g., *perseveration*: repeating the same step, wrong answer, or movement over and over



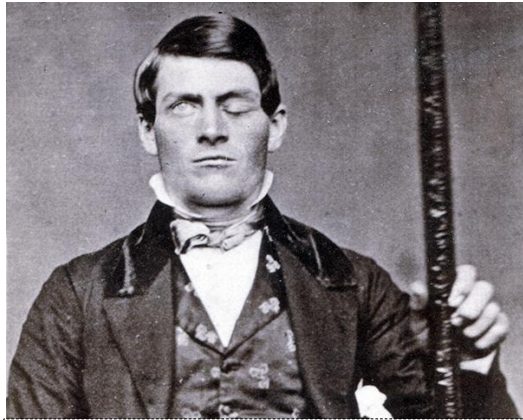
- Processes touch, temperature, and body position

- Processes visual information from the eyes

- Processes hearing and meaningful speech

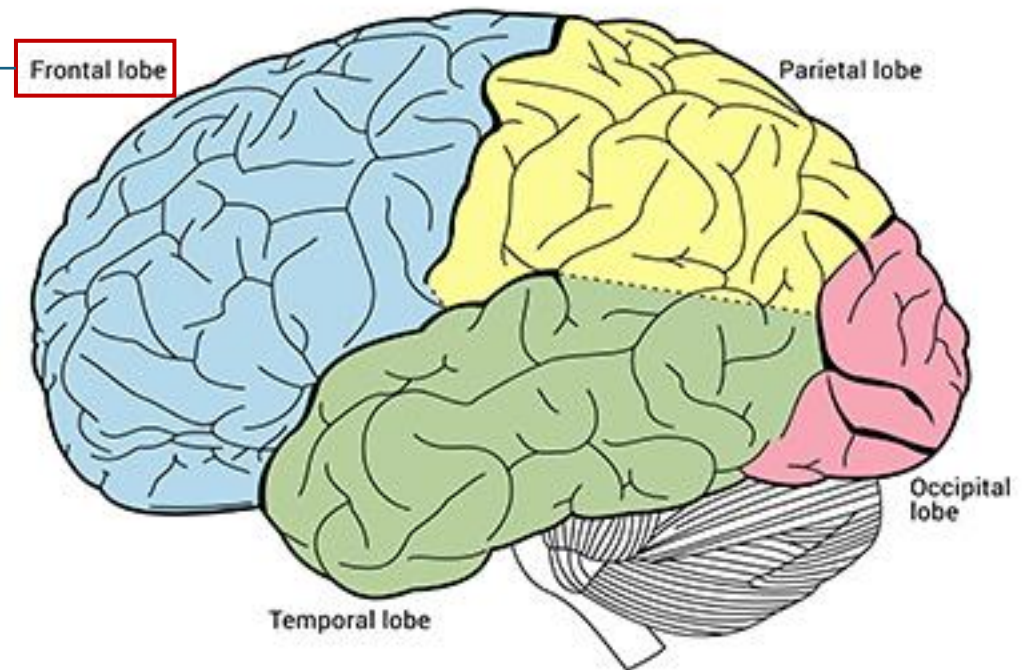
# Structure of the Brain: The Forebrain

Cortex, the basal ganglia, and the limbic system



- Changes after injury
- Loss of emotional control
- Change in personality
- Impulsive
- Poor judgment and decision making

Phineas Gage



# Structure of the Brain: The Forebrain

## Cortex, the basal ganglia, and the limbic system

- Left frontal lobe: Production of speech
- Broca's area: articulation and fluency of speech
- **Broca's aphasia**: expressive language disorder
- Affects fluency but not comprehension
- E.g., "Want water now" instead of "I want a glass of water"

"Dog dog park" instead of "There are two dogs at the park"



**B**roca = **B**roken speech

# Structure of the Brain: The Forebrain

Cortex, the basal ganglia, and the limbic system

- Left temporal lobe
- Wernicke's area: comprehension of words
- **Wernicke's aphasia**: impaired language comprehension
- Fluent speech that does not make sense
- E.g., you mean to say “chair,” but you say “table”

you mean to say “cat,” but you say “hat”



**Tactus**Therapy

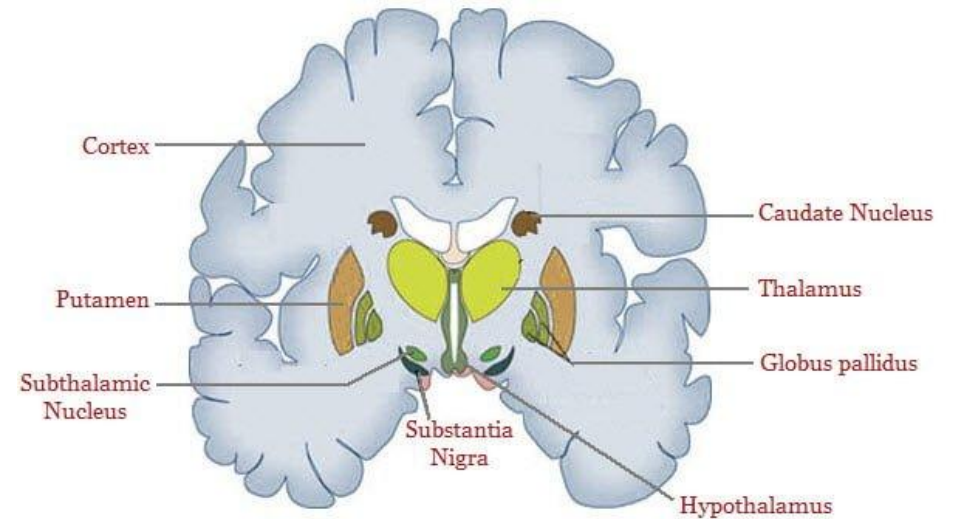


Wernicke = Word salad

# Structure of the Brain: The Forebrain

## Cortex, the basal ganglia, and the limbic system

- Group of deep brain structures called basal ganglia (or basal nuclei)
- Spread across the brain
- Regulate **voluntary motor movements** by fine-tuning and coordinating actions
- **Involved in decision-making** (evaluates risks and goals), emotions, motivation, learning habits, rewards
- Dysfunction linked to disorders like **addiction, Alzheimer's, Parkinson's, Huntington's** (Decay of brain nerve cells over time), and **Tourette syndrome**



# Structure of the Brain: The Forebrain

## Cortex, the basal ganglia, and the limbic system

### Thalamus

- Relay station
- Hearing, sight, touch, or taste
- Smell is the only sense that does not have to first pass through the thalamus
- Associated with dyslexia

### Hypothalamus

- Body temperature, thirst, hunger, sleeping and waking, sexual activity, and emotions
- Controls the pituitary gland, so hormone regulation ultimately depends on it
- Hormones influences: Depression

### Cingulate cortex

- Active during a variety of cognitive tasks such as selective attention, written word recognition, and working memory
- Associated with attention-deficit/hyperactivity disorder, major depressive disorder, and bipolar disorder

### Hippocampus

- ACh, the neurotransmitter involved in muscle control, is also involved in the memory function
- People who have Alzheimer's have much lower levels of ACh

### Amygdala

- Fear responses and memory of fear
- Sensory information reaches the amygdala first, enabling quick responses to danger before conscious awareness
- Damage: decreased fear response

# Structure of the Brain: The Forebrain

Cortex, the basal ganglia, and the limbic system

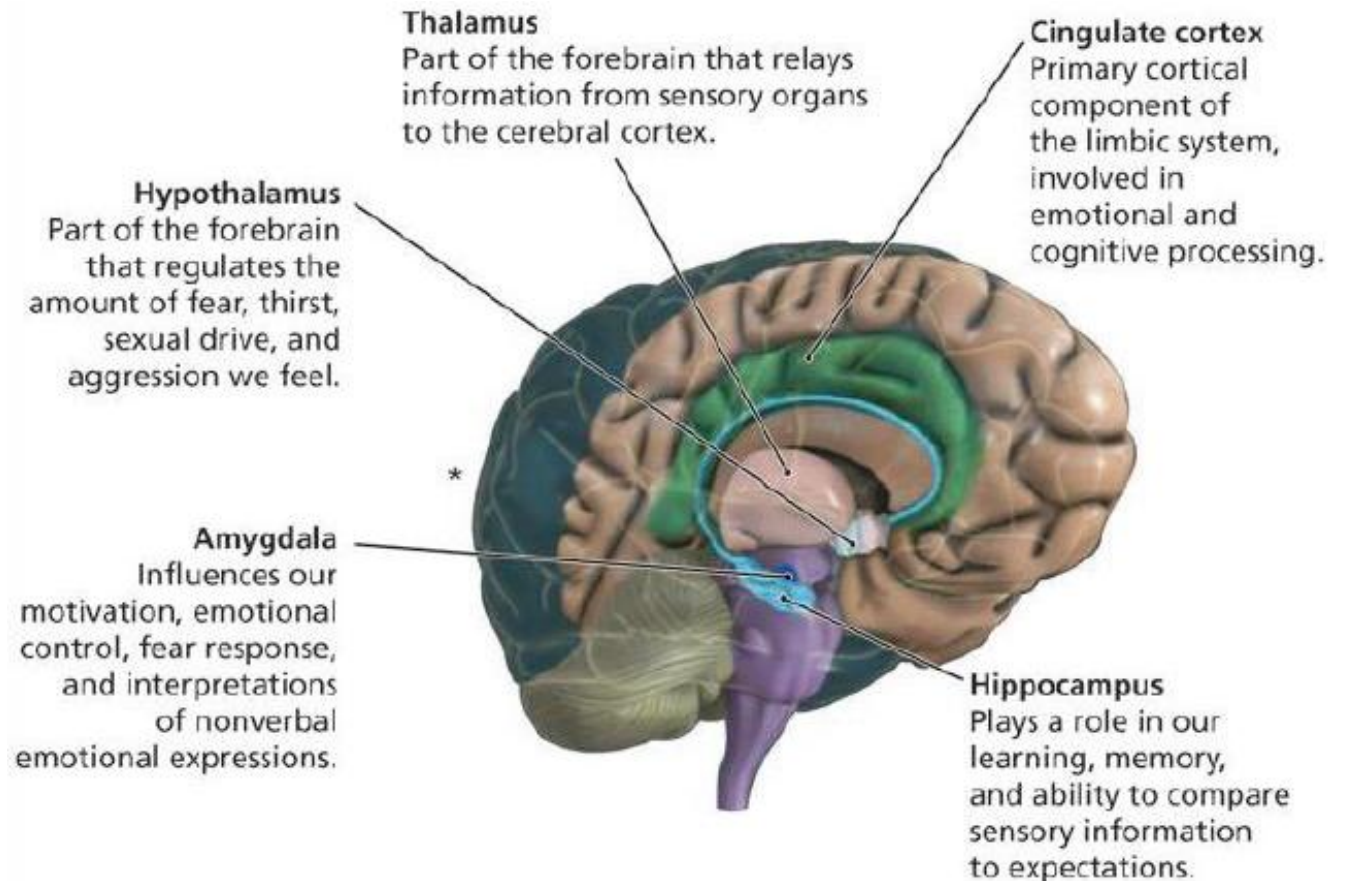
Emotions, motivation, memory, and learning

- **Klüver-Bucy syndrome**

**(Neuropsychiatry disorder):**

**temporal lobe and amygdala**

- Memory loss and behavioral issues
- Intense sex drive, compulsive eating, eating inedible objects, dementia etc.



# RECAP

- Cortex
  - Frontal lobe: Broca Aphasia (difficulty in using expressive language)
  - Temporal lobe: Wernicke's aphasia (word salad/ difficult in word comprehension)
- Basal Ganglia (decision making and coordination)
  - Addiction
  - Alzheimer's
  - Parkinson's
  - Huntington's (Decay of brain nerve cells over time)
  - Tourette syndrome
- Limbic System
  - Thalamus (relay station for sensory except smell): Dyslexia
  - Hypothalamus (controls emotions): Hormones issues/ Depression
  - Cingulate cortex: Dyslexia (selective attention)
  - Hippocampus: Alzheimer (memory)
  - Amygdala: Klüver-Bucy syndrome



# Biological Bases of Behaviour

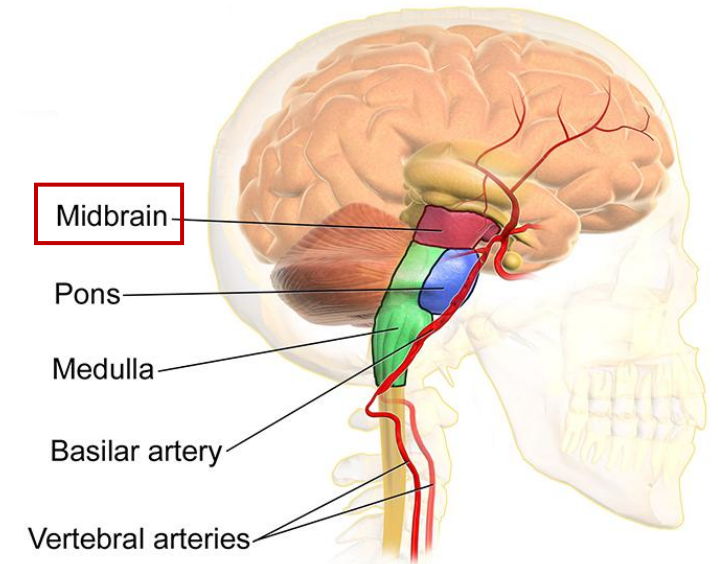
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Dr Shweta Chawak  
Assistant Professor  
Humanities and Social Sciences  
IIT Bombay

# RECAP

- Cortex
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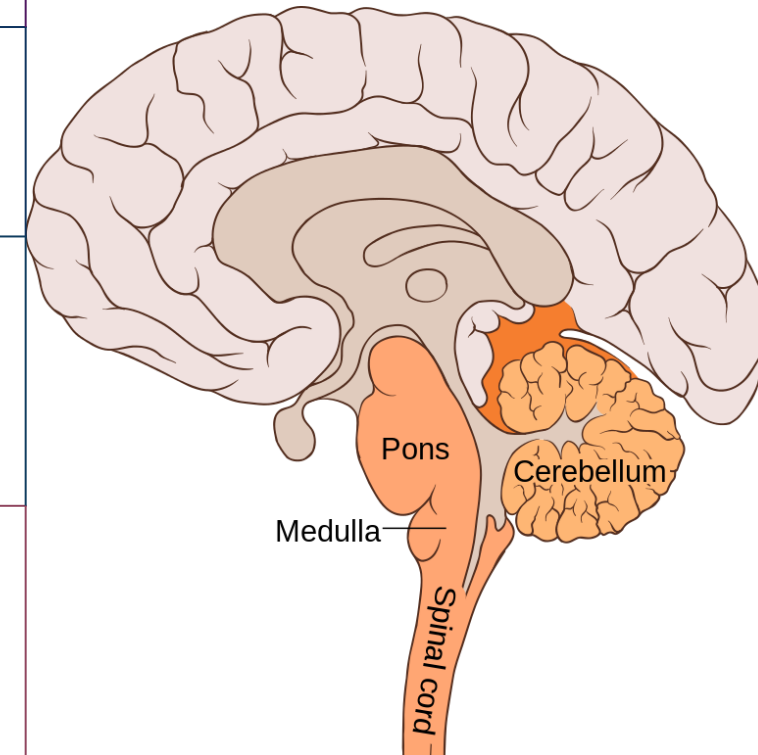
# Structure of the Brain: Midbrain

- Situated below Thalamus
- Processes visual and auditory signals
- Motor movement and reflexes
- Regulates alertness, wakefulness, sleep-wake cycles
- Pain signals
- Damage: Parkinson's disease, issues with sleep



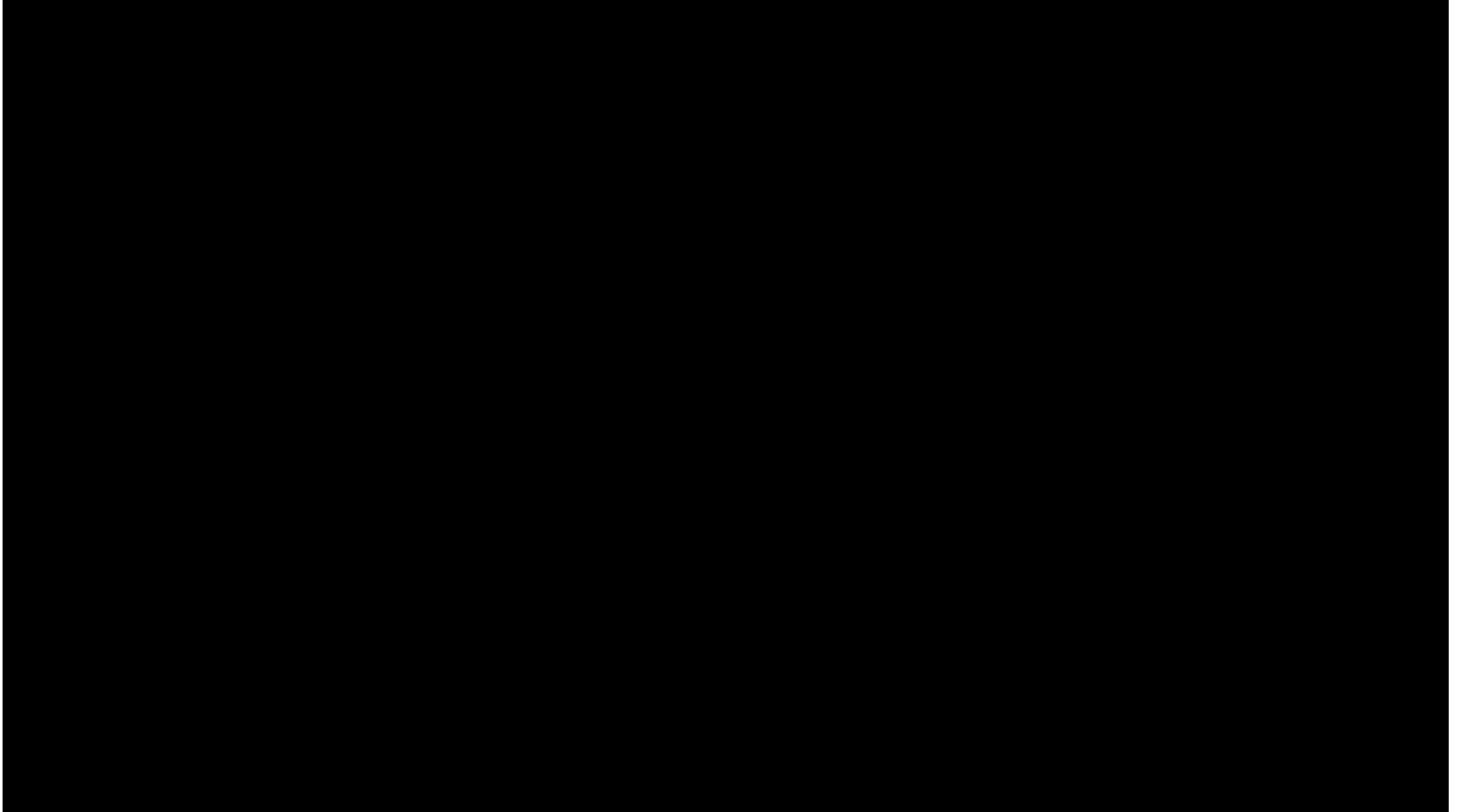
# Structure of the Brain: Hindbrain

| Hindbrain Structure | Location & Role                                                                                                                                                  | Consequences of Malfunction                                                                                                                                                                                                                                             | Mnemonic                                     |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| Medulla             | <ul style="list-style-type: none"> <li>Top of spinal column</li> <li>Heartbeat, breathing, swallowing</li> <li>Sensory nerves crossover</li> </ul>               | <ul style="list-style-type: none"> <li>Damage can be fatal; disrupts vital functions like heartbeat, breathing, swallowing</li> </ul>                                                                                                                                   | Medulla = Must survive (life functions)      |
| Pons                | <ul style="list-style-type: none"> <li>Above medulla</li> <li>Sleep, dreaming, arousal</li> <li>Motor nerve crossover (left-right body movements)</li> </ul>     | <ul style="list-style-type: none"> <li>Impaired coordination of movements</li> </ul>                                                                                                                                                                                    | Pons = Point to coordination (bridge, motor) |
| Cerebellum          | <ul style="list-style-type: none"> <li>Behind pons</li> <li>Controls (in)voluntary, rapid fine motor movement, balance, posture, learned motor skills</li> </ul> | <ul style="list-style-type: none"> <li>Tremors, unsteady walk, slurred speech, dizziness, muscle weakness</li> <li>(e.g., spinocerebellar degeneration)</li> <li>Symptoms similar to alcoholic intoxication</li> <li>Schizophrenia, autism spectrum disorder</li> </ul> | Cerebellum = Coordination centre             |



 Cerebellum aka little brain

**NEUROPLASTICITY: Brains ability to reorganise its structures, function, and neural connections**



## **Neuroplasticity**

Brains ability to change its structure and function in response to experience , learning, environment, and injury

### **Why it Matters?**

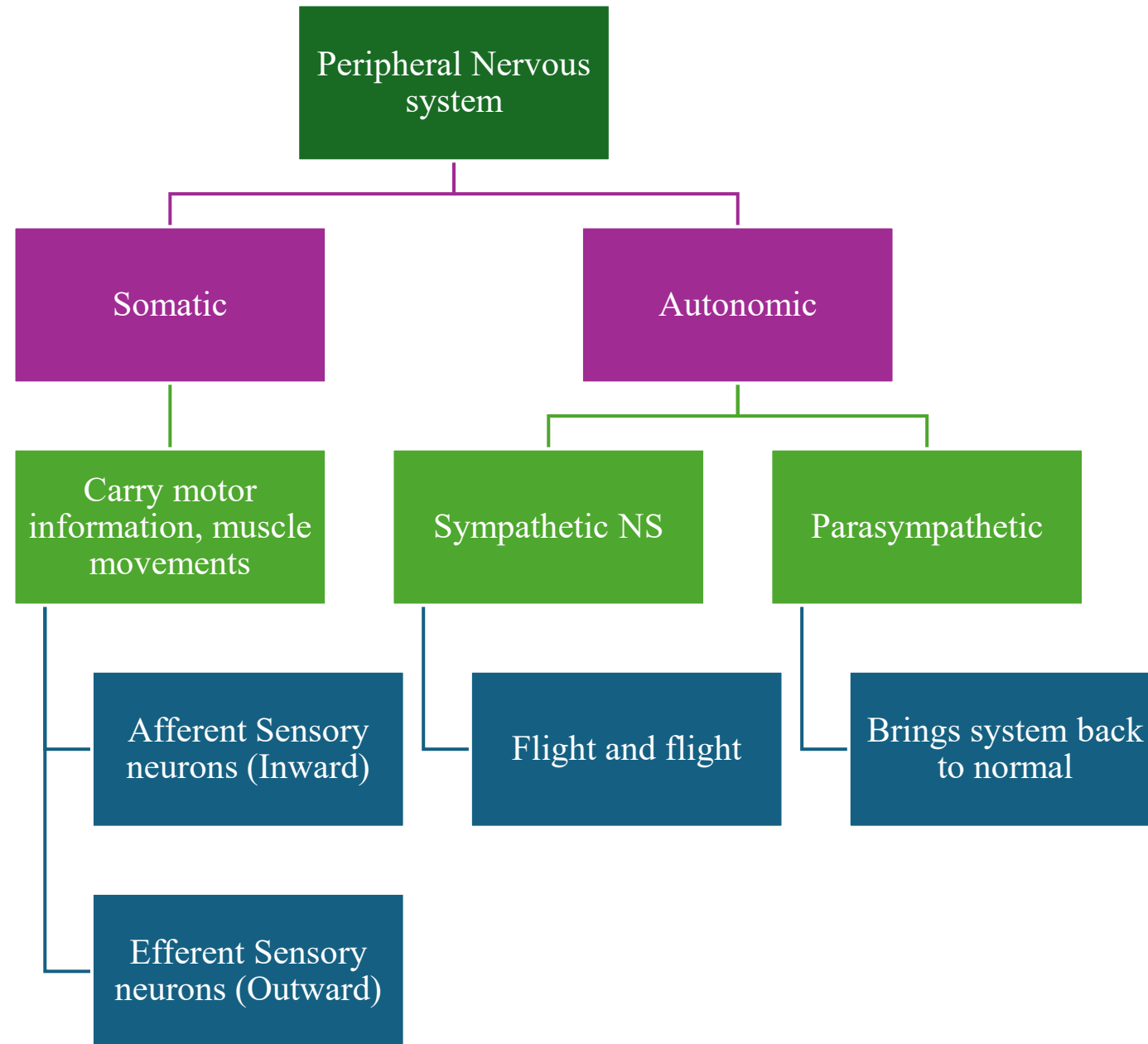
- Basic learning and memory
- Recovery from injury
- Mechanism behind rehabilitation and behaviour change

### **Types of neuroplasticity**

- Experience dependent plasticity (practice and exposure)
- Developmental plasticity: heightened plasticity during critical periods in childhood
- Injury-induced plasticity: Reorganisation following brain injury

### **Clinical and applied relevance**

- Rehabilitation: Stroke, traumatic brain injury, recovery rely heavily on neuroplastic mechanisms
- Education and skill learning: Teaching strategies leverage plasticity through spaced practice and multimodal learning



# The Autonomic Nervous System

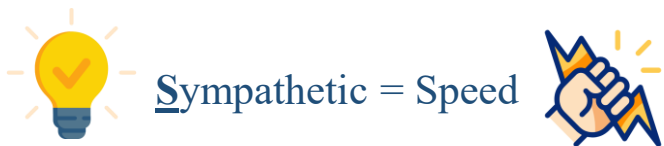
## Sympathetic division

- Most active when you are angry, afraid, or aroused
- **Fight-or-flight response**
- Increases heart rate and breathing
- Stops digestion

| Sympathetic division (arousing) |                | Parasympathetic division (calming)    |
|---------------------------------|----------------|---------------------------------------|
| Pupils dilate                   | EYES           | Pupils contract                       |
| Decreases                       | SALIVATION     | Increases                             |
| Perspires                       | SKIN           | Dries                                 |
| Increases                       | RESPIRATION    | Decreases                             |
| Accelerates                     | HEART          | Slows                                 |
| Inhibits                        | DIGESTION      | Activates                             |
| Secrete stress hormones         | ADRENAL GLANDS | Decrease secretion of stress hormones |

## Parasympathetic division

- **Calms body**
- Produces effects opposite to those of the sympathetic division
- Reduces heart rate and breathing
- Restores digestion





# The Somatic Nervous System

- Somatic Nervous System: voluntary nervous system
- Consists of neurons that communicate between the body and the brain
- **Afferent (sensory) neurons [inward]**
  - Neurons that carry messages from sense organs to spinal cord
- **Efferent (motor) neurons [outward]**
  - Neurons that carry messages from the spinal cord or brain to muscles and glands
- Also controls reflex which is an involuntary muscle response
  - **Interneurons** connect the afferent neurons to the efferent neurons



Afferent neurons access the spinal cord  
Efferent neurons exit

- Disease affecting motor neurons are motor neuron disease: progressive muscle weakness and waste



Knee-jerk<sup>10</sup> reflex

# Key Stuff to Lock In



- Brain
- Consciousness
- Nervous system
- Central Nervous system
- Brain
- Spinal cord
- Cranium or skull
- Cerebrum
- Cerebellum
- Diencephalon
- Midbrain
- Pons
- Medulla Oblongata

- Frontal
- Parietal
- Temporal
- Occipital
- Thalamus
- Hypothalamus
- Hippocampus
- Amygdala
- Afferent spinal nerves
- Efferent spinal nerves
- Peripheral Nervous system
- Somatic
- Autonomic

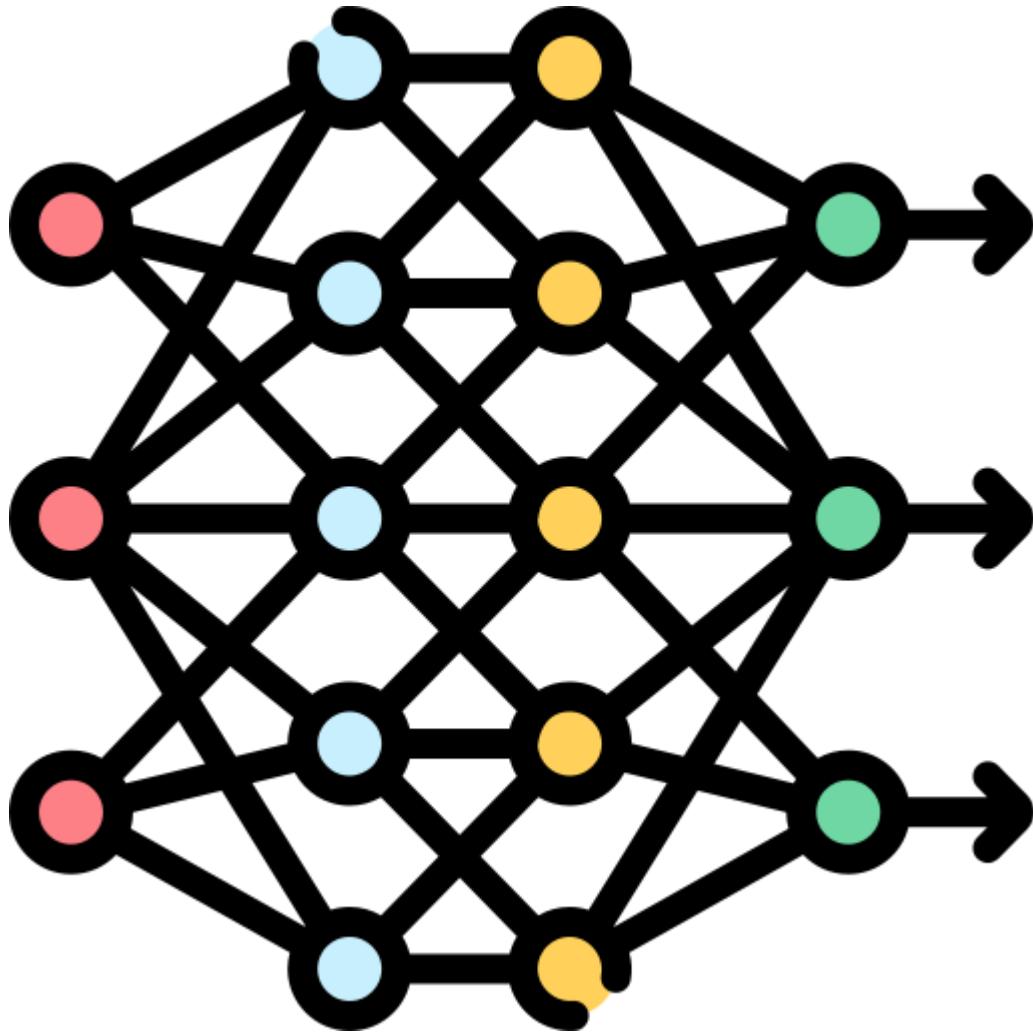
- Sympathetic
- Parasympathetic
- Broca Aphasia
- Wernicke's aphasia
- Addiction
- Alzheimer's
- Parkinson's
- Huntington's
- Tourette syndrome
- Dyslexia
- Cingulate cortex
- Klüver-Bucy syndrome
- Neuroplasticity

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# LEARNING

Course instructor:  
Dr Shweta Chawak  
Assistant Professor  
Humanities and Social Sciences  
IIT Bombay

**Write down one thing you know how to do very well (not academic). How did you learn it?**

**Discuss:**

- Learned by association (one event triggers a response to another)
- Learned by consequence (to seek reward or avoid punishment)
- Learned by watching others (Imitating or observing others)

# What is Learning?

- Learning is any relatively permanent change in behavior brought about by experience or practice
- Physical changes occurs in the brain
- Temporary changes or changes due to instinct or maturation are not learning
- Experience is the key feature of learning
- Behavior changes without practice or experience do not qualify as learning



Examples not considered learning:

- X Maturation changes (e.g., crawling, standing, walking at certain ages)
- X Illness or drug-induced temporary changes (e.g., lack of coordination, aggressive behaviour)

# Why Do We Need to Study about Learning?

- Learning builds essential skills and knowledge for personal and professional growth
- It helps individuals adapt to change and solve problems effectively
- Learning promotes critical thinking and informed decision-making





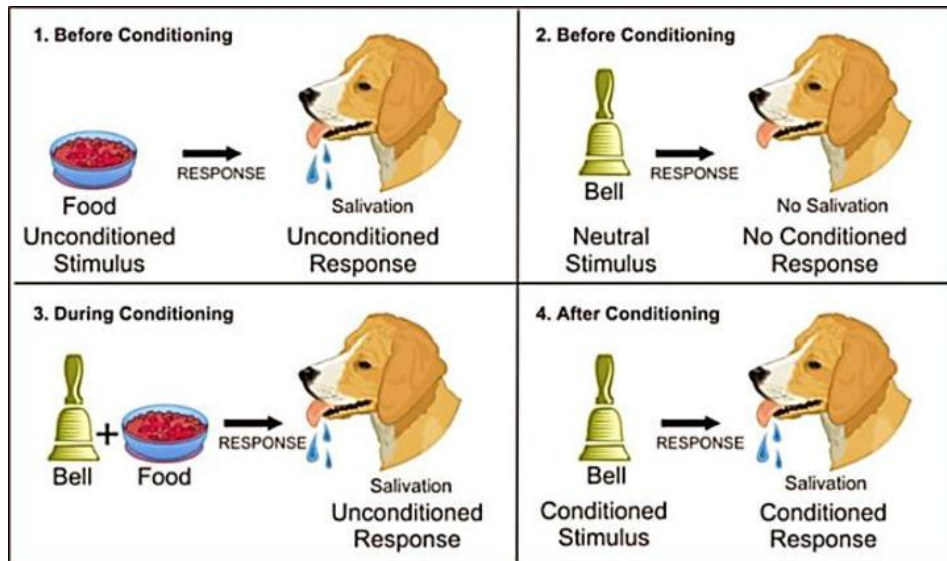
08/02/2026

HS 110 | Introduction to Psychology

# Theories of Learning: Classical Conditioning

- Ivan Pavlov
- **Reflex:** an involuntary response, one that is not under personal control or choice
- **Stimulus:** any object, event, or experience that causes a *response*, the reaction of an organism
- **Classical conditioning:** learning to make an **involuntary response** to a stimulus other than the original, natural stimulus that normally produces the response

# Theories of Learning: Classical Conditioning (cont'd...)



| Term                                                                                                             | Definition                                                                                      | Example                                    |
|------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|--------------------------------------------|
| Unconditioned Stimulus (UCS)  | A naturally occurring stimulus that leads to an involuntary and unlearned response              | Food causing salivation in Pavlov's dogs   |
| Unconditioned Response (UCR)  | Involuntary and unlearned response to a naturally occurring or unconditioned stimulus           | Salivation in response to food             |
| Neutral Stimulus              | A stimulus that has no effect on the desired response prior to conditioning                     | No response/no salivation                  |
| Conditioned Stimulus (CS)   | A previously neutral stimulus that, after association with the UCS, triggers a learned response | Bell causing salivation after conditioning |
| Conditioned Response (CR)   | A learned response to a conditioned stimulus                                                    | Salivation in response to bell alone       |

# Theories of Learning: Classical Conditioning (cont'd...)

## Some principles to consider:

- 1) The conditioned stimulus (CS) must be presented before the unconditioned stimulus (UCS) for conditioning to occur
- 2) The CS and UCS must be presented close together in time, ideally within 5 seconds
- 3) The neutral stimulus must be paired repeatedly with the UCS to create conditioning
- 4) The CS is usually a distinctive stimulus that stands out from other stimuli



# Theories of Learning: Classical Conditioning (cont'd...)



## Stimulus Generalisation

Responding to a stimulus similar to the original conditioned stimulus

- Different but similar stimulus evoke the same response

## Stimulus Discrimination

Learning to respond only to the specific conditioned stimulus, not similar ones

- Different but similar stimulus evoke the same response

# Theories of Learning: Classical Conditioning (cont'd...)

## Extinction

The weakening or disappearance of a learned response when the unconditioned stimulus or reinforcer is removed

**Example:** A teen's excitement to check their phone fades if the notification sound rings too often without any new messages

## Spontaneous Recovery

The reappearance of a learned response after extinction has occurred

**Example:** After ignoring phone notifications for a while, a teen suddenly reacts again with excitement to the notification sound





They taught him John Cena song means dinner time...😭



@doppammine

# Classical Conditioning: Application

## Student Life & Campus

- Notification sound -- anxiety/excitement
- Class bell-- alertness / disengagement

## Technology & Digital Behaviour

- App notification: urge to check phone
- Gaming sounds: excitement/arousal
- *Designers use sound, colour, and timing to create learned responses*

## Advertising & Consumer Behaviour

- Brand + celebrity / music positive emotion
- Food ads: hunger even when not hungry
- Festival visuals: *Emotion is conditioned to the product.*

## Health & Emotions

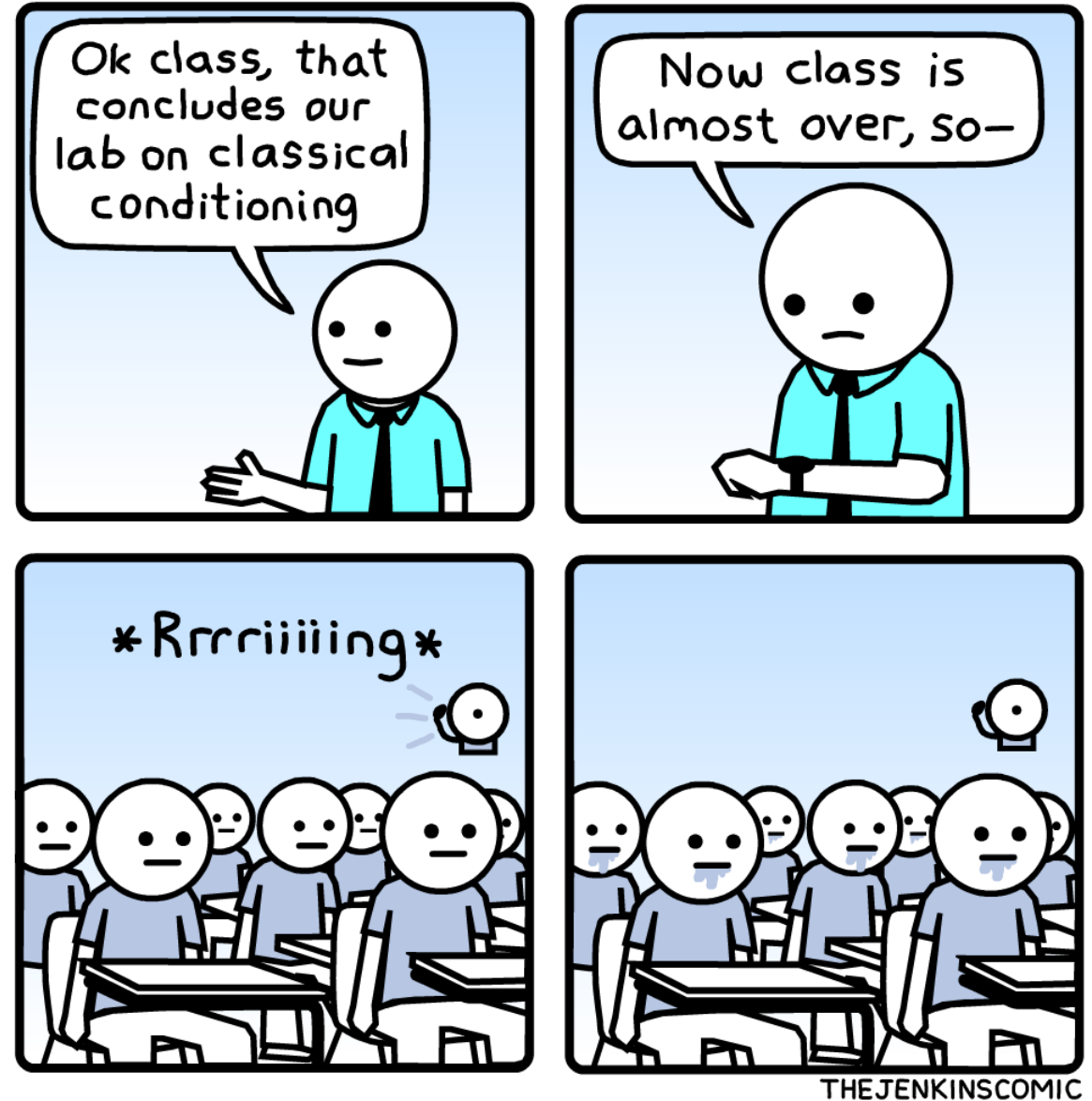
- Hospital smell: fear/anxiety
- White coat: nervousness

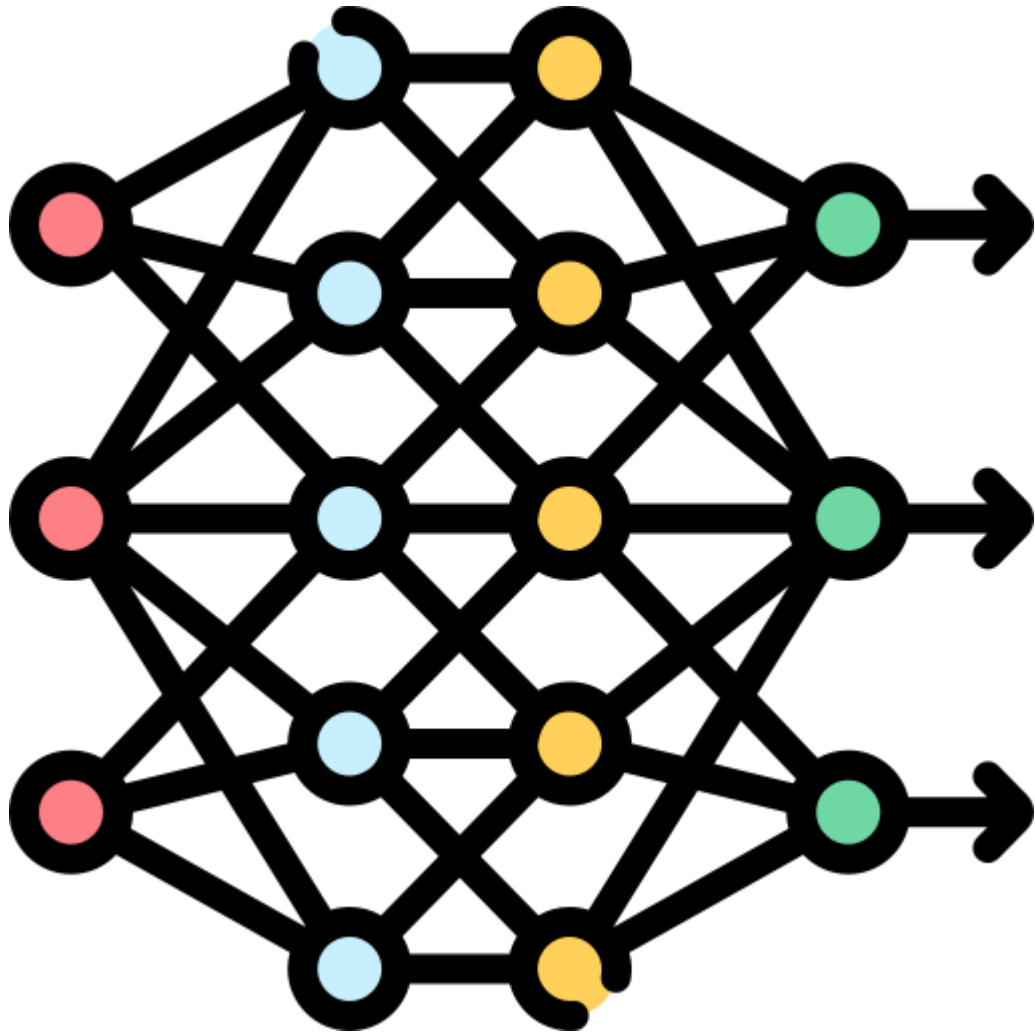


If you find yourself cringing at the mere sight of a hypodermic needle, your cringing is a CR to the CS of the needle. The pain of the shot would be the original UCS.

# RECAP

- Learned by
  - association (one event triggers a response to another)
  - consequence (to seek reward or avoid punishment)
  - watching others (Imitating or observing others)
- Classical conditioning
  - Unconditioned stimulus
  - Unconditioned Response
  - Neutral Stimulus
  - Conditioned stimulus
  - Conditioned response
- Stimulus Generalisation vs discrimination
- Stimulus Extinction vs spontaneous recovery
- Application of Classical conditioning in everyday life



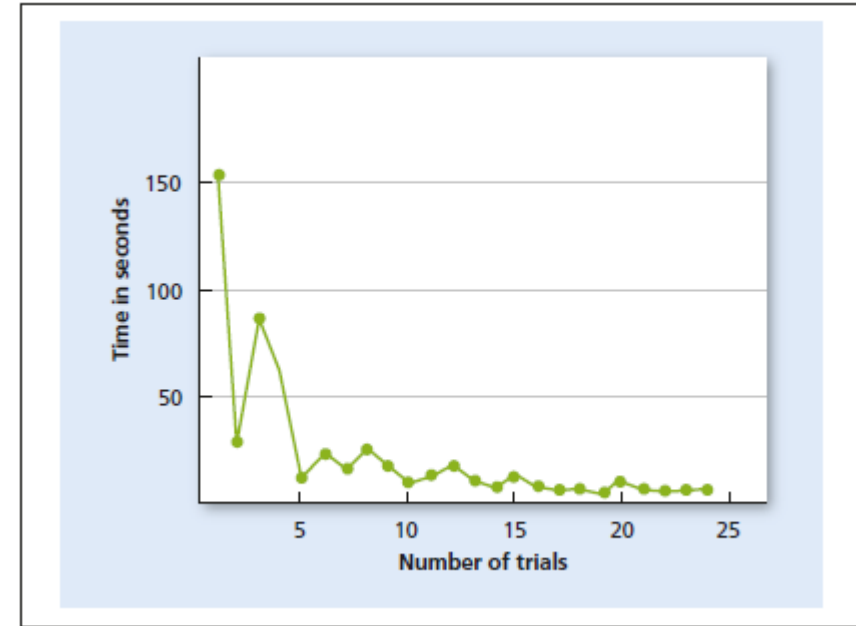


# LEARNING

Course instructor:  
Dr Shweta Chawak  
Assistant Professor  
Humanities and Social Sciences  
IIT Bombay

# Theories of Learning: Operant Conditioning

- Classical conditioning - learning of involuntary, automatic responses
- Operant conditioning - organisms learning of voluntary responses
- Edward L. Thorndike and B. F. Skinner
- Experiment of the 'puzzle box' by Thorndike
  - Trial-and-error learning



- Hungry cat placed in puzzle box – press lever to escape
- Trial and error
- Cat pushes the lever and opens the door
- Lever (stimulus), pushing the lever (response), consequence-escaping (good) and food (better)



# Theories of Learning: Operant Conditioning (cont'd...)

## Law of effect:

- If an action is followed by a pleasurable consequence, it will tend to be repeated
- If an action is followed by an unpleasant consequence, it will tend not to be repeated

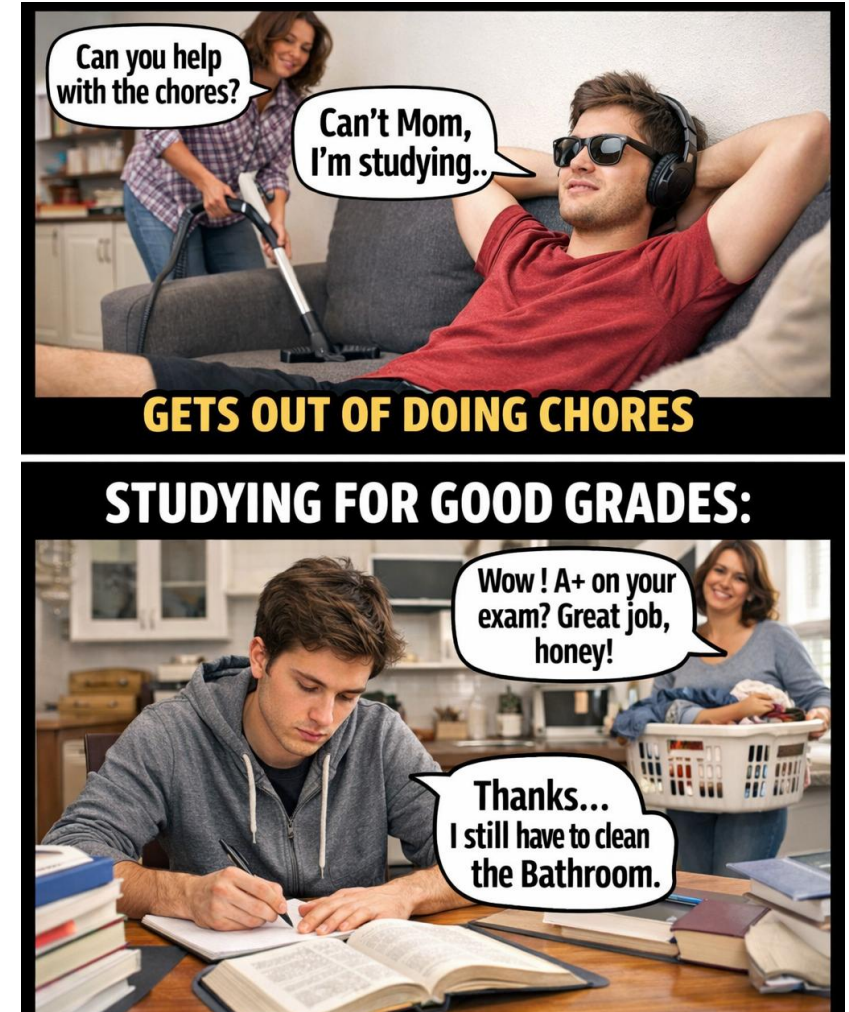
## Operant conditioning:

- The learning of voluntary behavior through the effects of pleasant and unpleasant consequences to responses
- Operant = any behavior that is **voluntary** and not elicited by specific stimuli
- B. F. Skinner
- Effect of consequences on behavior
- E.g., “If I do this, what’s in it for me?”

# Theories of Learning: Operant Conditioning (cont'd...)

“What’s in it for me?”

- **Reinforcement:** Any event or stimulus, that when following a response, increases the probability that the response will occur again
- “To strengthen”
- Related to the law of effect
- E.g., work and paycheck (pleasant consequence); avoiding doing chores (unpleasant consequence)



# Theories of Learning: Operant Conditioning (cont'd...)

## Types of Reinforcement

### Primary reinforcer:

Any reinforcer that is naturally reinforcing by meeting a basic biological need, such as hunger, thirst, or touch

### Positive reinforcement:

The reinforcement of a response by the addition or experiencing of a pleasurable stimulus

E.g., getting a shout-out on the university's social media page after volunteering at a campus event

### Secondary reinforcer:

Any reinforcer that becomes reinforcing after being paired with a primary reinforcer, such as praise, tokens, or gold stars

### Negative reinforcement:

The reinforcement of a response by the removal, escape from, or avoidance of an unpleasant stimulus

E.g., the professor stops sending reminder emails about missed assignments once a student begins submitting all work on time

BOTH positive AND negative reinforcement *increase* the likelihood of the behavior they follow—they *both* have the effect of strengthening, or reinforcing, the behavior

# Theories of Learning: Operant Conditioning (cont'd...)

| Schedule Type     | Definition                                                         | Example                                                                                                                                                          |
|-------------------|--------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fixed Ratio       | Reinforcement after a <u>set number of responses</u>               | Student receives a reward for every 5 questions answered correctly on an online quiz                                                                             |
| Variable Ratio    | Reinforcement after an <u>unpredictable number of responses</u>    | Extra credit is given after a random number of class participation points- could be the 2nd, 7th, or 10th time a student chimes in<br><br>Slot machines or games |
| Fixed Interval    | Reinforcement after a <u>fixed amount of time passes</u>           | Weekly review session: Students get bonus points if they submit weekly reflection papers on time<br><br>Salary                                                   |
| Variable Interval | Reinforcement after a varying, <u>unpredictable amount of time</u> | Surprise pop quizzes are given; students never know exactly when, sometimes after 3 days, sometimes after 8 days                                                 |

# Theories of Learning: Operant Conditioning (cont'd...)

## When you want to weaken the response

### Punishment by Application

Something unpleasant is added to the situation to reduce the response

For example: Scolding, extra homework,

### Punishment by removal

Behaviour is punished by removal of something pleasurable or desired after the behaviour occurs

For example: Removing freedom by grounding, taking away phone (going out with friends or talking on phone)

|                                              | Reinforcement                                                      | Punishment                                                 |
|----------------------------------------------|--------------------------------------------------------------------|------------------------------------------------------------|
| <b>Positive<br/>(Adding)</b>                 | Positive Reinforcement<br>Something valuable or desirable is added | Punishment by Application<br>Something unpleasant is added |
| <b>Negative<br/>(Removing/<br/>avoiding)</b> | Negative reinforcement<br>Something unpleasant is removed          | Punishment by removal<br>Something desirable is removed    |

# Theories of Learning: Operant Conditioning (cont'd...)

**Which of the following examples would be reinforcement or punishment and which would be negative reinforcement:**

1. P's father nags him to wash his car. P hates being nagged, so he washes the car so his father will stop nagging
2. N learns that talking in a funny voice gets him lots of attention from his classmates, so now he talks that way more often
3. A student keeps talking during lectures. The teacher responds by assigning extra homework each time the student talks. Over time, the student talks less in class
4. R repeatedly watches television and procrastinating. After several warnings, her mother takes away her television privileges for three days. After this, R stops procrastinating and is doing her homework

1. Negative reinforcement: P is being negatively reinforced for washing his car because the nagging (unpleasant stimulus) stops when he does so
2. Positive reinforcement: N is getting positive reinforcement in the form of his classmates' attention
3. Punishment by application: Student is punished by adding extra homework hence, student talks less
4. Punishment by removal: TV privileges were removed leading reducing procrastination

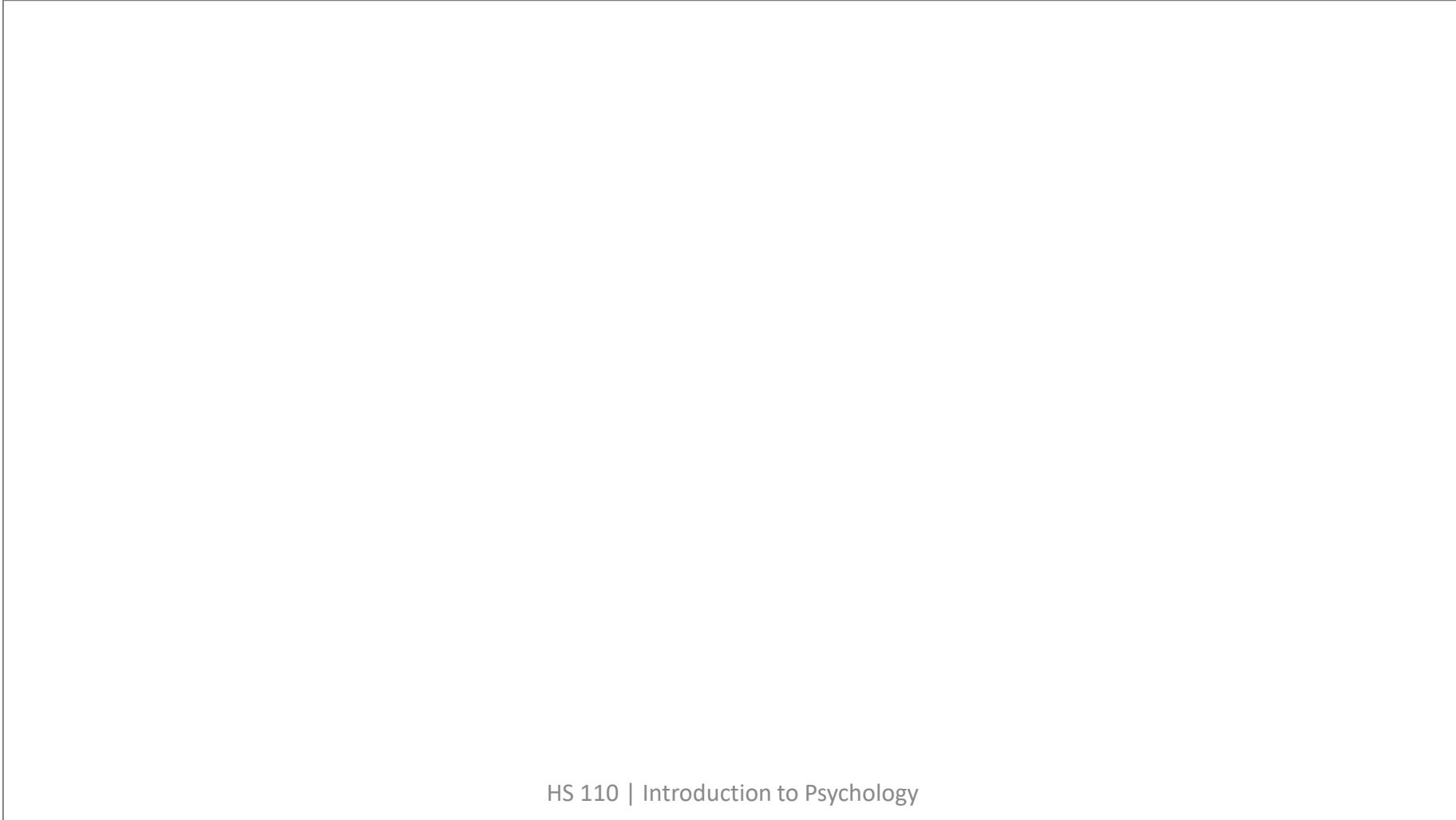
What type of reinforcement or punishment worked best for you when you were in school?

# Theories of Learning: Operant Conditioning (cont'd...)



# Theories of Learning: Observational Learning

Learning new behavior by watching a model perform that behavior



# Theories of Learning: Observational Learning (cont'd...)

## Four elements of observational learning:

**Attention:** To learn anything through observation, the learner must first pay *attention* to the model

E.g., watching a senior student answer questions confidently in discussion

**Memory:** The learner must also be able to retain the *memory* of what was done

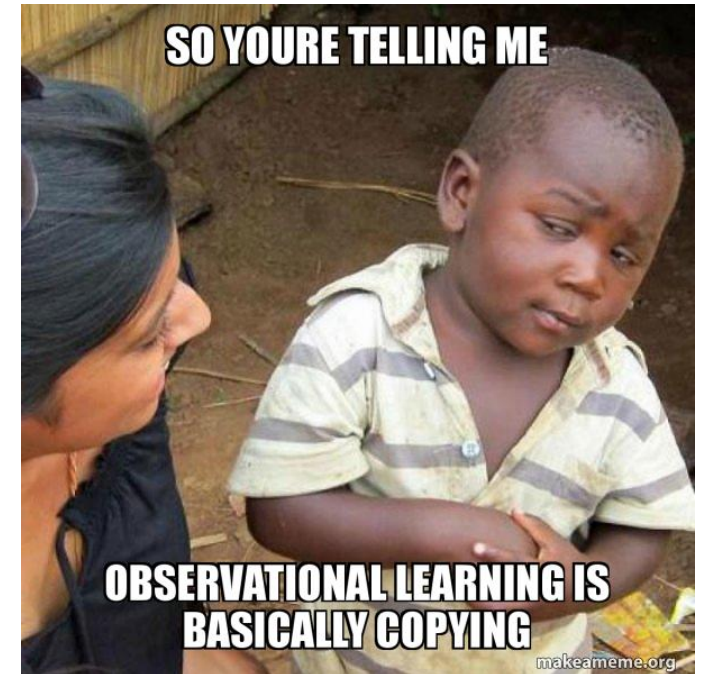
E.g., remembering the key arguments and style used

**Imitation:** The learner must be capable of reproducing, or *imitating*, the actions of the model

E.g., using a similar approach in their own discussion response

**Desire:** Finally, the learner must have the desire or *motivation* to perform the action

E.g., driven by wanting peer approval and good participation marks



# Conclusion

- Operant conditioning
  - Positive reinforcement
  - Negative reinforcement
  - Punishment by application
  - Punishment by removal
- Observational learning
  - Attention
  - Memory
  - Imitation
  - Desire

# RECAP

Introduction, Research Methods, Biological basis, Learning

Course instructor:  
Dr Shweta Chawak  
Assistant Professor  
Humanities and Social Sciences  
IIT Bombay

# What is Psychology

- Psychology
  - Greek word: psyche = mind
  - Logos = knowledge or study
- Scientific study of behaviour and mental processes
  - Overt behaviours: directly observable actions and responses
  - Covert behaviours: internal
- Study of ABC
  - Affect: emotions; feelings; urges
  - Behaviour: (un)observable actions and responses taken by the individual
  - Cognitions: our inner thoughts; our rationales; our beliefs
- Not common sense:
  - Focuses on evidence
  - Systematic and intersubjective

# Approaches to Psychology

- Father of Psychology: Wilhem Wundt (Structuralism)
- **Approaches:**
  - Biological: dopamine, reward pathway, prefrontal cortex
  - Behavioral: reinforcement, habit learning, cues
  - Humanistic: needs, meaning, self-growth
  - Cognitive: beliefs, attention, interpretation
  - Sociocultural: norms, peer influence, family context

# Types of psychological professionals

🌟 Who is a psychiatrist? Who is a psychologist? 🌟

| Type of professional   | Psychiatrist                                                                                                                                          | Psychologist                                                                                                                                                                                  | Psychiatric social worker                                                                                                  |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Educational degree     | Doctor of Medicine (MD)                                                                                                                               | Doctorate degree                                                                                                                                                                              | Master of Social Work (MSW)                                                                                                |
| Training               | Medical doctor (MBBS)                                                                                                                                 | Academic                                                                                                                                                                                      | Academic                                                                                                                   |
| Specialisation         | <ul style="list-style-type: none"> <li>▪ Diagnosis and treatment of psychological disorders</li> <li>▪ May provide therapy and counselling</li> </ul> | <ul style="list-style-type: none"> <li>▪ Master's degree: counsellor, therapist, researcher</li> <li>▪ Clinical psychologist: license by the Rehabilitation Council of India (RCI)</li> </ul> | <ul style="list-style-type: none"> <li>▪ Environmental conditions</li> <li>▪ E.g., poverty, stress, drug misuse</li> </ul> |
| Prescribe medications? | Yes                                                                                                                                                   | No                                                                                                                                                                                            | No                                                                                                                         |
| Workplace setting      | Private practice or hospital settings                                                                                                                 | Multiple                                                                                                                                                                                      | Clinics, hospitals, social service organisations                                                                           |

|                                       |                                    |                                                                                   |
|---------------------------------------|------------------------------------|-----------------------------------------------------------------------------------|
| Foundational Fields                   | Biological Psychology/Neuroscience | Examines the biological bases of behavior and mental processes                    |
|                                       | Cognitive Psychology               | Studies mental processes like memory, perception, and problem-solving             |
|                                       | Development Psychology             | Examines psychological growth and changes throughout the lifespan                 |
|                                       | Social Psychology                  | Explores how people influence and relate to each other                            |
|                                       | Behavioural Psychology             | Analyses observable behavior and its modification                                 |
|                                       | Personality Psychology             | Investigates individual differences in personality traits                         |
| Applied Fields                        | Clinical                           | Focuses on diagnosing and treating mental health disorder                         |
|                                       | Counselling                        | Helps individuals cope with personal and interpersonal challenges                 |
|                                       | Health                             | Studies psychological factors affecting health and illness                        |
|                                       | Industrial/Organisational          | Focuses on human behavior in the workplace to improve productivity and well-being |
|                                       | Educational                        | Focuses on how people learn and the effectiveness of educational interventions    |
|                                       | Forensic                           | Applies psychology to legal and criminal justice issues                           |
|                                       | Sports                             | Optimises athletic performance and psychological well-being in sports             |
| Interdisciplinary and emerging fields | Cultural Psychology                | Cultural context shapes cognition, emotion, and behaviour                         |
|                                       | Environmental                      | Studies the interaction between people and their physical environment             |
|                                       | Positive                           | Focus on well-being, strengths, resilience, flourishing                           |
|                                       | Psycholinguistics                  | Focus on how we acquire and use language                                          |
|                                       | Neuropsychology                    | Brain and behaviour in a clinical setting                                         |

# Research Methods

- Scientific Method
- Perceiving the question – Hypothesis – Hypothesis Testing – Conclusion – Report
- Types of research methods
  - Descriptive
  - Correlational
  - Experimental
- Descriptive types
  - Naturalistic observation: Observer bias
  - Laboratory observation
  - Clinical case study
  - Survey: Biases (Courtesy bias, Wording bias, Question order bias)

## Cross sectional research

- Observational study that collects data from diverse population at a single point in time
- Helps understand prevalence and describe characteristics

## Longitudinal research

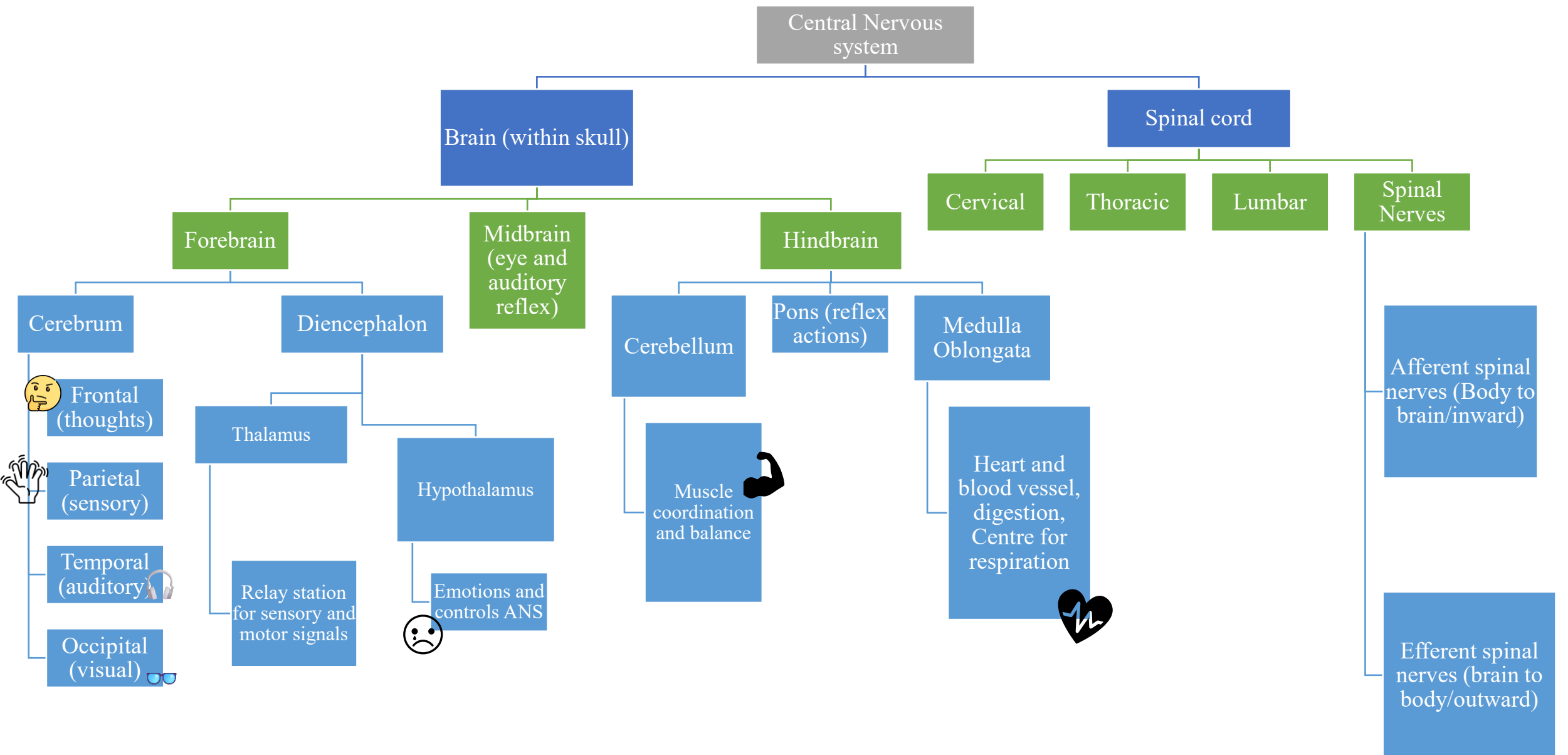
- A study method that involves collecting data from same individual or group over extended period of time from weeks to decades

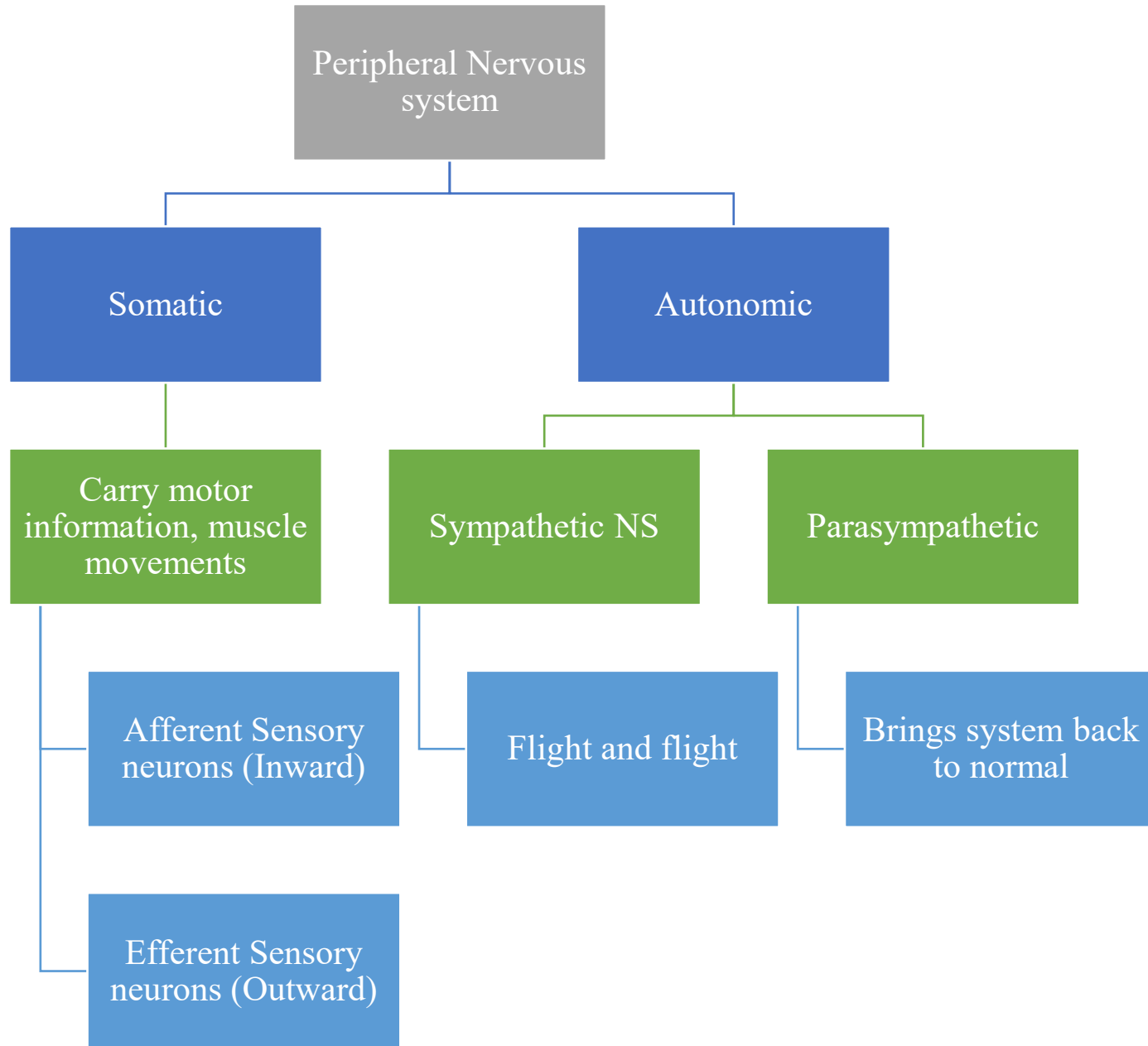
## Correlational research

- Non experimental method that measures the statistical relationship between two or more variables
- It seen if change in one corresponds change in another without manipulation

# Experimental Research

- Experiment: Direct manipulation of variables
- Variables
  - Independent
  - Dependent
  - Confounding
  - Extraneous
- Experimenter effect: unintentional influence
- Single anonymous study
- Double anonymous study





# Forebrain

- Cortex
  - Frontal lobe: Broca Aphasia (difficulty in using expressive language)
  - Temporal lobe: Wernicke's aphasia (word salad/ difficult in word comprehension)
- Basal Ganglia (decision making and coordination)
  - Addiction
  - Alzheimer's
  - Parkinson's
  - Huntington's (Decay of brain nerve cells over time)
  - Tourette syndrome
- Limbic System
  - Thalamus (relay station for sensory except smell): Dyslexia
  - Hypothalamus (controls emotions): Hormones issues/ Depression
  - Cingulate cortex: Dyslexia (selective attention)
  - Hippocampus: Alzheimer (memory)
  - Amygdala: Klüver-Bucy syndrome

# Midbrain and Hindbrain

- Midbrain: below Thalamus
  - Motor movements and reflexes
  - Damage: Parkinson's disease, issues with sleep
- Hindbrain
  - Medulla: Damage can be fatal
  - Pons: impaired coordination of movements
  - Cerebellum: Tremors unsteady walk (Alcoholic intoxication, Schizophrenia, Autism)

# The Autonomic Nervous System

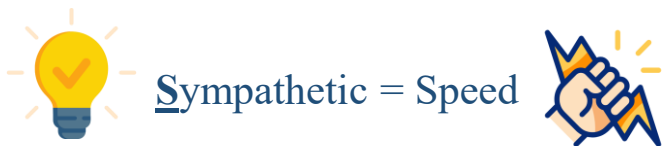
## Sympathetic division

- Most active when you are angry, afraid, or aroused
- **Fight-or-flight response**
- Increases heart rate and breathing
- Stops digestion

| Sympathetic division (arousing) |                | Parasympathetic division (calming)    |
|---------------------------------|----------------|---------------------------------------|
| Pupils dilate                   | EYES           | Pupils contract                       |
| Decreases                       | SALIVATION     | Increases                             |
| Perspires                       | SKIN           | Dries                                 |
| Increases                       | RESPIRATION    | Decreases                             |
| Accelerates                     | HEART          | Slows                                 |
| Inhibits                        | DIGESTION      | Activates                             |
| Secrete stress hormones         | ADRENAL GLANDS | Decrease secretion of stress hormones |

## Parasympathetic division

- **Calms body**
- Produces effects opposite to those of the sympathetic division
- Reduces heart rate and breathing
- Restores digestion



Sympathetic = Speed

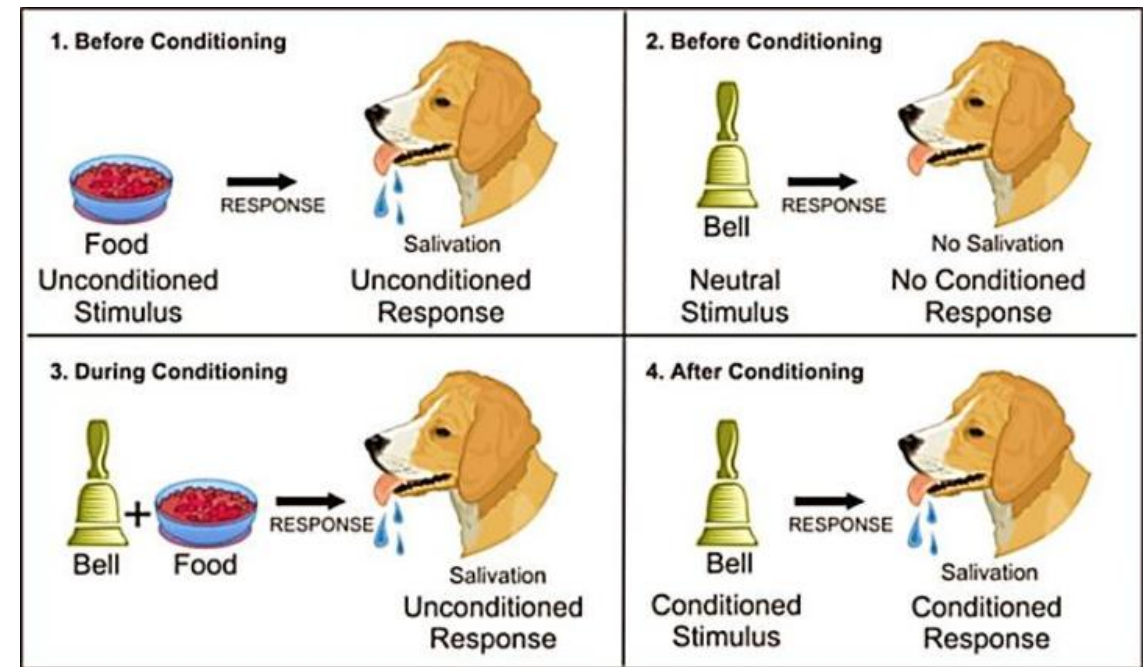


# Neuroplasticity

- Brains ability to change the structure and function in response to experience, learning, environment, and injury
- Types
  - Experience dependent plasticity
  - Developmental plasticity
  - Injury-induced plasticity

# Learning

- Classical conditioning
  - Involuntary response
  - Stimulus generalization ( similar stimuli)
  - Stimulus discrimination (respond to specific stimuli)
- Operant conditioning
  - Voluntary response
- Observational learning
  - Learning by observation
  - Attention, Memory, Imitation, Desire



|                                     | Reinforcement                                                             | Punishment                                                        |
|-------------------------------------|---------------------------------------------------------------------------|-------------------------------------------------------------------|
| <b>Positive (Adding)</b>            | <b>Positive Reinforcement</b><br>Something valuable or desirable is added | <b>Punishment by Application</b><br>Something unpleasant is added |
| <b>Negative (Removing/avoiding)</b> | <b>Negative reinforcement</b><br>Something unpleasant is removed          | <b>Punishment by removal</b><br>Something desirable is removed    |

# MEMORY

## Sensory memory and short-term memory

Course instructor:

Dr Shweta Chawak

Assistant Professor

Humanities and Social Sciences

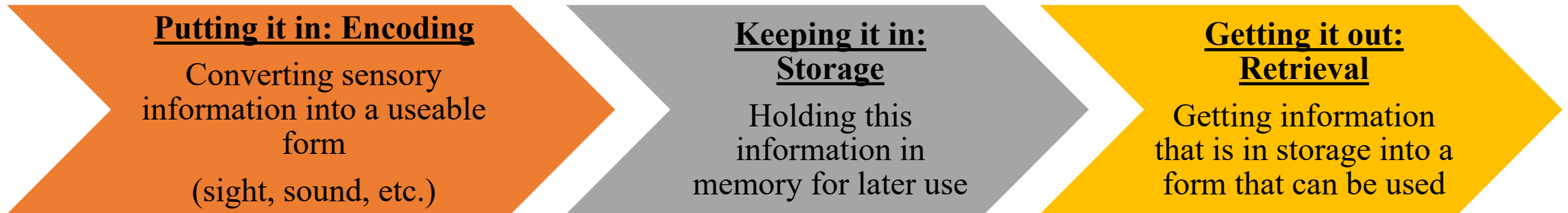
IIT Bombay



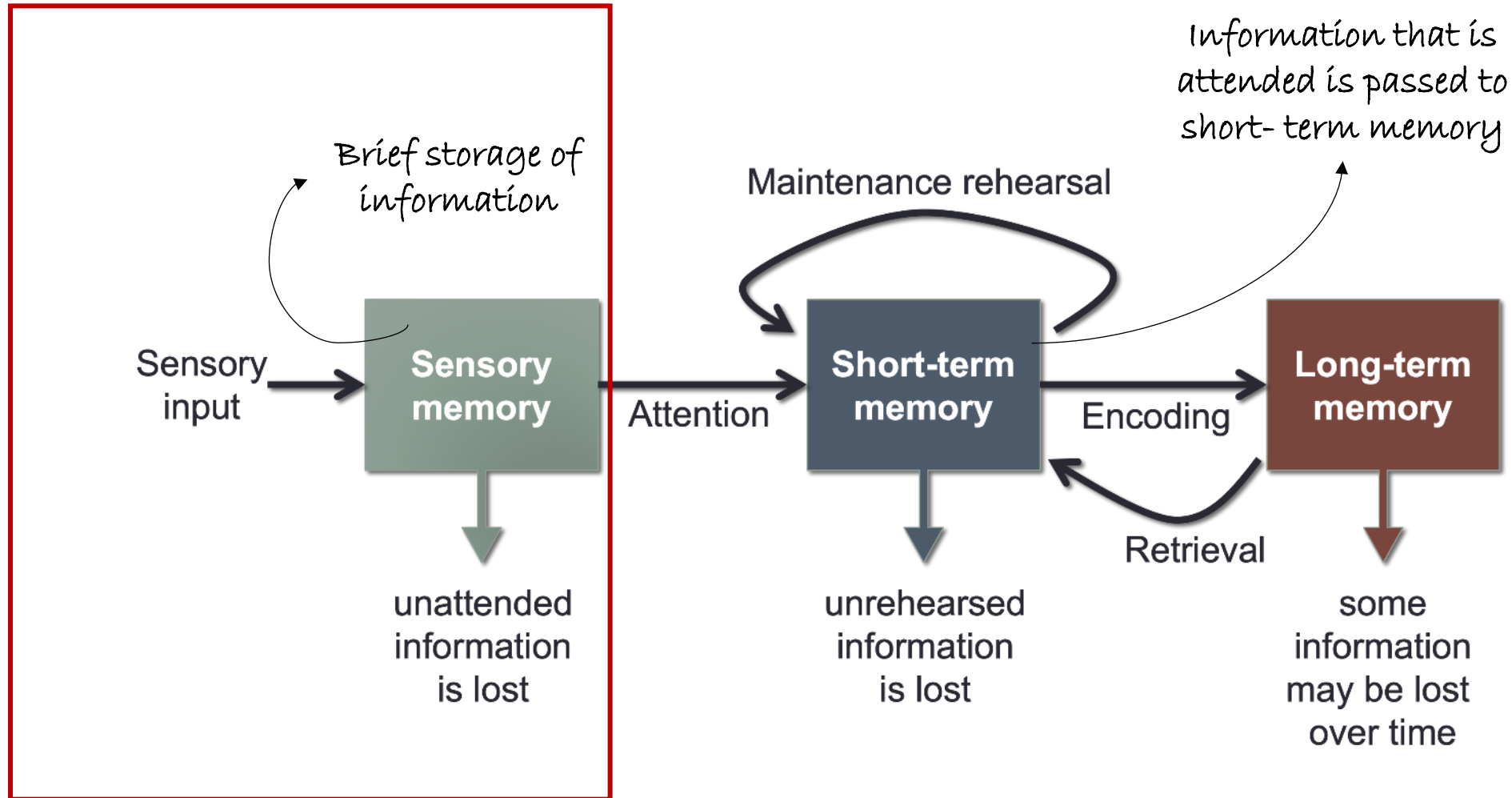
# What is Memory?

**Memory** is an active system that receives information from the senses, puts that information into a usable form, organizes it as it stores it away, and then retrieves the information from storage

## Three processes of memory

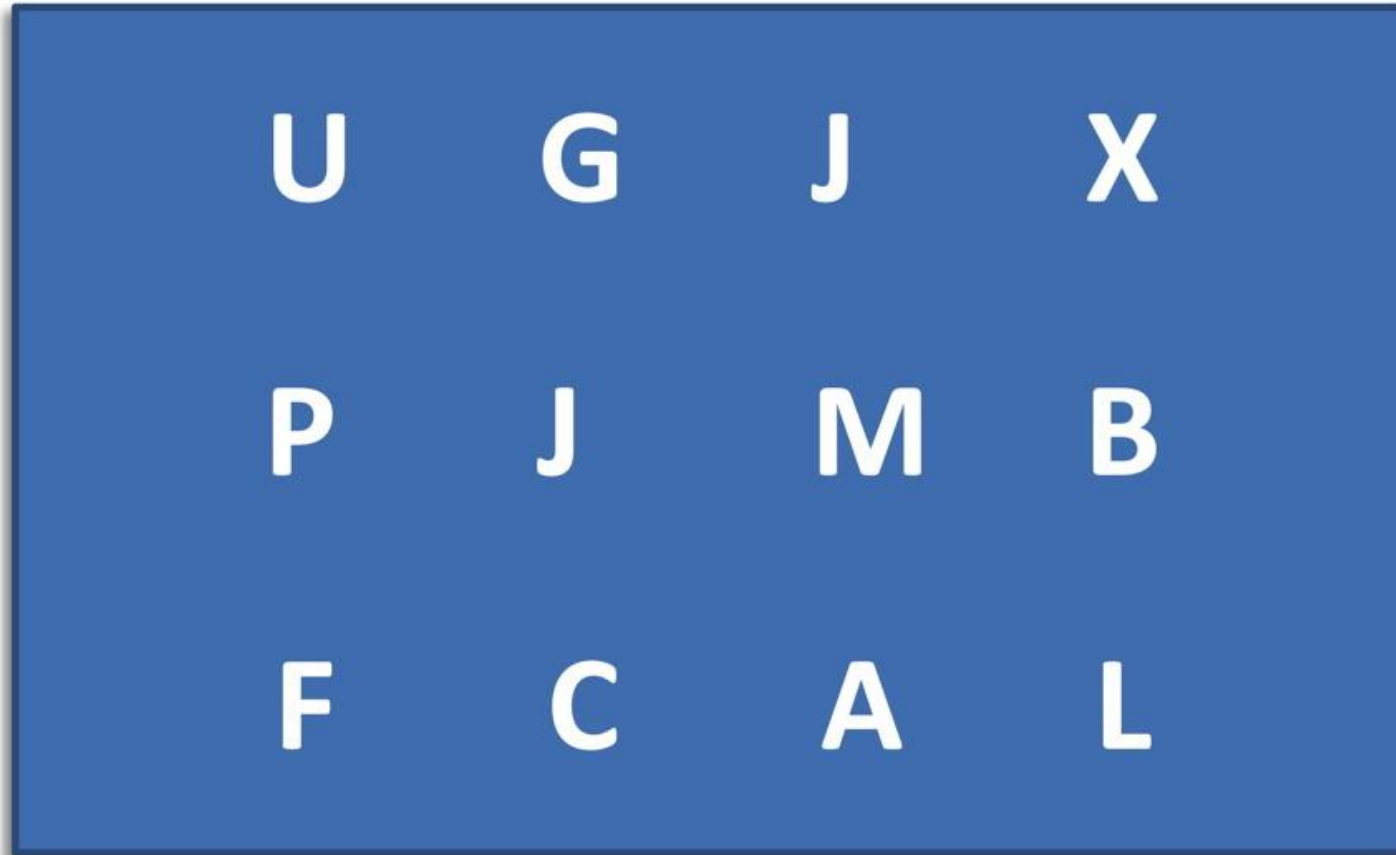


# Information processing model

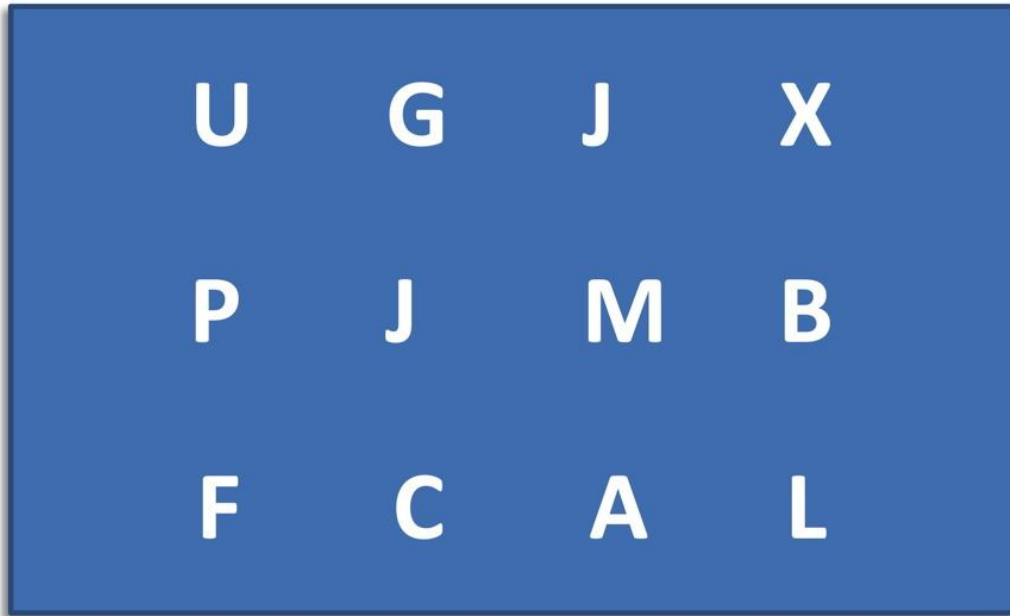


Pay attention to the next slide...

# How do we store memories?! Sensory Memory

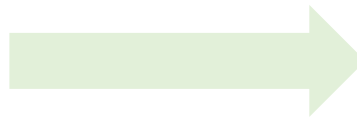


# How do we store memories?! Sensory Memory



## Sensory memory:

- raw information from the senses is held
- the very first stage of memory
- very brief period of time (within a second)



## Why Sensory memory

- Follow a conversation
- Read smoothly
- Perceive motion as continuous
- Understand spoken language
- Recognize visual scenes
- Experience the world as stable and coherent

### Iconic memory

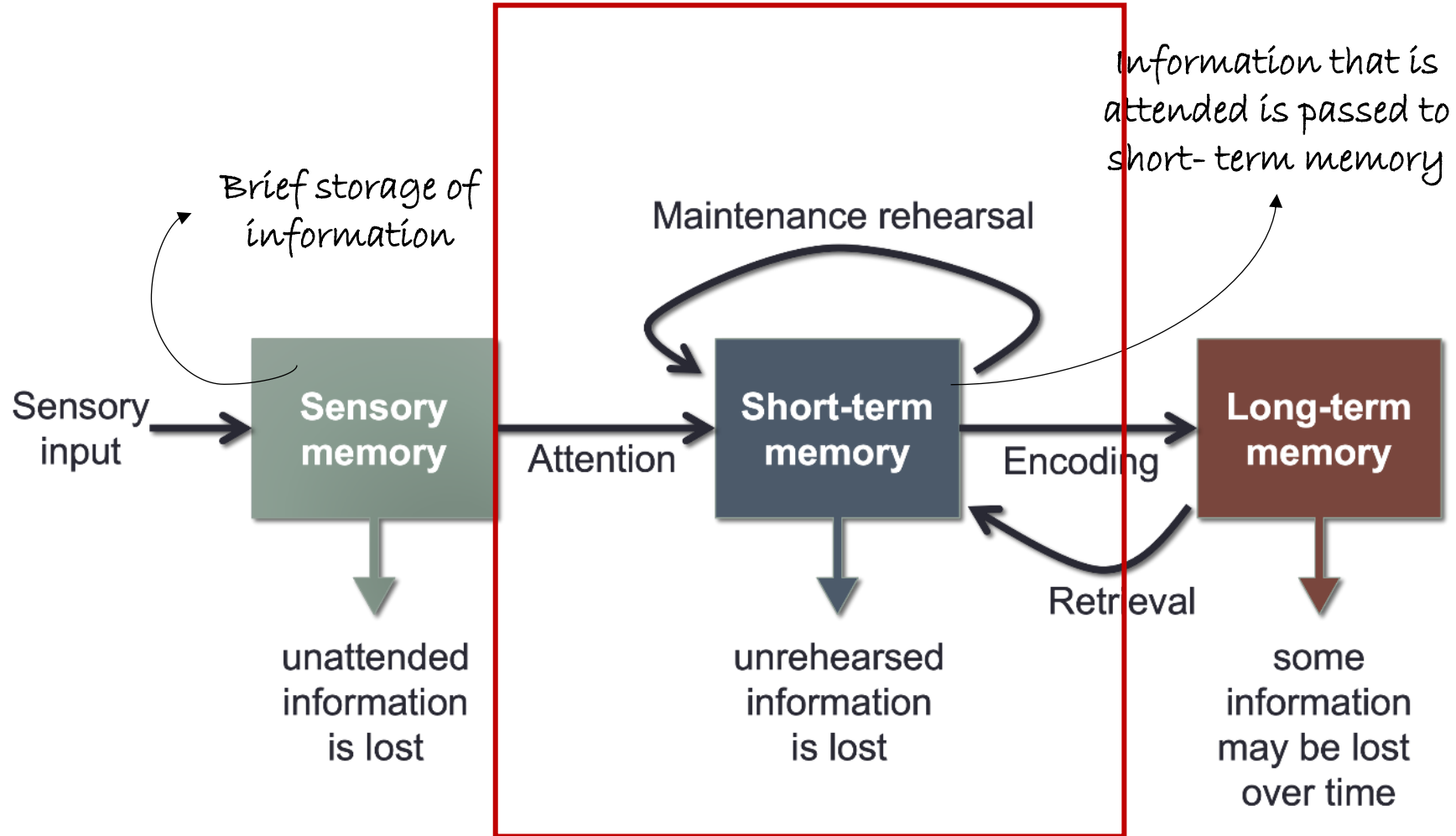
- visual sensory memory
- (fraction of a second)

### Echoic memory

- Auditory sensory memory
- (2-4 seconds)



# Information processing model



# The Digit-span test

<https://www.psychtoolkit.org/experiment-library/digitspan.html>



# How do we store memories?! Short-term Memory

## Short-term memory (STM):

- Information is held for brief periods of time while being used
  - Also called working memory
  - Mostly encoded in auditory form
  - Lasts around 30 seconds
- 
- George Miller (1956)
  - Digit-span test: memory test
  - Magical number 7, plus or minus two:  $7 \pm 2$
  - Selective Attention: Focusing (voluntarily) on a selected portion of sensory input (e.g., selective hearing; cocktail-party effect)

(Atkinson & Shiffrin, 1968)



# How do we store memories?! Short-term Memory

## Maintenance rehearsal

- Practice of saying some information to be remembered over and over in one's head in order to maintain it in short-term memory
- Interference causes memory loss
- STM hindered when it exceeds capacity
  - Chunking helps remember more information

E.g.,

- Reading a definition or fact again and again right before a quiz, but not linking it to other information or personal experiences
- Memorizing a locker combination by repeating it constantly until opening the lock
- Chunking: 99-22-53-41...



# RECAP

- Memory is an active system that receives information from the senses, puts that information into a usable form, organizes it as it stores it away, and then retrieves the information from storage
  - Putting it (Encoding)
  - Keeping in storage (holding information)
  - Getting it out (Retrieval)
- Information processing model
  - Sensory memory – short term memory – long-term memory
  - Sensory memory: raw sensory information hold [iconic (visual) & echoic (audio)]
  - Short term memory: information held for shorter duration ( $7\pm2$ )
- How to hold information: maintenance rehearsal or chunking

# MEMORY

Long-term memory, Recall vs recognition, is memory reliable?

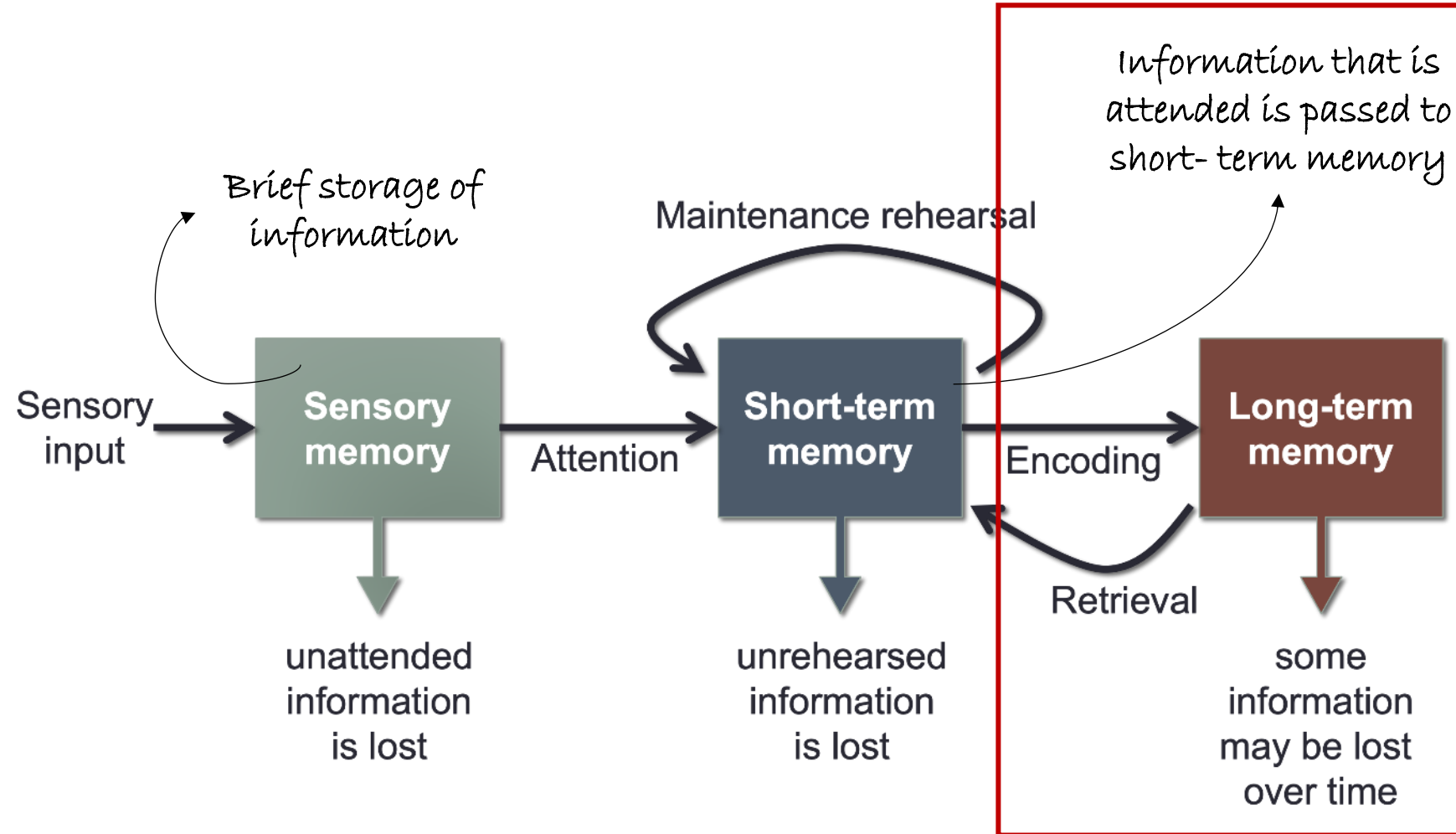
Course instructor:  
Dr Shweta Chawak  
Assistant Professor  
Humanities and Social Sciences  
IIT Bombay



# RECAP

- Memory is an active system that receives information from the senses, puts that information into a usable form, organizes it as it stores it away, and then retrieves the information from storage
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# Information processing model



# ACTIVITY: MEMORY GAME!

- Remember your first day on this campus!
- Remember the words we discussed in last class?

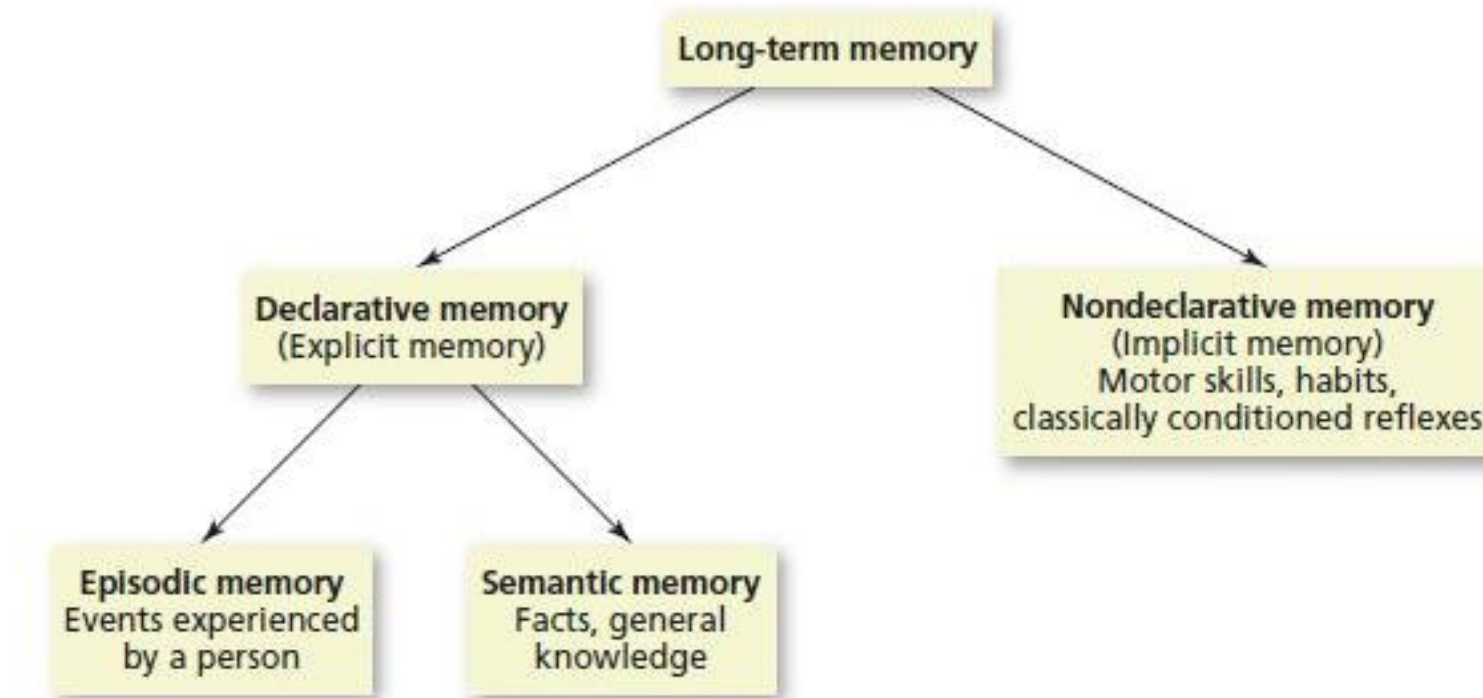
# How do we store memories?! Long-term Memory

## Long-term memory (LTM):

- The system of memory into which all the information is placed to be kept more or less permanently
- Encoded in terms of images, sounds, smells, or tastes
- Stored as long-lasting memories of events and concepts that are meaningful and important



# How do we store memories?! Long-term Memory



In thinking about a typical day, how do you use each type of memory: nondeclarative, episodic, and semantic?

# How do we store memories?! Long-term Memory

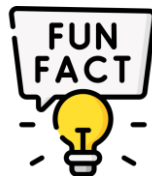
## Types of Long-term Information (LTM): Declarative (Explicit) memory

- All the things that people **can know**—the facts and information that make up knowledge
- “What was for brunch?”
- Conscious and known



### Semantic memory:

- The word *semantic* refers to meaning
- Awareness of the meanings of words, concepts, and terms as well as names of objects, math skills, and so on



### Episodic memory:

- Memories of what has happened to people, such as certain birthdays, childhood events that were particularly special
- Represent episodes from their lives



# How do we store memories?! Long-term Memory

## Types of Long-term Information (LTM): Nondeclarative (Implicit) memory

- Memories for things that people know **how to do**, like tying shoes and riding a bicycle
- Emotional associations (amygdala), habits, and simple conditioned reflexes (hind brain)
- These memories are not conscious but are implied to exist because they affect conscious behaviour
- The kind of memories people “never forget”

- **Anterograde amnesia:** new long-term declarative memories cannot be formed
- Loss of memory from the point of injury or trauma forward
- Damage to the hippocampal area of the brain
- Alzheimer’s disease, who also suffer from anterograde amnesia, do not forget how to walk, talk, fasten clothing, or even tie shoes





# How do we store memories?! Long-term Memory

## Elaborative rehearsal

- Way of increasing the number of retrieval cues (stimuli that aid in remembering) for information by connecting new information with something that is already well known
- Information processed according to its **meaning**, not mere sounds or words
- Remembered more efficiently and for a longer period of time

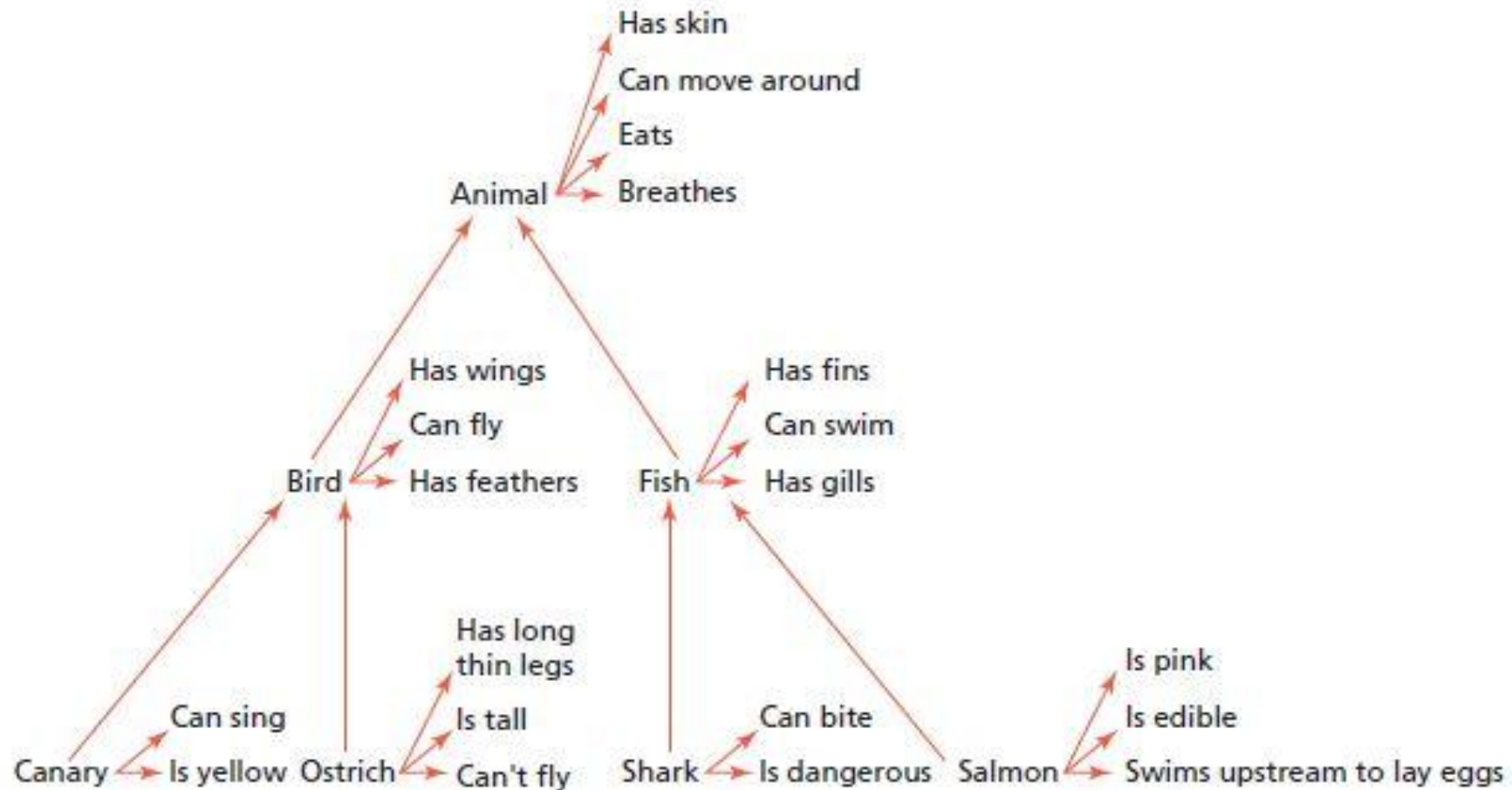


E.g.,

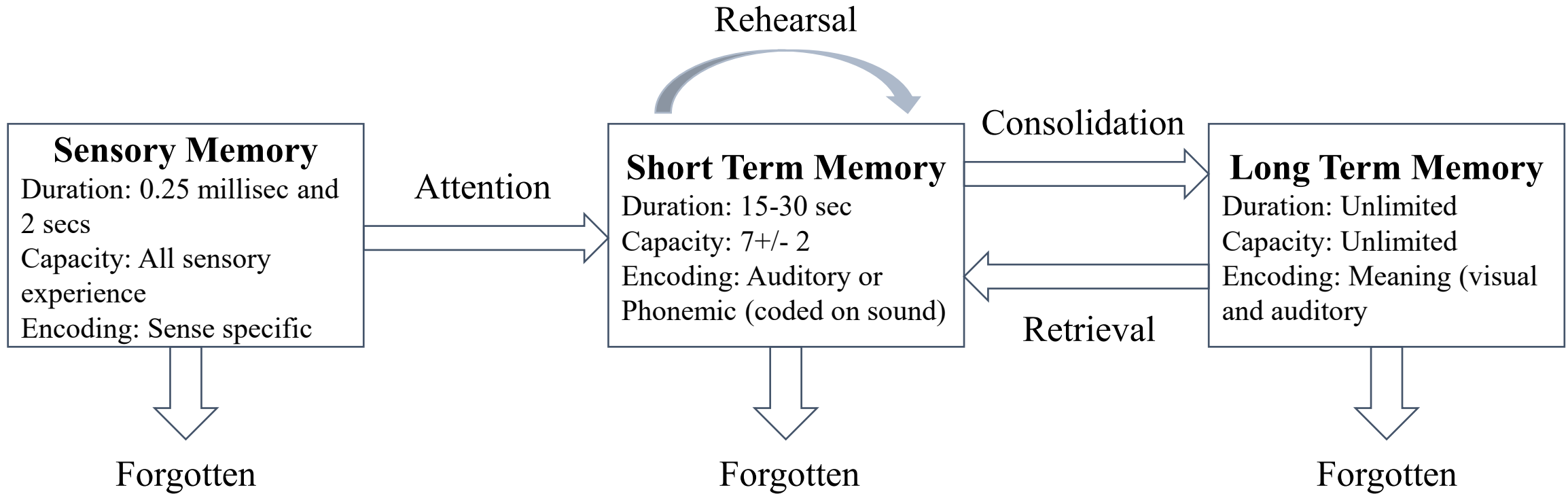
- **Personal analogy**: Connect the concept of “operant conditioning” to getting more likes on Instagram posts after posting funny memes, encouraging more meme-sharing
- **Using familiar songs**: Set the structure of the brain to the tune of a favorite pop song for easier recall during exams
- **Password creation**: Use a sentence or a familiar phrase with meaningful substitutions—“MyDogBen2019!” combines a pet’s name and the year adopted
- **Remembering someone’s name**: If introduced to a person named Lily, visualize a lily flower or connect her name to a “Lily” character from a TV show watched regularly



# How do we store memories?! Long-term Memory



**An Example of a Semantic Network**



# How do we retrieve memories? Recall versus Recognition

## Recall

- Memories are retrieved with few or no external cues
- E.g., filling in the blanks on an application form

*Tip of the tongue (TOT) phenomenon*

## Serial position effect

- Tendency of information at the **beginning and end** of a body of information to be remembered more accurately than information in the middle of the body of information

## Primacy effect

- Words at the beginning of a list are remembered better
- Early words get more rehearsal time in short-term memory
- Middle words are replaced quickly by new incoming words, so they are less rehearsed

## Recency effect

- The last words on a list are remembered easily
- These words are still in short-term memory (STM)
- No new words come in to push them out, making retrieval easier

**Are multiple-choice tests easier than essay tests? If so, why?**

# How do we retrieve memories? Recall versus Recognition

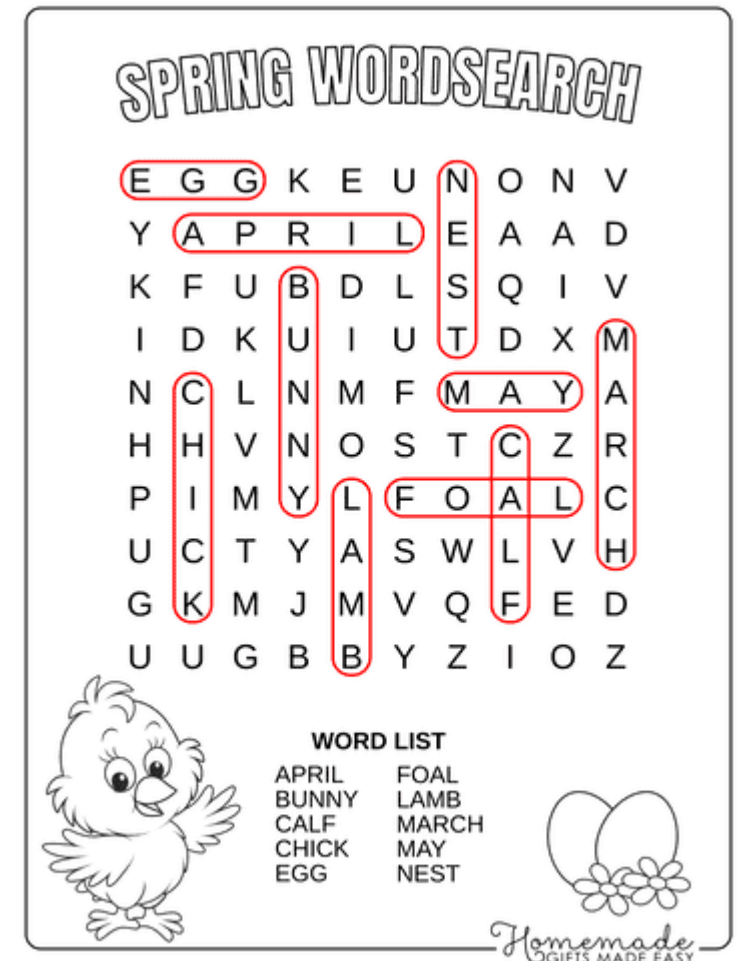
If you're given a list of 12 words: "table, chair, window, door, spoon, lamp, book, pen, apple, orange, milk, water," you might recall:

- **Primacy Effect:** Words at the beginning, like "table," "chair," and "window"
- **Recency Effect:** Words at the end, like "milk" and "water"

# How do we retrieve memories? Recall versus Recognition

## Recognition

- Involves looking at or hearing information and **matching** it to what is already in memory
- E.g., word-search puzzle and multiple-choice
- Recognition is accurate for images, especially human faces
- Susceptible to false positive: incorrectly recognize something as correct



# How reliable are our memories?

**Hello.**

# How reliable are our memories? (cont'd...)

## Constructive processing of memory:

- Memories are **built or reconstructed** from stored information, not simply retrieved as exact copies
- Each time a memory is recalled, it can be changed; new information may be added or some details left out, depending on what is available at the time of reconstruction
- E.g., after learning new details about an event, people often revise their memories to match what they now know, discarding previous mistakes or inaccurate information
- “*Hindsight bias*” where people mistakenly believe they “knew it all along” after learning the actual outcome
- E.g., after receiving a graded test, a student might say, “I knew I should have studied that topic more,” even though before the test, they actually were uncertain or didn’t think it was important



# Why do we forget?

| Concept              | Definition                                                                                                                                                                                                                                                                          | Example                                                                                                                                                                                                                                |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Encoding Failure 🚫   | When information is never processed into long-term memory                                                                                                                                                                                                                           | Not remembering someone's name because attention was on the phone during introductions                                                                                                                                                 |
| Storage Failure 📉    | Loss of information from memory over time due to decay or non-use                                                                                                                                                                                                                   | Forgetting details of last semester's lectures because never reviewed or used them                                                                                                                                                     |
| Retrieval Failure ❓  | Inability to access stored information due to missing cues                                                                                                                                                                                                                          | Experiencing "tip-of-the-tongue" when trying to recall a familiar password or a celebrity's name                                                                                                                                       |
| Interference 🔄       | <p>When other memories block retrieval</p> <p><u>Proactive interference</u>: older information already in memory interferes with the learning of newer information</p> <p><u>Retroactive interference</u>: newer information interferes with the retrieval of older information</p> | <p><u>Proactive interference</u>: Remembering an old address makes it difficult to recall a newly moved-to address</p> <p><u>Retroactive interference</u>: Learning a new phone number makes it harder to remember your old number</p> |
| Transfer of Memory 🔄 | Applying or failing to apply learned information across contexts/situations                                                                                                                                                                                                         | Struggling to use memory strategies from psychology class when memorizing a grocery list for the first time                                                                                                                            |

Retroactive affects the *past* (old memories), proactive affects the *future* (new memories)

# Memory Techniques

- ☐ Recitation: repeating information verbally until it is encoded into long term memory
- ☐ Rehearsal: silently repeating information until it is encoded into long term memory
- ☐ Selection: highlighting important information instead of reading an entire page
- ☐ Organization (Chunking): separating information into smaller chunks of information to remember
- ☐ Spaced practice: studying a small amount of information for a short time, then taking a break before studying more
- ☐ Mnemonic: any kind of memory system or aid (e.g., James-Lange theory = Jumpy Legs)



# RECAP

- The system of memory into which all the information is placed to be kept more or less permanently
- Information processed according to its **meaning**, not mere sounds or words
- Remembered more efficiently and for a longer period of time
- Types of LTM:
  - Declarative:
    - Semantic
    - Episodic
  - Non-declarative
- Elaborate rehearsal – Retrieval cues – Semantic network
- Recall vs Recognition
- Constructive process of memory
- Why do we forget?
- Memory techniques

# References

- Ciccarelli, S. K., White, J. N., & Misra, G. (2017). *Psychology* (5th ed.). Pearson Education India.

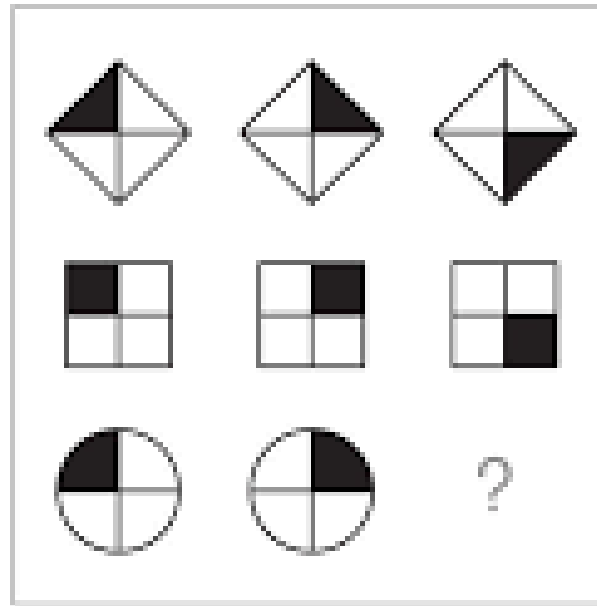
**Gifs, memes, images, supplementary reading resources, & icons taken from:**

- <https://www.flaticon.com>

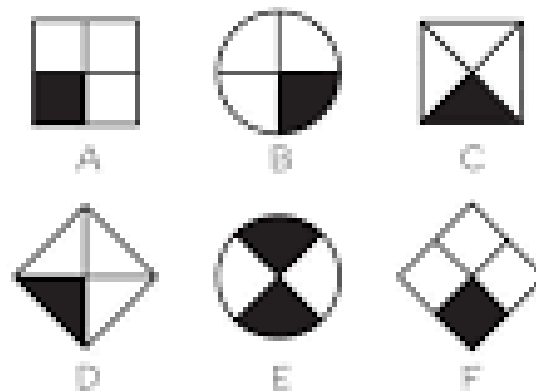


# Intelligence

Course instructor:  
Dr Shweta Chawak  
Assistant Professor  
Humanities and Social Sciences  
IIT Bombay



CHOOSE ANSWER



- Is intelligence just a number on a test, or is it the savvy to navigate life's challenges?
- Is it about acing exams, or about thriving socially, financially, and personally?

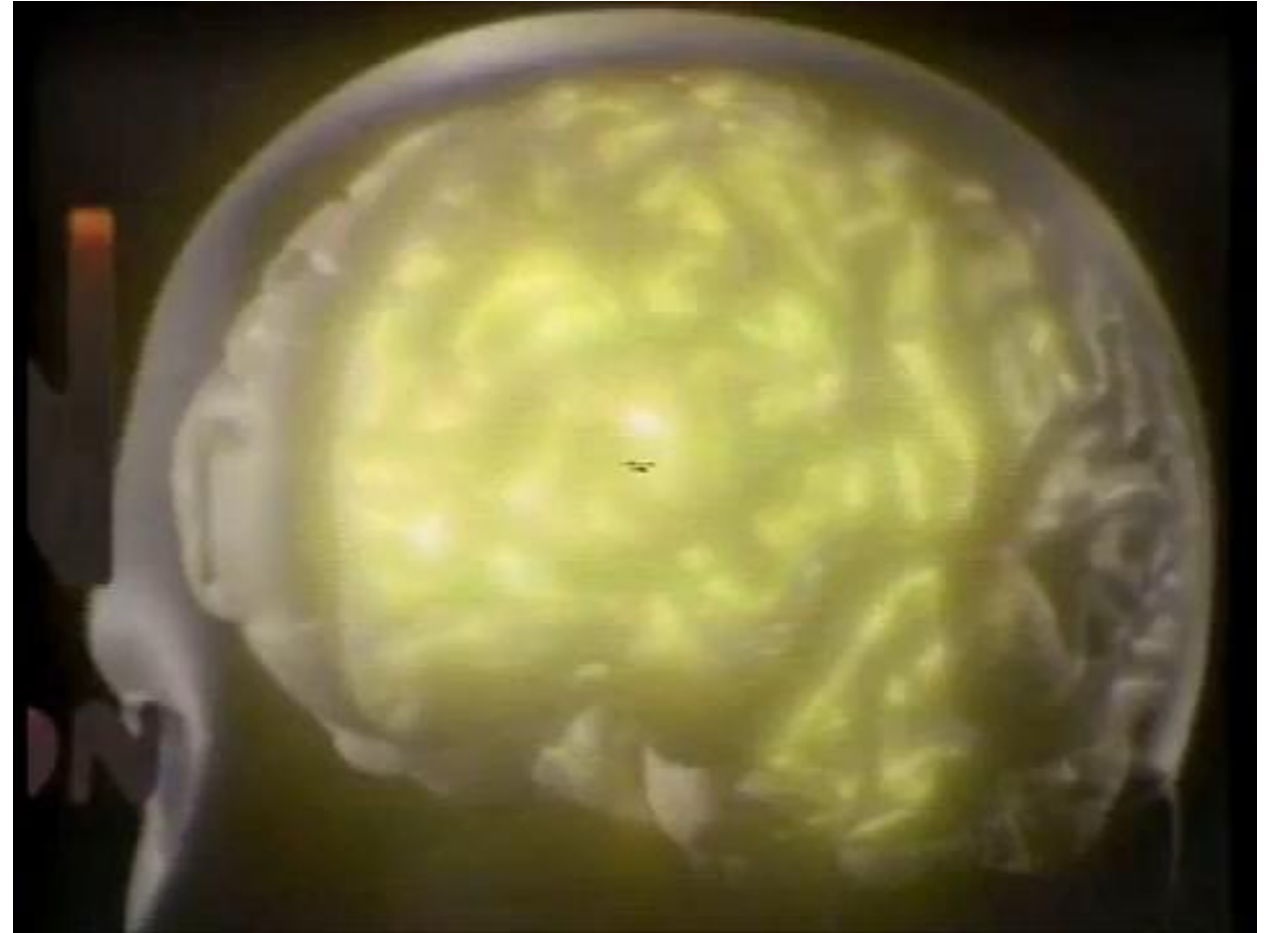
## SHARE YOUR THOUGHT!

*In your Field, what does it mean to be intelligent or smart?*



# What is Intelligence?

- The ability to **learn** from one's experiences, **acquire** knowledge, and **use resources** effectively in **adapting** to new situations or solving problems
- Shaped by one's culture
- Collectivist culture: interpersonal skills, adaptability to social and relational context
- Individualist culture: problem-solving, logical reasoning, and analytical thinking



Judy Kearins

(Feldman, 2009; Sternberg & Kaufman, 1998; Wechsler, 1975)

# Theories of Intelligence

# Theories of Intelligence: Binet's Single Factor Theory

- Alfred Binet: French psychologist
- Focused on a single or a uni - factor theory
- All intellectual abilities stems from a single, capability to judge, understand, and reason well
- Developed to identify children needing specific education attention
- One factor universal for all activities of the intelligence

## Criticisms:

- Does not account for unevenness of abilities- people often excel in some areas and struggle in others
- Contradicts common observation that individuals have different strengths in domains like math, language, or music
- Viewed as overly simplistic; fails to recognize the multidimensional nature of intelligence

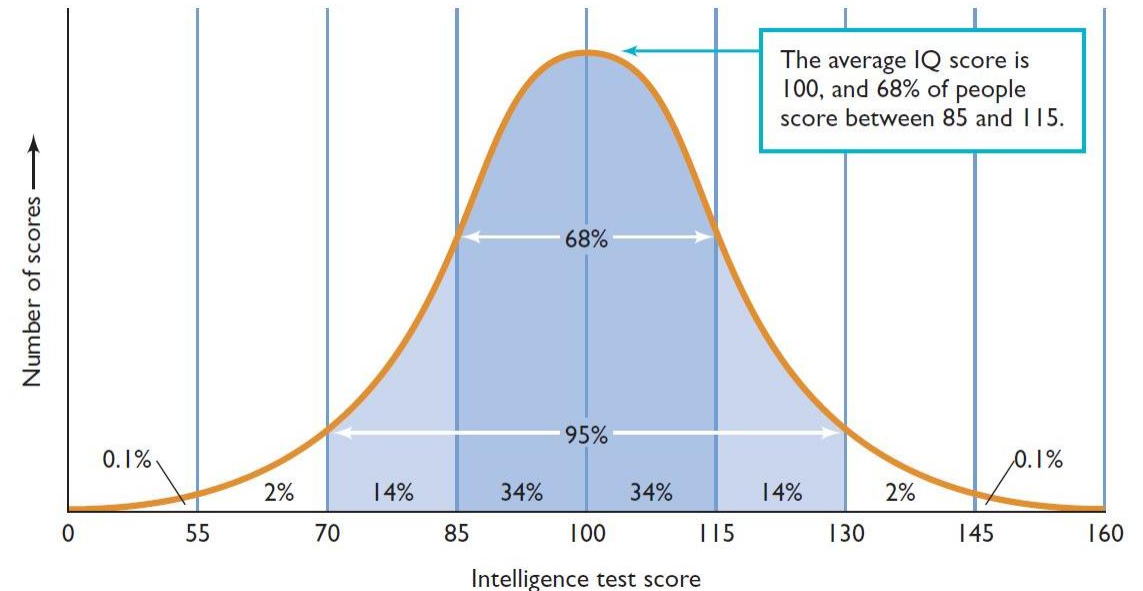
# Theories of Intelligence: Binet's Single Factor Theory

## Binet-Simon Intelligence test

- Developed the first intelligence test: Binet-Simon Intelligence Scale (1900s)
- Presently, Stanford-Binet Intelligence Scale: popular test in IQ
- Focused on memory, attention, and problem-solving, designed to predict academic potential

$$\text{IQ score} = \frac{\text{MA}}{\text{CA}} \times 100$$

MA = Mental Age; CA = Chronological Age



| Full Scale IQ | Classification      | Percentile |
|---------------|---------------------|------------|
| 130 and above | Very gifted         | 98 – 99.5  |
| 120-129       | Gifted              | 91 – 97    |
| 110-119       | High Average        | 75 – 90    |
| 90-109        | Average             | 25 – 73    |
| 80-89         | Below Average       | 9 – 23     |
| 70-79         | Border Line         | 2 – 8      |
| 69 and below  | Intellectually Poor | 0.01 – 2   |

# Theories of Intelligence: Spearman's Two Factor Theory

- Spearman's two factor theory (1904)
- Intelligence as two different abilities
- **g factor** for *general intelligence*, innate ability: Ability to reason and solve problems
- Reasoning, problem-solving, and abstract thinking
- It is the foundation for successful performance on all types of mental activities and tasks
- **s factor** for *specific intelligence*; acquired from the environment, such as learning and experience
- Task-specific abilities in certain areas such as music, business, art, mathematics, or language
- For instance, solving a math problem draws on both overall intelligence (g) and math-specific skills (s)

## Criticism:

- Oversimplification of the complexity of human cognitive abilities
- Cultural and individual differences: It does not account well for cultural influences or individual experience shaping specific abilities and intelligence expression





# Intelligence

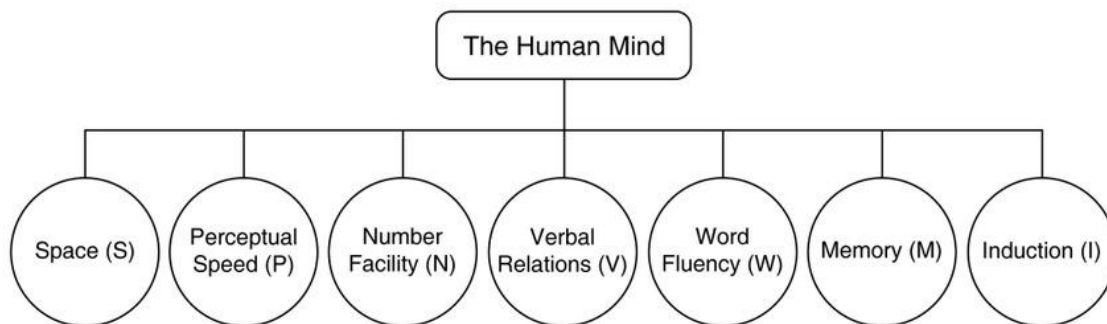
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Dr Shweta Chawak  
Assistant Professor  
Humanities and Social Sciences  
IIT Bombay

# RECAP

- Intelligence: The ability to learn from one's experiences, acquire knowledge, and use resources effectively in adapting to new situations or solving problems
- Binet's uni-factor theory: Mental age
- Spearman G factor: general intelligence
- Spearman s factor: specific intelligence

# Theories of Intelligence: Thurstone's Primary Mental Abilities

- Thurstone's primary mental abilities or the group factor theory (1938)
- Proposed multiple distinct mental abilities or factors rather than the 'g' factor
- Seven primary mental abilities:** Relatively independent of each other
- Later studies: interconnection between the abilities (hinting on g factor)



(Thurstone, 1938)



LL Thurstone, an American. Of

# Theories of Intelligence: Thurstone's Primary Mental Abilities (cont'd...)

## **Space (S)**

Ability to mentally visualise and manipulate visual images and spaces  
E.g., spatial reasoning and spatial relationships;  
interior designer, graphic designer, aerospace engineering

## **Perceptual Speed (P)**

Ability to quickly and accurately perceive visual details and discriminate between different stimuli  
E.g., video game tester, radiologist, and quality control inspector

## **Number Facility (N)**

Capacity to work with numbers, perform mathematical calculations  
E.g., data scientist, engineer, accountant

## **Verbal Relations (V)**

Ability to understand and use language effectively  
E.g., reading comprehension, verbal reasoning, vocabulary

## **Word Fluency (W)**

Ability to generate words and ideas quickly and fluently  
E.g., spontaneous storytelling, improvisation

## **Memory (M)**

Ability to encode, store, retrieve information, and make connections between ideas and form associations  
E.g., academic researcher, medical professionals, librarian

## **Induction (I)**

Ability to identify patterns, make generalisations, and draw conclusions based on incomplete information  
E.g., problem-solving, critical thinking, inference-making

# Theories of Intelligence: Thurstone's Primary Mental Abilities (cont'd...)

## Criticism:

- Interdependence of Abilities: abilities often overlap or correlate, suggesting the influence of an underlying general intelligence factor (g)
- Oversimplification: Simplifies intelligence by focusing on seven factors only

# Gardner's multiple intelligence

- Intelligence is not single entity but distinct modalities
- Selectively independent faculties
  - Linguistic Intelligence
  - Logical-mathematical Intelligence
  - Spatial Intelligence
  - Bodily Kinesthetics Intelligence (Body smart)
  - Musical Intelligence
  - Interpersonal intelligence
  - Intrapersonal intelligence (Self smart)
  - Naturalistic intelligence
  - Existential intelligence (possible 9th )

29/03/2026



# Theories of Intelligence: Cattell's Theory of Intelligence

- Raymond Cattell (1963)
- *Crystallized intelligence*: acquired knowledge and skills; accumulated through life's experiences
- *Fluid intelligence*: information-processing capabilities, such as, reasoning and problem solving and adaptability in unfamiliar situations
- Fluid intelligence declines after adolescence, while crystallized intelligence grows through adulthood

Crystallized = Grows like a crystal  
Fluid = Fades like flowing water

## Crystallized intelligence examples:

- A chef who has memorized many recipes and cooking techniques over years of experience uses crystallized intelligence
- Playing trivia games where recalling facts and information is key

## Fluid Intelligence examples:

- Figuring out how to fix a new type of gadget without instructions
- Using abstract reasoning to solve logic problems or pattern sequences on the spot

# Theories of Intelligence: Cattell's Theory of Intelligence (cont'd...)

## **Criticism:**

- The theory oversimplifies human intelligence by dividing it into only two types
- Fluid and crystallized intelligence are interrelated, not completely separate
- It focuses on intellectual abilities, ignoring creativity, emotional, and practical intelligence

Crystallized intelligence: 'I studied this for weeks!'

Fluid intelligence: 'Come on brain, figure this out in 5 minutes!'



# Theories of Intelligence: Sternberg's Triarchic Theory

- Robert Sternberg (1988)
- Triarchic theory of intelligence
- Three kinds of intelligence
- **Analytical, creative, and practical intelligence**

## **Analytical intelligence**

- Ability to break problems down into component parts, or analysis, for problem solving
- Measured by intelligence tests and academic achievement tests
- “Book smarts”

## **Practical intelligence**

- Ability to use information to get along in life
- Tactful, manipulate situations to their advantage, use inside information to increase their odds of success
- “Street smarts”

## **Creative intelligence**

- Ability to deal with new and different concepts
- Find new ways of solving problems (divergent thinking)

# Theories of Intelligence: Sternberg's Triarchic Theory (cont'd...)

## Example:

- Analytical: Understands theories and analyzes data
- Creative: Comes up with new research ideas or ways to explain concepts
- Practical: Manages time, organizes study groups, and applies knowledge in real life

# Creativity

The process of solving problems by **combining ideas or behavior in new ways**

## Convergent thinking

Type of thinking in which a problem is seen as having only one answer, and all lines of thinking will eventually lead to that single answer, using previous knowledge and logic

### Examples:

- Solving a multiple-choice test where there is only one correct answer
- A detective using evidence to identify the single culprit in a crime, like Sherlock Holmes

## Divergent thinking

Type of thinking in which a person starts from one point and comes up with many different ideas or possibilities based on that point

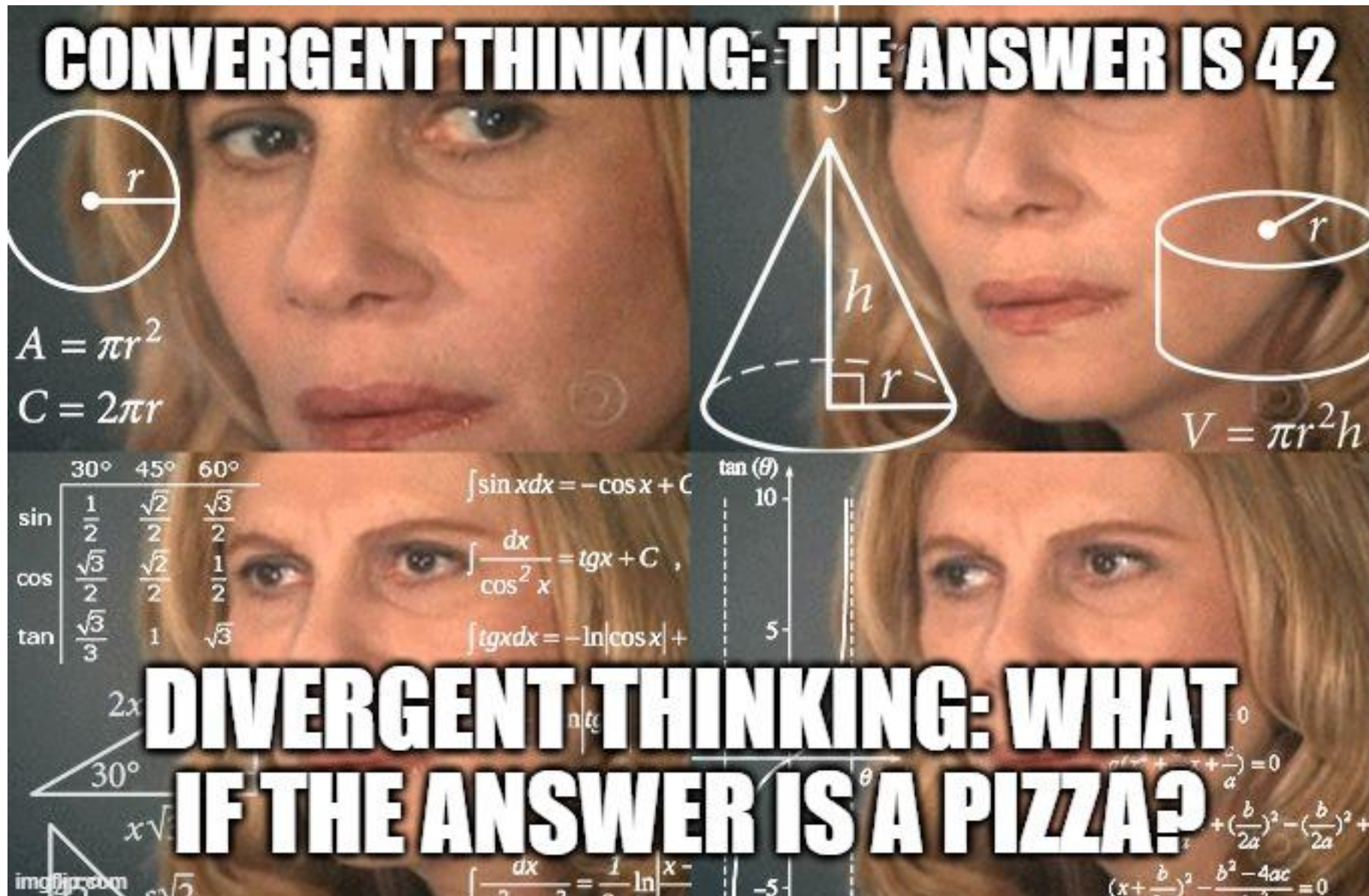
### Examples:

- Brainstorming a variety of possible uses for a common object, like a paperclip
- Composing music or writing creative lyrics- producing numerous novel ideas without a single right answer

(Csikszentmihalyi, 1996)

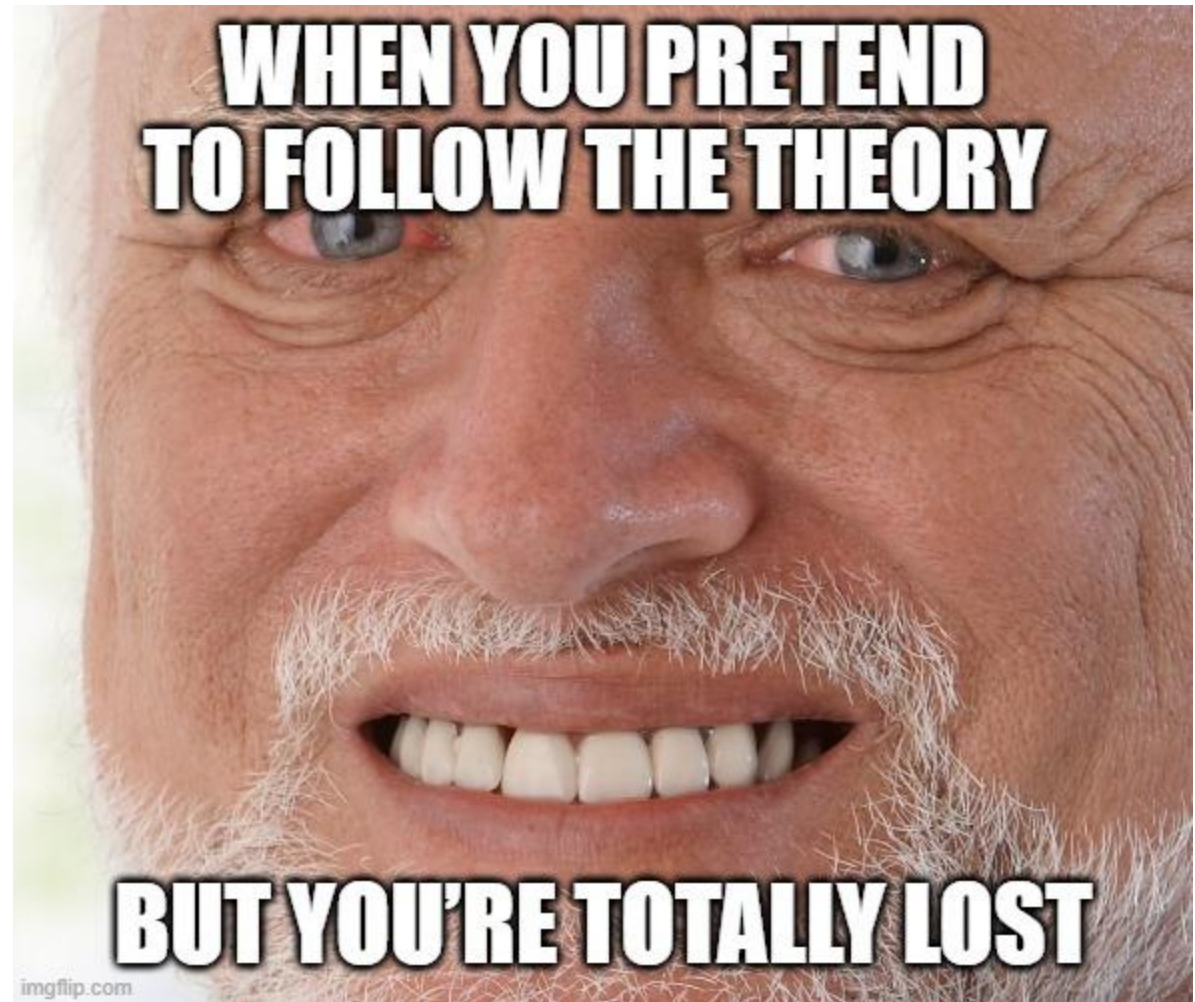
Divergent thinking = different and diverse kinds of responses

Convergent thinking = common sense kinds of responses



# RECAP

- Thurstone Primary Mental Abilities
  - Seven Primary mental ability
- Gardner's multiple intelligence
  - Independent faculties
- Cattell's theory of intelligence
  - Crystallized intelligence
  - Fluid intelligence
- Sternberg's triarchic theory
  - Analytical
  - Practical
  - Creative intelligence
- Creativity
  - Convergent thinking
  - Divergent thinking



# References

- Ciccarelli, S. K., White, J. N., & Misra, G. (2017). *Psychology* (5th ed.). Pearson Education India.
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# Intelligence

Course instructor:  
Dr Shweta Chawak  
Assistant Professor  
Humanities and Social Sciences  
IIT Bombay

# Mapping your intelligence journey!

- Plot key moment when different intelligence were important for you
- Quick theory reference:
  - Spearman: Single g factor
  - Thurstone: verbal comprehension, word fluency, numerical ability, spatial visualisation, associative memory, Perceptual speed, Reasoning
  - Gardner: Linguistic Intelligence, Logical-mathematical Intelligence, Spatial Intelligence, Bodily Kinesthetics Intelligence (Body smart), Musical Intelligence, Interpersonal intelligence, Intrapersonal intelligence (Self smart), Naturalistic intelligence, Existential intelligence (possible 9th )
  - Cattell's theory of intelligence: Crystalized intelligence, Fluid intelligence
  - Sternberg: Analytical, Creative, Practical
  - Creativity: Convergent and divergent

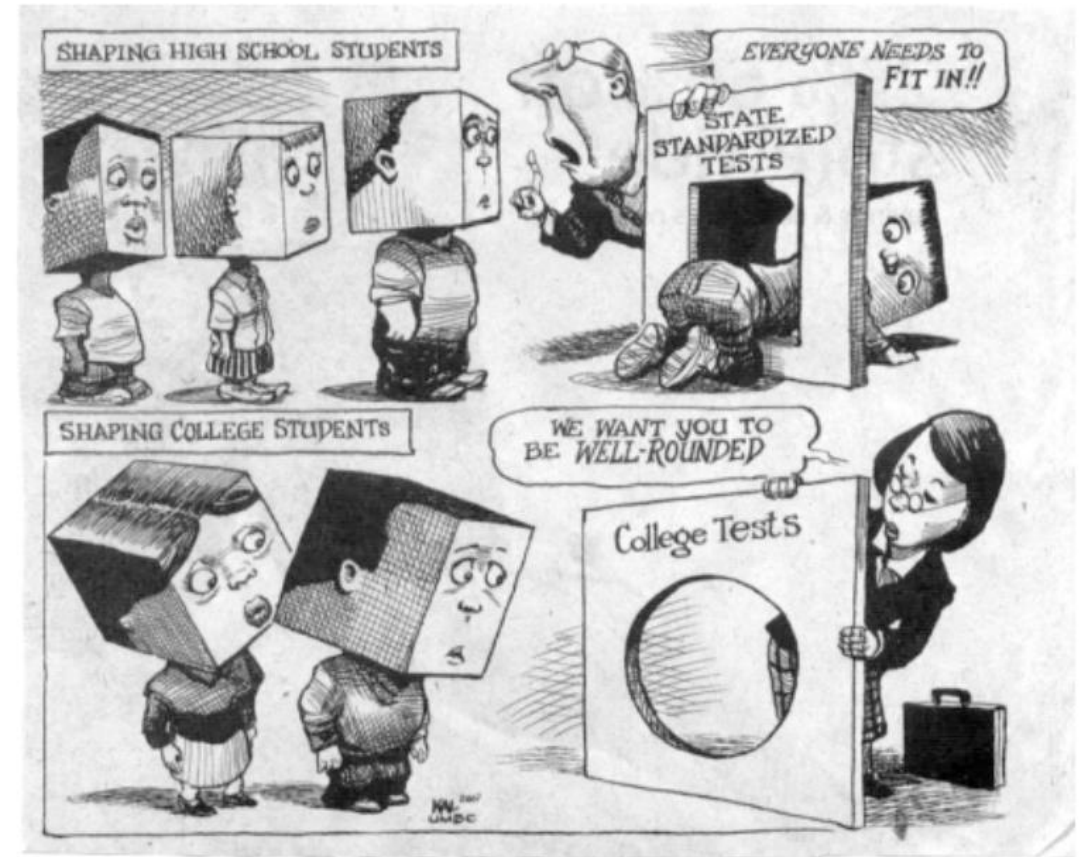


# History of testing



# Types of Cognitive ability testing

- Ability tests (Present):
  - Assess a person's innate or general cognitive capacity
  - Natural capability in specific skills
  - For example: Wechsler Adult Intelligence Scale (WAIS)
- Aptitude tests (Future):
  - Measures the potential and capacity to learn new skills
  - Persons ability to acquire knowledge or skills that have not been learned
  - For example: DAT
- Achievement tests (Past):
  - Evaluate what individuals have already learned or accomplished
  - Knowledge and skill gained through instructions or experience
  - For example: College exams



Speed test  
(time limit)

Power test  
(increasing  
difficulty)

# **Discussion: GATE as an Aptitude test**

# Insights!

- Does this test reflect aptitude required today to pursue career?
- Cultural Bias
- Narrow focus
- Coaching effect
- Learning potential in future?
- Relevance today!

MAYBE STANDARDIZED TESTING  
HAS GONE A LITTLE TOO FAR.

IS SHE  
PROFICIENT?

NEEDS  
IMPROVE-  
MENT.

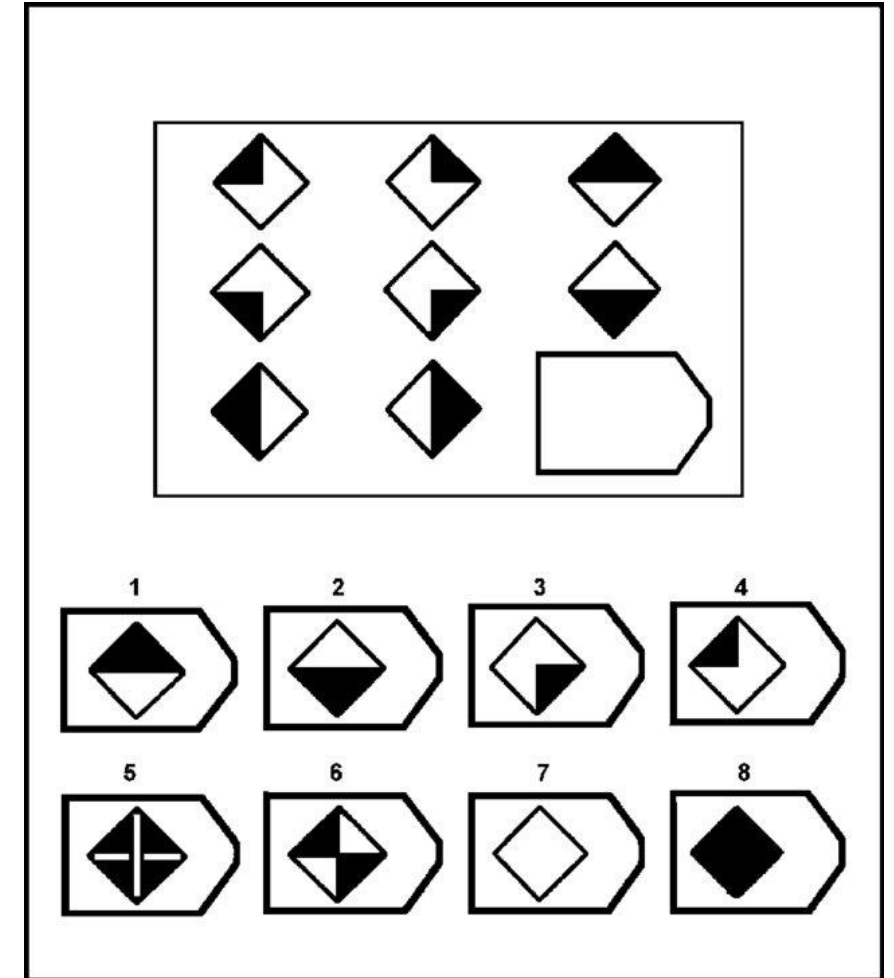
REPEAT

## Culture- fair test!



# RAVEN'S PROGRESSIVE MATRICES

- Developed by John Raven (1938)
- Designed to understand non-verbal intelligence
- Key feature:
  - Culture-fair test
  - Power test: items increase in difficulty
  - Some may include time limit but focus is on quality of answer and ability to solve complex patterns



# THOUGHTS?

- Any intellectual ability required?
- Is it culture-fair?
- Does it study ability?

# CONCLUSION

- History of testing
- Current intelligence testing
  - Ability, aptitude, achievement testing
- Culture-fair test
  - Raven's progressive matrices

# MOTIVATION



Course instructor:  
Dr Shweta Chawak  
Humanities and Social Sciences  
Indian Institute of Technology Bombay

# A Heads-up!

- This presentation includes topics related to mental health disorders, such as eating disorders, stress, and emotional well-being. Some content may be sensitive or emotionally challenging
- If you feel uncomfortable at any point, please feel free to pause, take a break, or reach out for support
- Your well-being is important

**Stand up and Stretch!**

# Why we do the things we do?



# What is Motivation?

- Latin word *movere*, which means “to move”
- Motivation is what “moves” people to do the things they do
- The process by which activities are **started**, **directed**, and **continued** so that physical or psychological needs or wants are met



# Types of Motivation?

## Extrinsic motivation

A person performs an action because it leads to an outcome that is separate from the person



## Intrinsic motivation

A person performs an action because the act itself is rewarding or satisfying in some internal manner



# How did Studying Motivation Begin?

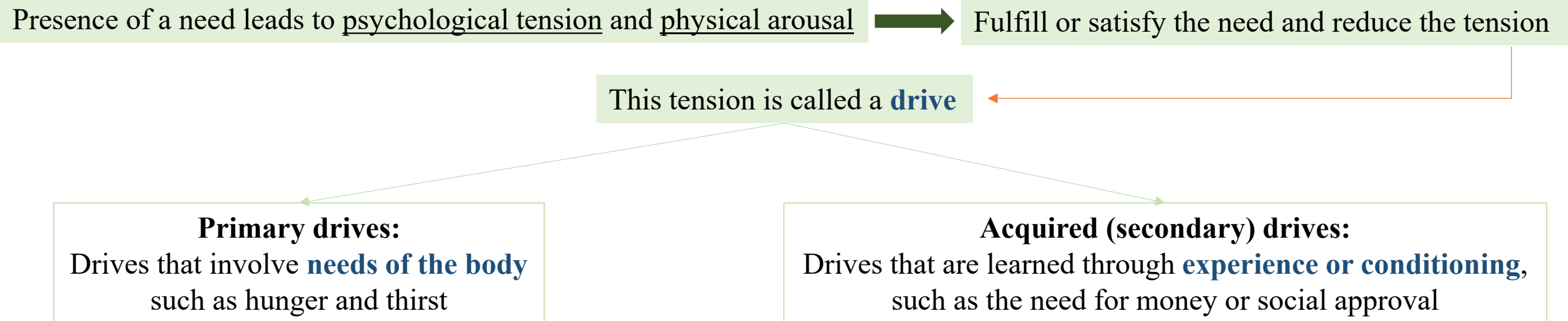
## Evolutionary Approach

- **Instincts**: biologically determined and **innate patterns** of behavior that exist in both people and animals
- Unlearned
- E.g., in animals: nest building, mating, territory protection
- In humans, reproduction, fear, survival, anger, love, laughter, curiosity etc.
- Hereditary factors may account for more than 50% variance in aspects of human cognition, temperament, and personality

# How did Studying Motivation Begin? (cont'd...)

## Drive-reduction theory (Clark Hull)

- Focusses on the concept of needs and drives
- A need is a requirement of some material that is **essential for survival** of the organism
- E.g., food or water



# How did Studying Motivation Begin? (cont'd...)

## Primary drives:

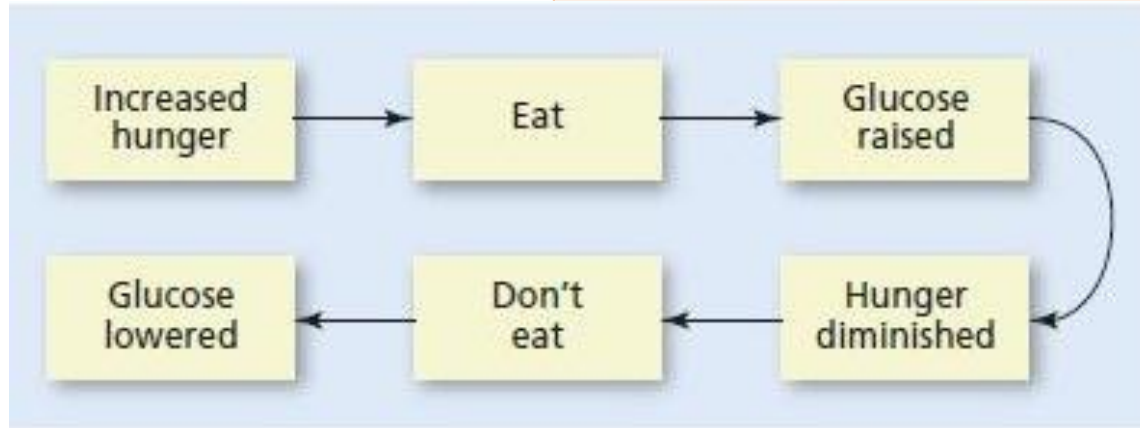
Drives that involve **needs of the body** such as hunger and thirst

E.g., after staying up late to complete assignments or scroll social media, students experience fatigue and drowsiness, which signal the body's need for sleep

## Acquired (secondary) drives:

Drives that are learned through **experience or conditioning**, such as the need for money or social approval

E.g., the desire for high grades is a secondary drive because it is learned through cultural and educational experiences, and reinforced by rewards like praise, scholarships, or job prospects

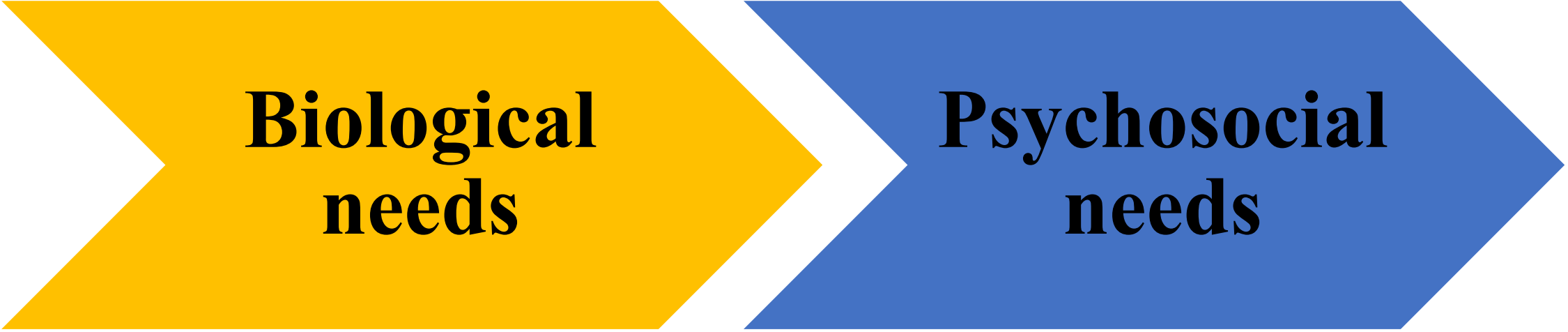


**Homeostasis:** The tendency of the body to maintain a steady state (body's version of thermostats)



- Drive-reduction theory explains the actions people take to reduce tension created by needs
- However, why do people eat when they are not really hungry?
- People don't always seek to reduce their inner arousal either; sometimes they seek to increase it
- Motivation is complex and multi-dimensional
- Influenced by: needs, level of arousal, external incentives, personal meaning & importance

# Types of Needs



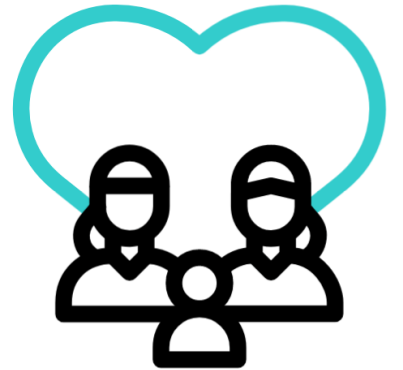
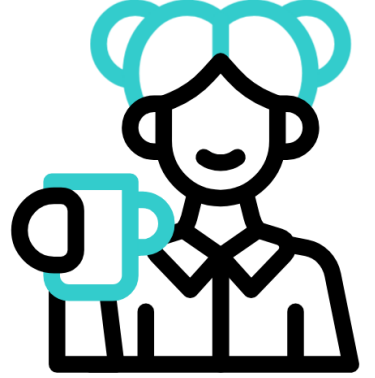
The diagram consists of two large, stylized arrows pointing from left to right. The first arrow is yellow and contains the text 'Biological needs'. The second arrow is blue and contains the text 'Psychosocial needs'. The two arrows are positioned side-by-side, with a small gap between them, suggesting a sequence or relationship between the two types of needs.

**Biological  
needs**

**Psychosocial  
needs**



## Biological Needs



# Biological Needs: Hunger

- Satisfying hunger is one of our most primary needs
- Eating is not only a basic survival behavior that reduces a primary drive
- Contraction of stomach where there a lack of food

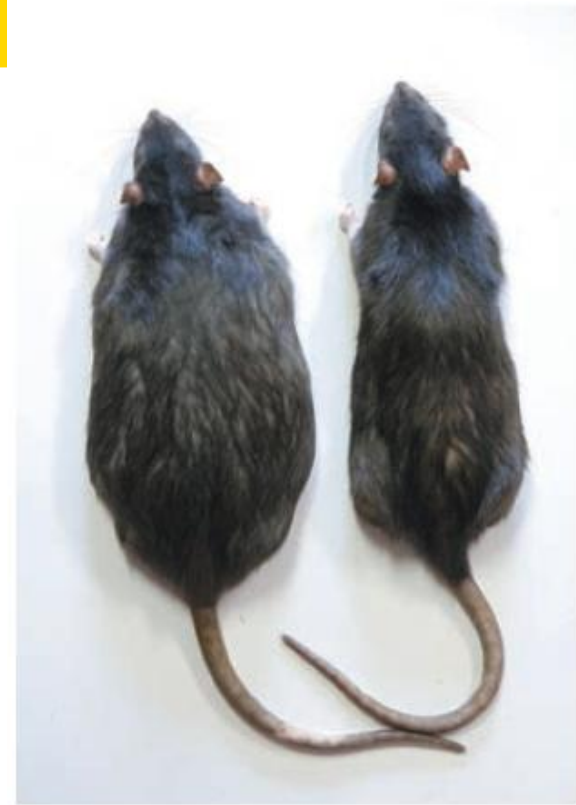
## Physiological factors:

## Hormonal influences

- Insulin: helps absorb glucose from the bloodstream for energy
- Leptin: controls appetite; signals body to stop; satiated

## The Brain

- Hypothalamus
- Ventromedial hypothalamus (VMH): stops eating response when glucose levels go up
- Lateral hypothalamus (LH): influence the onset of eating when insulin goes up



**Figure 9.4 Obese Laboratory Rat**

The rat on the left has reached a high level of obesity because its ventromedial hypothalamus has been deliberately damaged in the laboratory. The result is a rat that no longer receives signals of being satiated, and so the rat continues to eat and eat and eat.

# MOTIVATION



Course instructor:  
Dr Shweta Chawak  
Humanities and Social Sciences  
Indian Institute of Technology Bombay

# RECAP

- Motivation is what “moves” people to do the things they do
- Types of Motivation
  - Extrinsic
  - Intrinsic
- Evolutionary – Instincts
- Drive reduction theory: Needs – psychological tension called drive – reduce tension
  - Primary drive: bodily needs
  - Acquired (secondary) drive: experience
- Types of needs
  - Biological
  - Psychological
- Hunger: Most primary needs
  - Physical factors: hormones and brain
  - Social components

# Biological Needs: Hunger (Cont'd...)

## Social components:

- Conditioning
- Appeal of food: external features of food rather than the internal hunger
- Cultural factors: food portion differences
- Gender norms: practices where women eat last and least
- Escape or a comfort in times of stress: immediate pleasure

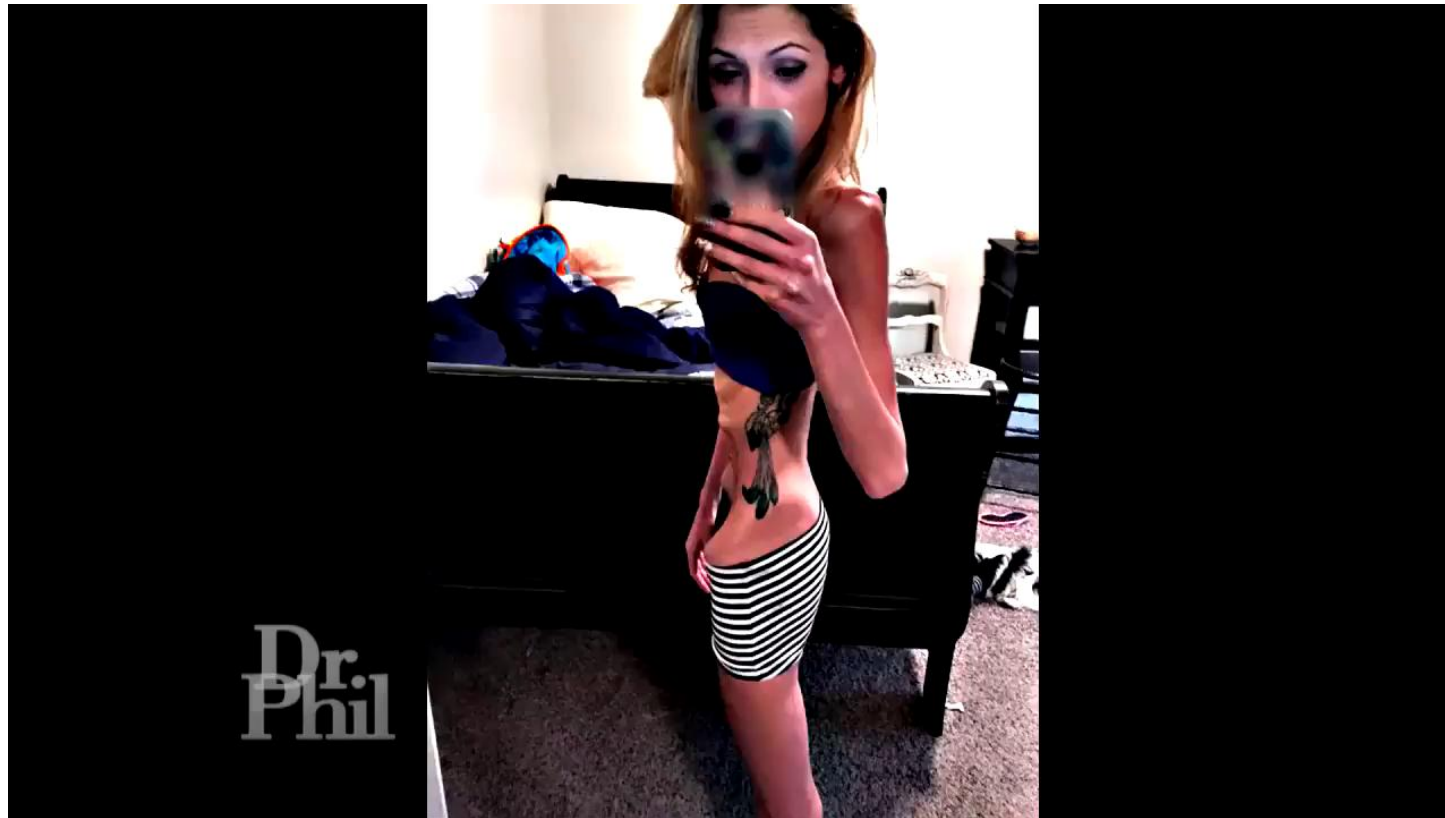




# Biological Needs: Hunger (Cont'd...)

## Eating disorders:

- **Anorexia nervosa:** severe eating disorder in which people may refuse to eat while denying that their behavior and appearance—which can become skeleton-like—are unusual



@Dr Phil

# Biological Needs: Hunger (Cont'd...)

## Eating disorders:

- **Bulimia:** A disorder in which a person binges on large quantities of food, followed by efforts to purge the food through vomiting or other means
- Brain scans show that individuals with eating disorders process information about food differently from healthy individuals
- Genetic factors that cause chemical imbalance
- Social beauty norms
- Overly demanding parents or family issues

**Arousal Theory**



# Psychosocial Needs



**Hierarchy of Needs**

# Psychosocial Needs: Arousal Theory

- People have an **optimal** (best or ideal) level of tension
- The tension is maintained by increasing or decreasing stimulation
- Task performance: severe anxiety versus boredom
- Moderate level of arousal is the best

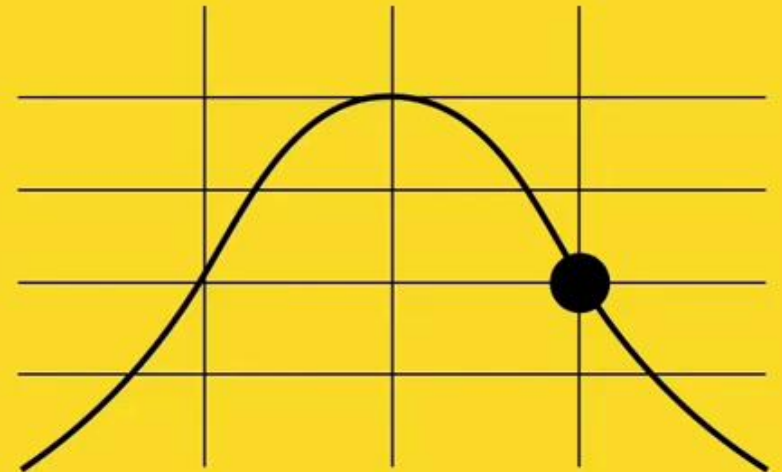
## Yerkes-Dodson law

- When tasks are simple, a higher level of arousal leads to better performance
- When tasks are difficult, lower levels of arousal lead to better performance

Easy tasks = “high–moderate” level for optimal performance  
Difficult tasks = “low–moderate” level for optimal performance



# The Yerkes-Dodson Law



# Psychosocial Needs: Arousal Theory (cont'd...)

Romantic partners who are under aroused may pick a fight with their spouse



Highly anxious students may try to reduce anxiety to improve test performance



Students with no anxiety may lack motivation to study, lowering performance



A **sensation seeker** is someone who requires higher arousal and craves more complex, varied sensory experiences than others



Presence of others can enhance or hinder an individual's performance  
Adolescents often take riskier decisions in groups due to peer pressure compared to when they are alone



# Psychosocial Needs: Arousal Theory (cont'd...)

## Strengths

- **Clear explanation of basic motivation:** Explains how biological needs create drives that push behavior to reduce tension and restore balance (homeostasis)
- **Applicable across species:** Relevant for both humans and animals, providing a general framework for understanding motivated behavior

## Drawbacks

- **Cannot explain all behaviors:** Fails to account for behaviors that increase arousal or tension, such as curiosity-driven acts, or long-term goal pursuits that temporarily increase discomfort (e.g., studying hard)
- **Ignores cognitive and social factors:** Overlooks the role of thoughts, beliefs, social motivations, and cultural influences on human behavior

# MOTIVATION

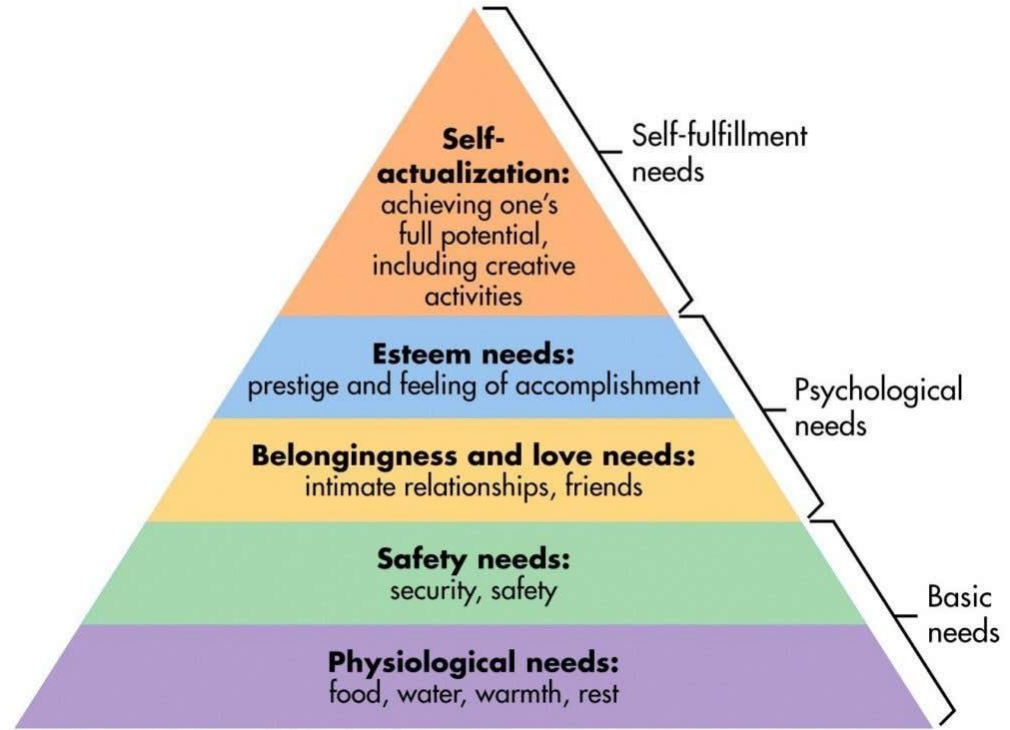
LOADING



Course instructor:  
Dr Shweta Chawak  
Humanities and Social Sciences  
Indian Institute of Technology Bombay

# Psychosocial Needs: Maslow's Hierarchy of Needs

- Human needs are arranged in a **hierarchy**
- Needs lower down in the hierarchy must be satisfied before individuals can attend to needs higher up
- Basic needs: physiological and short term
- Higher needs: psychological and long-term
- Higher the hierarchy, more difficult to satisfy



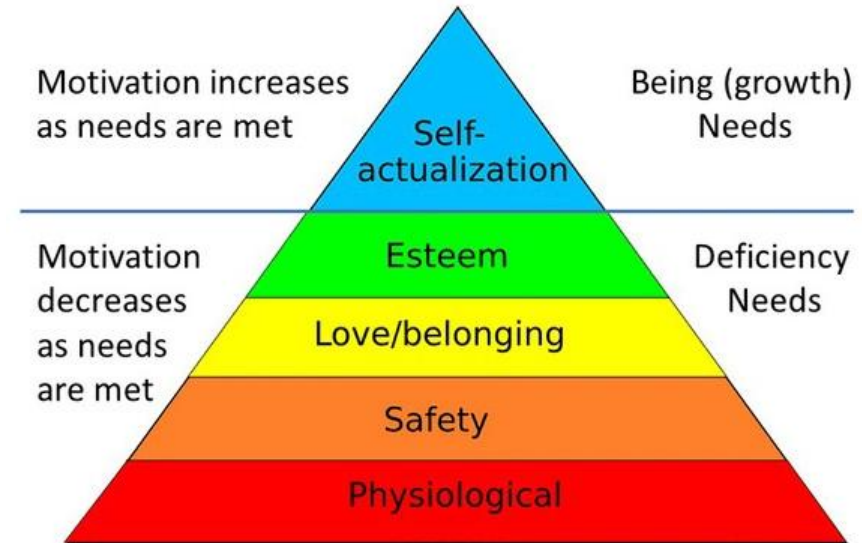
# Psychosocial Needs: Maslow's Hierarchy of Needs (cont'd)

## Growth needs:

- Maslow called the needs at the top of the pyramid growth needs
- These needs don't stem from a lack of something, but rather from a desire to grow as a person

## Deficiency needs:

- Deprivation; means to an end
- Physiological, security, social, and esteem needs are deficiency needs, which arise due to deprivation
- Satisfying these lower-level needs is important to avoid unpleasant feelings or consequences



# Psychosocial Needs: Maslow's Hierarchy of Needs (cont'd)

- **Physiological Needs:** Eating regular meals, getting enough sleep, staying hydrated during exams.
- **Safety Needs:** Feeling safe on campus, having financial aid or scholarships, access to healthcare services.
- **Love and Belongingness Needs:** Making friends in class or dorms, joining clubs or study groups, having supportive roommates.
- **Esteem Needs:** Receiving good grades, getting praise from professors, feeling confident in presentations or projects.
- **Self-Actualization:** Exploring interests through electives, engaging in meaningful research, developing talents, setting career goals

Basic Stuff Like Eating Spaghetti

# Psychosocial Needs: Maslow's Hierarchy of Needs (cont'd)

## Strengths:

- **Widely Applicable:** Fields like education, business, and healthcare to understand and improve human motivation and well-being
- **Focus on Positive Potential:** Highlights human growth and self-actualization rather than just problems or deficiencies

## Criticisms

- **Cultural Bias:** The focus on individual self-actualization reflects Western, individualistic values and may not apply well in collectivist cultures that prioritize social needs differently
- **Rigid Hierarchy Questioned:** People don't always follow a strict order; sometimes higher needs are pursued even if lower needs are unmet (e.g., artists pursuing creativity despite hardship)
- **Oversimplification:** Human motivation is complex and fluid

# Psychosocial Needs: McClelland's Acquired Needs Theory

## Need for affiliation

- Need for friendly social interactions and relationships with others
- Need to be liked by others, high regard by people

## Need for power

- Need for control or influence over other people
- Status and prestige: own expensive stuff; money

## Need for achievement

- Strong desire to succeed
- Desire for challenges and achievement for attaining goals

# Psychosocial Needs: McClelland's Acquired Needs Theory (cont'd...)

## **Strengths:**

- **Flexible:** Acknowledges that different people prioritize different needs (achievement, power, affiliation) which can change over time

## **Drawbacks:**

- **Simplifies human motivation:** Reduces motivation to three main needs, ignoring other possible motivational factors like security or self-actualization
- **Cultural limitations:** The dominance of certain needs may vary across cultures, limiting universal applicability

# References

- Ciccarelli, S. K., White, J. N., & Misra, G. (2017). *Psychology* (5th ed.). Pearson Education India.

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- <https://www.simplypsychology.org/>

# The Psychology of Emotion: I Feel, Therefore I Am

**Dr Shweta Chawak**  
Assistant Professor,  
Humanities and Social Sciences  
Indian Institute of Technology Bombay





# What if we didn't have emotions?

- **Decision-Making**
  - Without emotion– no value weighting
  - Decisions become computational but indecisive
- **Social Relationships**
  - No empathy, love, guilt
  - Relationships become purely functional
  - Breakdown of bonding and trust
- **Motivation**
  - No desire, fear, reward anticipation
  - Why study? Why avoid harm?
- **Survival**
  - Fear– threat detection
  - Disgust– disease avoidance
  - Without these– reduced survival

# What are emotions

**Emotion** is the "feeling" aspect of consciousness, characterized by three elements: physiological arousal, expressive behavior, and inner awareness of the feeling.

## **Three components of emotions:**

- Physiological experience of emotions
- Cognitive experience of emotions
- Behavioral expression of emotions

(Nevid, 2009; Myers, 2010)

# Bodily Arousal

- Activation of ANS
- Strong emotions – nervous system activation
- Release of hormones (epinephrine & norepinephrine) – increase alertness



**Autonomic Nervous System Controls Physiological Arousal**

| Sympathetic division (arousing) |                | Parasympathetic division (calming)    |
|---------------------------------|----------------|---------------------------------------|
| Pupils dilate                   | EYES           | Pupils contract                       |
| Decreases                       | SALIVATION     | Increases                             |
| Perspires                       | SKIN           | Dries                                 |
| Increases                       | RESPIRATION    | Decreases                             |
| Accelerates                     | HEART          | Slows                                 |
| Inhibits                        | DIGESTION      | Activates                             |
| Secrete stress hormones         | ADRENAL GLANDS | Decrease secretion of stress hormones |

# Cognition (appraisal and experience)

- Interpreting the subjective feeling by giving it a label:
  - Anger, fear, disgust, happiness, sadness, shame, interest, etc.
- *Cognitive element* involves retrieving memories of previous similar experiences, perceiving the context of the emotion, and coming up with a solution, **a label**.
- Interpretation of situation (threat vs harmless)
  - Example:** Rubber snake – startle – its fake – no fear
- The label a person applies to a subjective feeling is at least in part a learned response influenced by their language and culture.
- Labeling is learned: Labels for emotions may differ in people of different cultural backgrounds, shaped by language and social context.

**Research shows that even the subjective feeling of happiness varies by culture:**

## **Japanese Students**

Associated positive emotional state with friendly or socially engaged feelings (collectivistic)

## **US Students**

Associated positive state with socially disengaged feelings like pride (individualistic)

# Behavioural expression

- Approach: Joy, love, anger (move toward/  
attack)
- Avoidance: fear disgust (withdraw/ flee)
- Expressed through: facial expression, tone of  
voice, body posture

(Nevid, 2009; Myers, 2010)



**Anger**

**Furrowed brows, clenched jaw**



**Happiness**

**Smile, eye crinkle**



**Fear**

**Wide eyes, raised brows**



**Surprise**

**Raised brows, open mouth**



**Disgust**

**Nose wrinkle, lip curl**



**Sadness**

**Downturned mouth, tears**



**Contempt**

**One-sided smile**



**Love**

**Soft eyes, smile**

# Universal emotions

- These universalists believed that emotions evolved as a response to the environments of our primordial ancestors, so they are the same across all cultures.
- **Charles Darwin (1898):** Emotions are a product of evolution and therefore universal.
- Even children blind from birth can produce appropriate facial expressions without ever having witnessed them.

# Role of culture in emotions

**Display rules:** learned ways of controlling displays of emotion in social settings

## Cultural variation

- Regulation of emotional expression: Asian culture (suppression) versus Western culture (openness)
- E.g., biting one's tongue: India versus United States of America

## Gender differences

- Expression of emotions: Women reported extreme negative affect to grief compared to men
- However, when autonomic arousal levels were compared, no significant gender differences
- Men restrained their outward emotions



When is emotional restraint beneficial?  
When is open emotional expression more effective?



(Elfenbein & Ambady, 2002; Haidt & Keltner, 1999; Khosla, 1994)

# Functions of an emotion

## *Prepares us for action*

- If you saw an angry dog charging toward you, your emotional reaction (fear) would be associated with physiological arousal of the sympathetic division of the autonomic nervous system, the activation of the “fight-or-flight” response

## *Shapes our future behaviour*

- Emotions promote learning that will help us make appropriate responses in the future
- For instance, your emotional response to unpleasant events teaches you to avoid similar circumstances in the future

## *Helps us interact more effectively with others*

- We often communicate the emotions we experience through our verbal and nonverbal behaviors, making our emotions obvious to observers
- These behaviors can act as a signal to observers, allowing them to understand better what we are experiencing and to help them predict our future behavior.

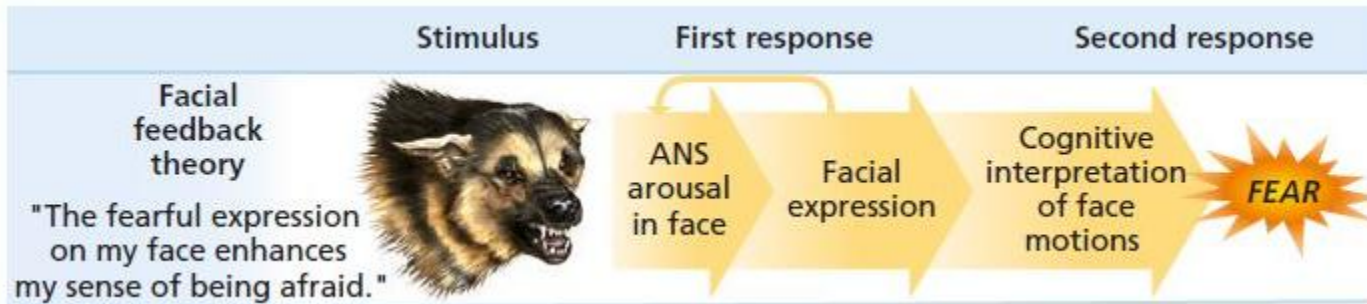
(Frederickson & Branigan, 2005; Frijda, 2005; Gross, 2006)

# Theories of emotion

## Facial Feedback Hypothesis

- Emotions can be influenced by the facial expressions we project
- Provides feedback to the brain
- Facial expression = emotional experience
- However, unable to explain people with a paralysed face

**E.g., we smile when we feel happy, but smiling can also make us feel happy**







@themurkyofficial

# Theories of emotion: Which comes first? The thought or the feeling?

## James-Lange Theory

- Bodily reactions or sensations before emotions
- Focusses on autonomic nervous system (ANS)

### Drawbacks

- People with spinal cord injuries still experience emotions despite lack of sympathetic feedback
- Vagus nerve provides alternative feedback pathway, complicating the theory



Mnemonic tip: Jumpy Legs, then emotion



E.g., I see, I run, I become afraid

# Theories of emotion (con'd...)

## Cannon-Bard Theory

- Emotional experience and bodily reactions occur simultaneously
- Focusses on the thalamus

### Strengths

- Accounts for complex emotions

### Drawbacks

- Quick versus slow cognitive appraisals
- Overemphasis on the thalamus



Mnemonic tip: Cannon's Both at once!



E.g., I see, I experience both fear and a pounding heart,  
and then I run

(Lindquist et al., 2012; Nevid, 2009; Söderkvist et al., 2018)

# Theories of emotion (con'd...)

## Schachter-Singer Theory

- Two-factor model:
  - i) physiological arousal;
  - ii) cognitive interpretation or labelling from Environmental cues

### Strengths

- Accounts for biological and cognitive component

### Drawbacks

- Cognitive appraisal involves more than the external cues



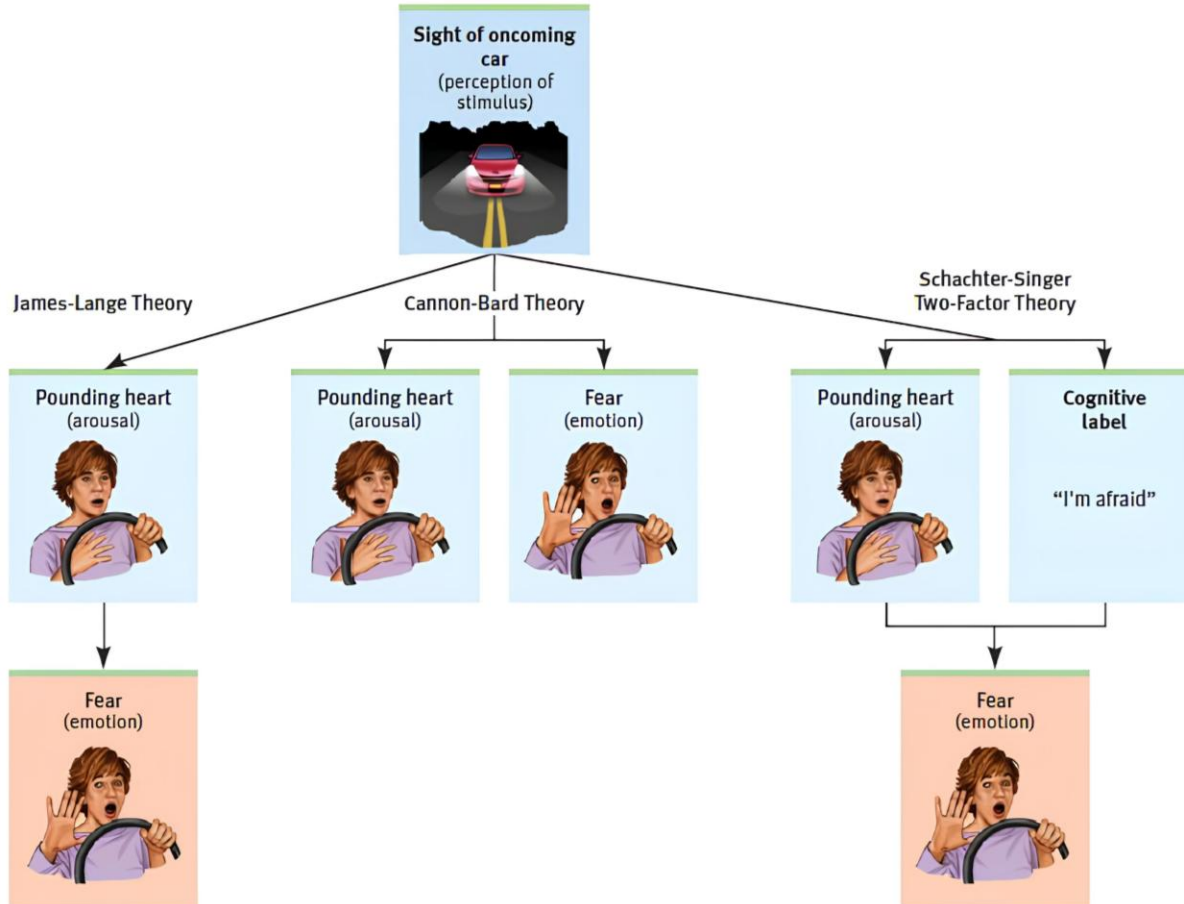
Mnemonic tip: Sweaty and Smart!



E.g., I see, I run, I think, and I become afraid

(Nevid, 2009; Myers, 2010)

# Comparison of the theories of emotion



My daughter saw the anatomy of  
an ant then this happened...😭



(Kyle Winther via ViralHog)

# The Psychology of Emotion: I Feel, Therefore I Am

**Dr Shweta Chawak**  
Assistant Professor,  
Humanities and Social Sciences  
Indian Institute of Technology Bombay



# Managing difficult emotions

- You respond instead of react
- To respond is to exercise **emotional intelligence**. To react is to be **emotional**
- So how do you consistently respond instead of react? Begin by slowing down the process



## RESPONDING



Responding means you think through what you want to have happen in an interaction or conflict

## REACTING



Reacting is immediate, without thought, and often results in a negative outcome

# How do I identify my emotion(s)?

1. What is the context/situation/person?

2. What am I feeling?

3. Allow yourself to feel your feelings. Sit with it

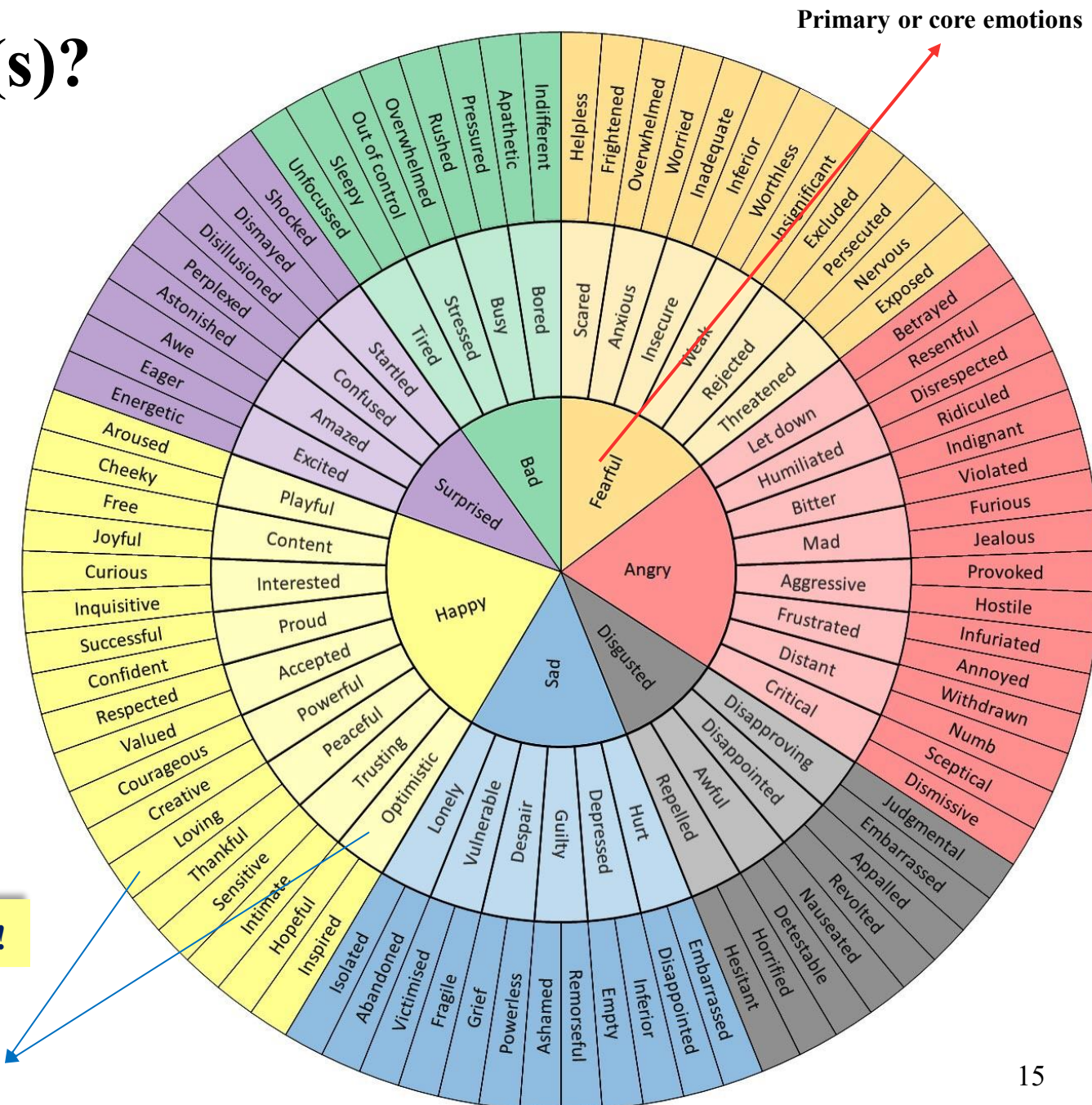
4. Notice the physical sensations in your body  
(i.e., pressure on chest, stomachache, headache)

5. Then identify whether you are experiencing a  
primary or secondary emotion

**Emotion identification is a skill related to emotional intelligence!**

(Salovey & Mayer, 1990)

Secondary emotions



# Breathing exercise



# 5 - 4 - 3 - 2 - 1

## GROUNDING EXERCISE



**5**

Things you can  
**SEE**



**4**

Things you can  
**FEEL**



**3**

Things you can  
**HEAR**



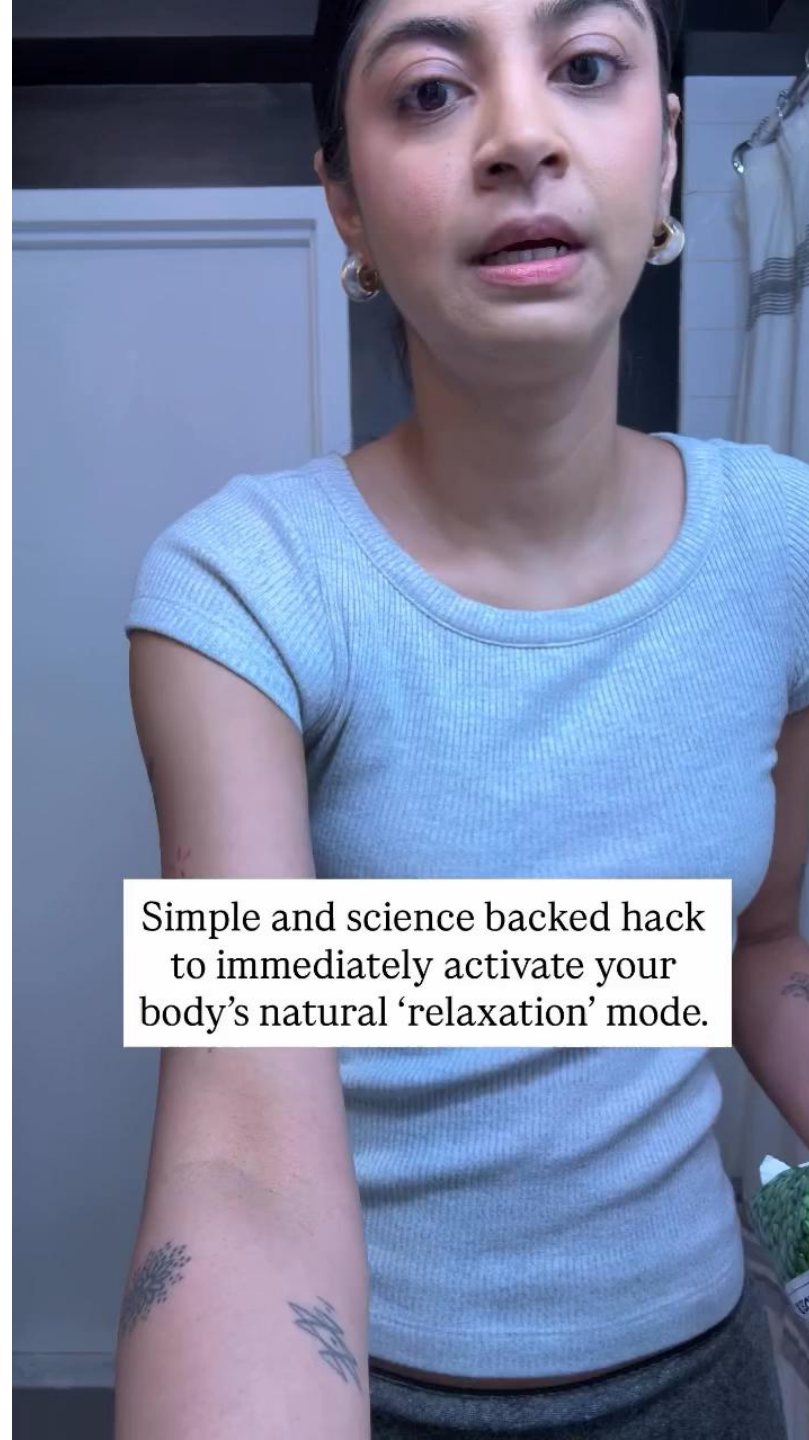
**2**

Things you can  
**SMELL**



**1**

Things you can  
**TASTE**



Simple and science backed hack  
to immediately activate your  
body's natural 'relaxation' mode.

# Other ways to manage your emotions to build your resilience

Be kind to yourself

Take care of yourself



Stay connected to people for support

Disconnect from social media

# RECAP

- Emotions have a physiological, thought, and behavioural component
- Universal and unique
- Theories: James-Lange, Cannon-Bard, and Schachter-Singer, facial feedback hypothesis
- Managing and identifying emotions – respond and not react

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# End-Semester Exam

- 20 Marks: Multiple choice
- 10 Marks: True/False and Match the following
- 10 Marks: Answer in one word (terminology)
- 10 Marks: Answer in Brief (Any 2 of 3)
- Full syllabus